PCOS ULTRASOUND CLASSIFICATION — SEEDED MODELS (REPRODUCIBLE RESULTS)

All reproducible experiments: fixed-seed runs of MobileNetV2, EfficientNet-B0 and EfficientNet-B3 (CNN fine-tuned end-to-end, and the same CNN's features classified with AdaBoost), and the soft-voting ensembles of those models. Test set: 1,468 images (1,052 Appears abnormal, 416 Appears normal).

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Section 1 — Single-seed folder

Results of the two notebooks of the single-seed folder: “(seeded) MobileNet and EfficientNet B0-Adaboost .ipynb” (MobileNetV2 and EfficientNet-B0, seed 2622715460) and “(seeded) EfficientNet B3 with Adaboost.ipynb” (EfficientNet-B3, seed 512). Each notebook trains the CNN with a fixed seed, then classifies its features with AdaBoost. Ranked by test accuracy.

Rank	Model	Classifier	Seed	Test Acc. (%)	Prec. (%)	Recall (%)	F1 (%)	Source notebook (.ipynb)
1	MobileNetV2	AdaBoost	2622715460	94.01	93.96	94.01	93.95	(seeded) MobileNet and EfficientNet B0-Adaboost .ipynb (cell 4)
2	EfficientNet-B3	AdaBoost	512	92.71	92.64	92.71	92.64	(seeded) EfficientNet B3 with Adaboost.ipynb (cell 4)
3	EfficientNet-B0	AdaBoost	2622715460	92.64	92.62	92.64	92.49	(seeded) MobileNet and EfficientNet B0-Adaboost .ipynb (cell 8)
4	MobileNetV2	Direct (fine-tuned CNN)	2622715460	92.44	92.57	92.44	92.20	(seeded) MobileNet and EfficientNet B0-Adaboost .ipynb (cell 4)
5	EfficientNet-B0	Direct (fine-tuned CNN)	2622715460	89.03	88.91	89.03	88.69	(seeded) MobileNet and EfficientNet B0-Adaboost .ipynb (cell 8)
6	EfficientNet-B3	Direct (fine-tuned CNN)	512	87.60	87.90	87.60	86.85	(seeded) EfficientNet B3 with Adaboost.ipynb (cell 4)

1. MobileNetV2 + AdaBoost (single-seed notebook) — Seed 2622715460

Source notebook: (seeded) MobileNet and EfficientNet B0-Adaboost .ipynb, cell 4 (single-seed folder), fixed SEED = 2622715460

Accuracy: 94.01%

Precision: 93.96%

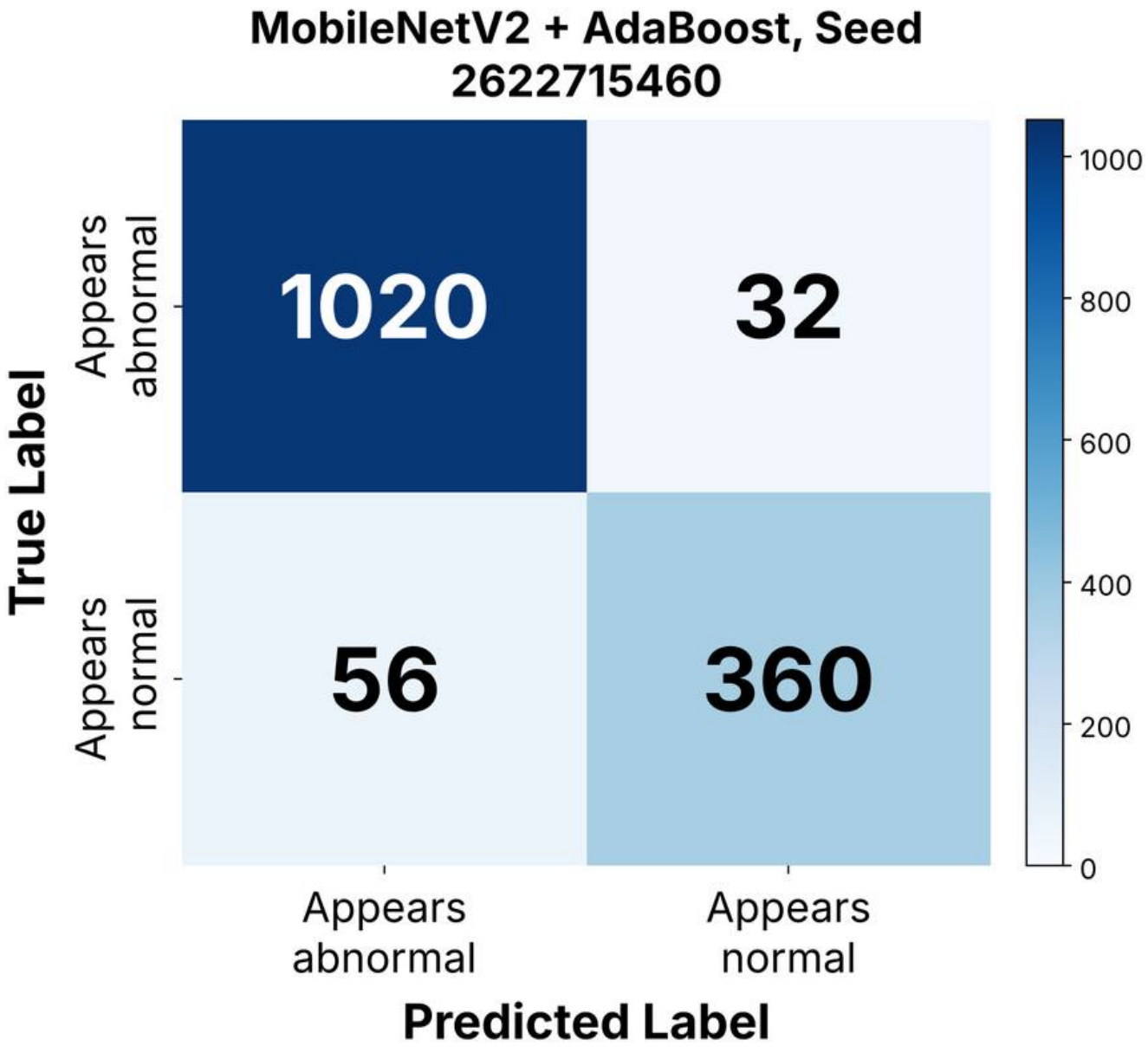
Recall: 94.01%

F1-score: 93.95%

AdaBoost validation accuracy: 67.29%

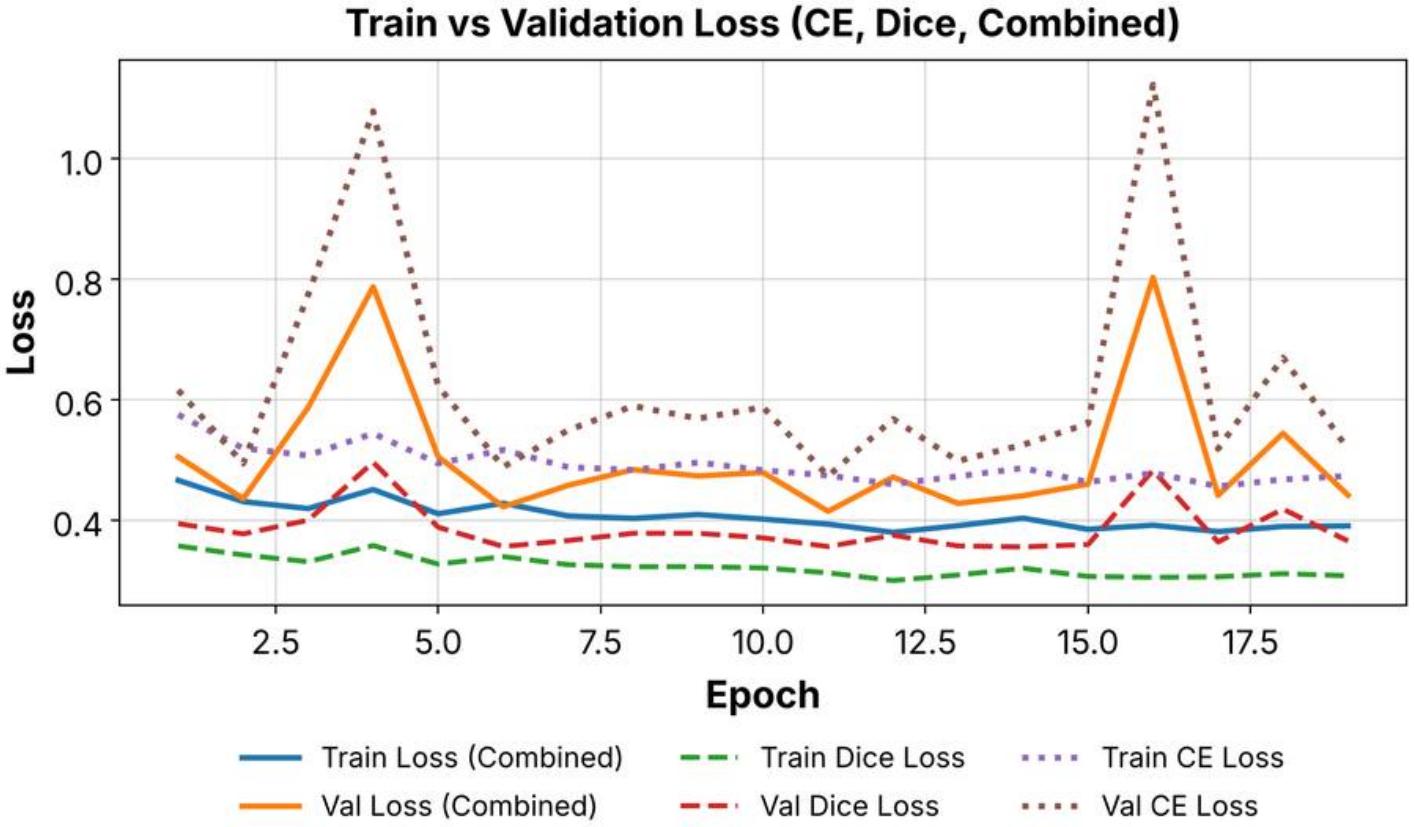
The same seed and model were also run inside 3_Seed_exp.ipynb (Section 2): the numbers on this page are identical to that seed's numbers there.

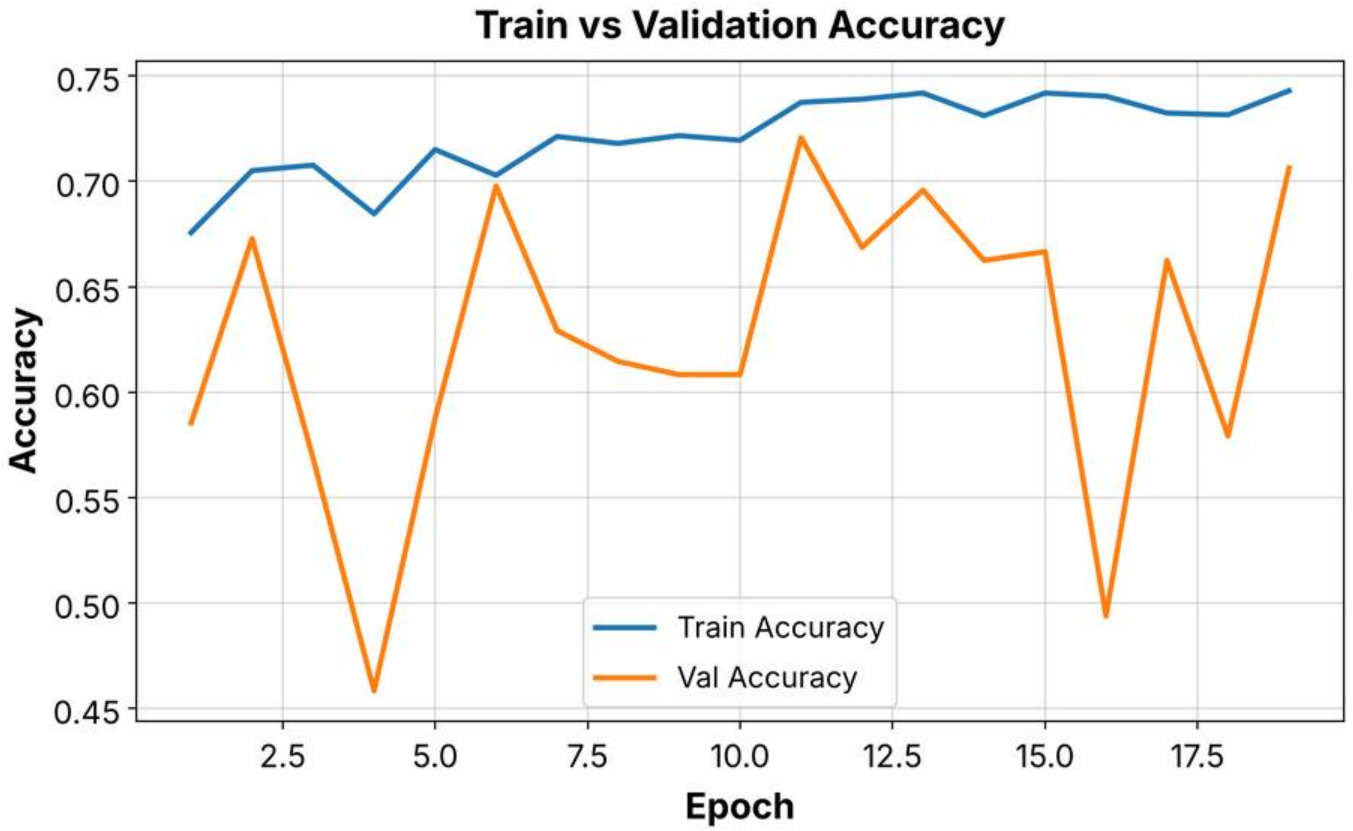
Confusion Matrix



1. MobileNetV2 + AdaBoost (single-seed notebook) — graphs

Training / Validation Graphs





2. EfficientNet-B3 + AdaBoost (single-seed notebook) — Seed 512

Source notebook: (seeded) EfficientNet B3 with Adaboost.ipynb, cell 4 (single-seed folder), fixed SEED = 512

Accuracy: 92.71%

Precision: 92.64%

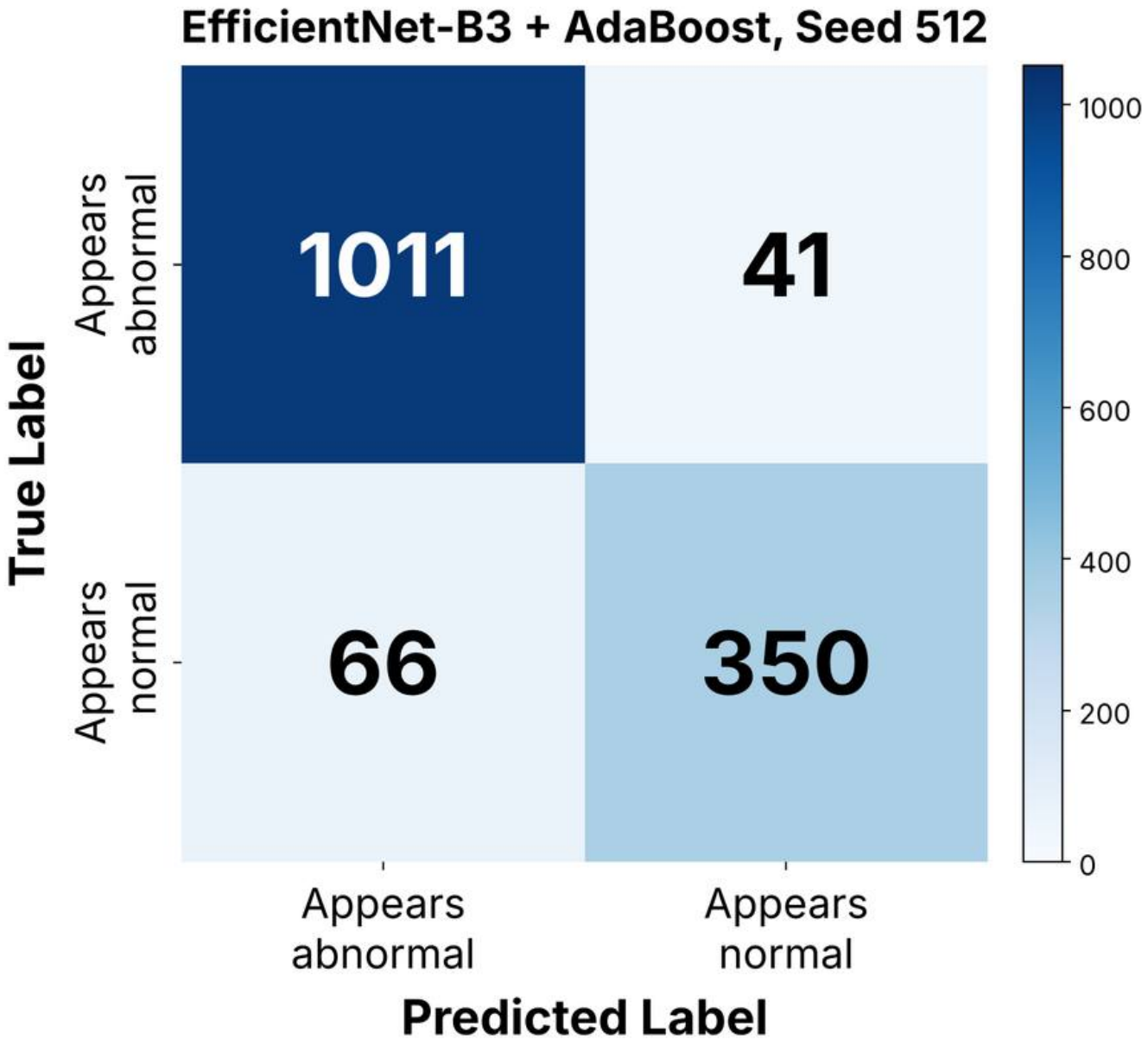
Recall: 92.71%

F1-score: 92.64%

AdaBoost validation accuracy: 69.17%

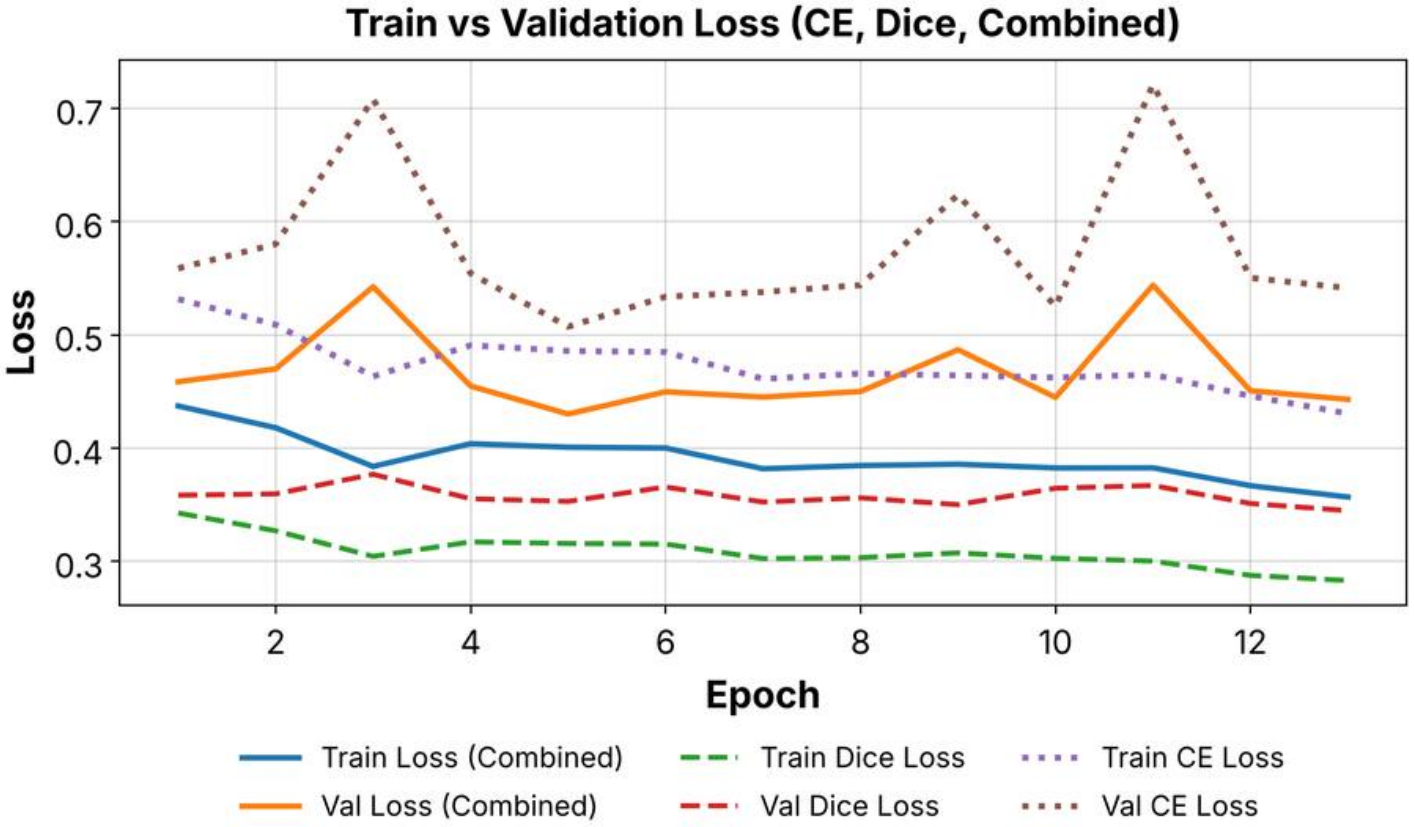
The same seed and model were also run inside 3_Seed_exp.ipynb (Section 2): the numbers on this page are identical to that seed's numbers there.

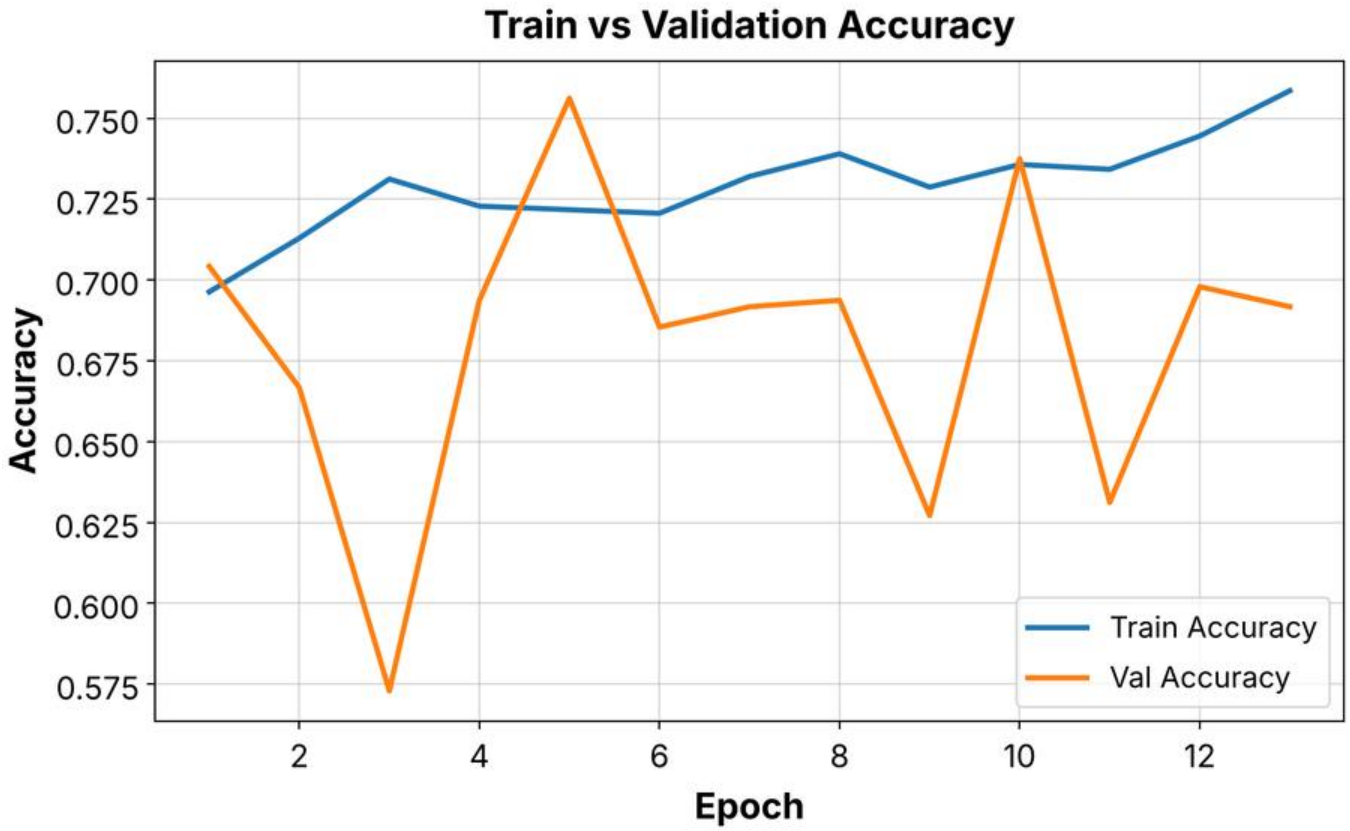
Confusion Matrix



2. EfficientNet-B3 + AdaBoost (single-seed notebook) — graphs

Training / Validation Graphs





3. EfficientNet-B0 + AdaBoost (single-seed notebook) — Seed 2622715460

Source notebook: (seeded) MobileNet and EfficientNet B0-Adaboost .ipynb, cell 8 (single-seed folder), fixed SEED = 2622715460

Accuracy: 92.64%

Precision: 92.62%

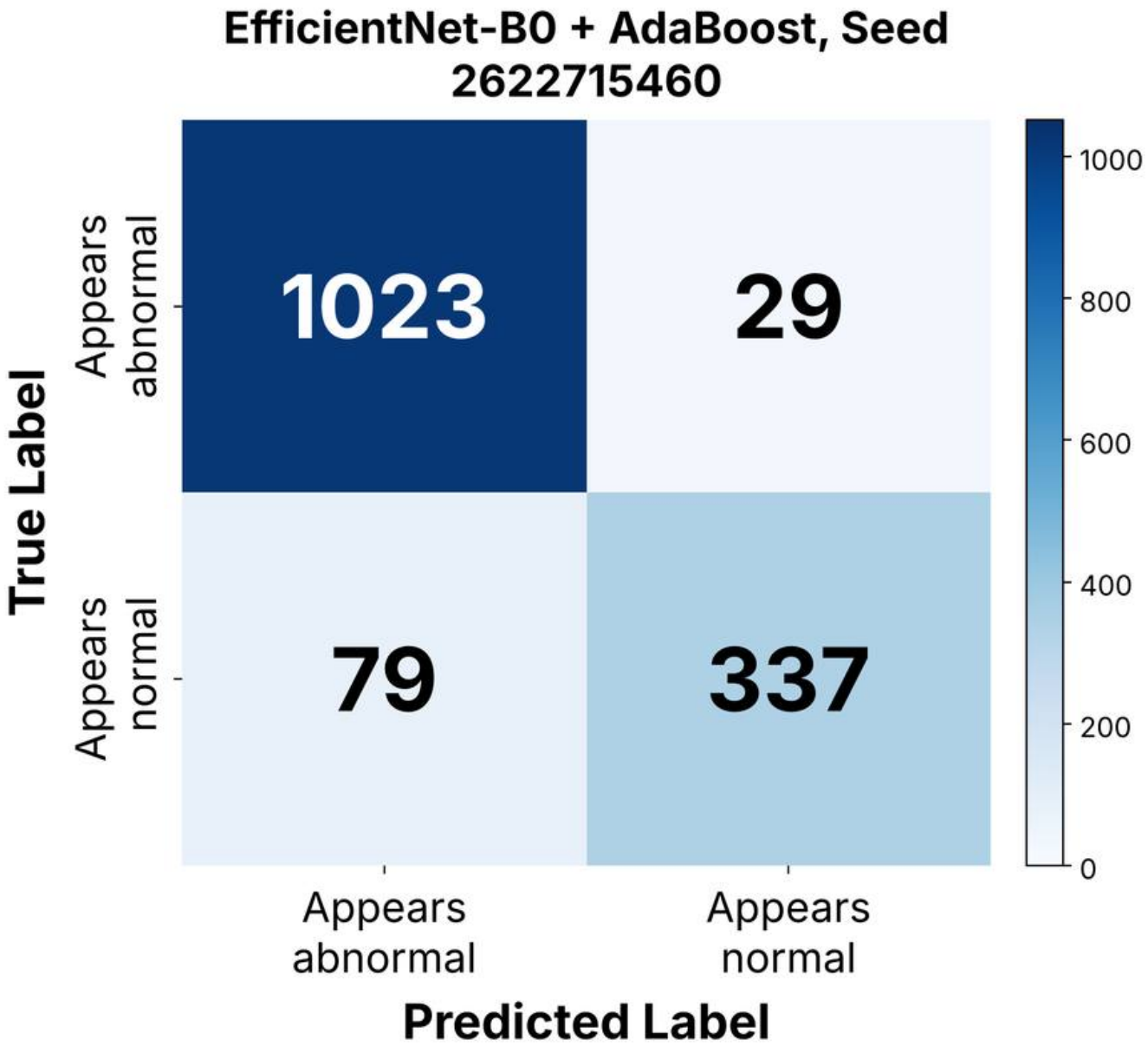
Recall: 92.64%

F1-score: 92.49%

AdaBoost validation accuracy: 71.46%

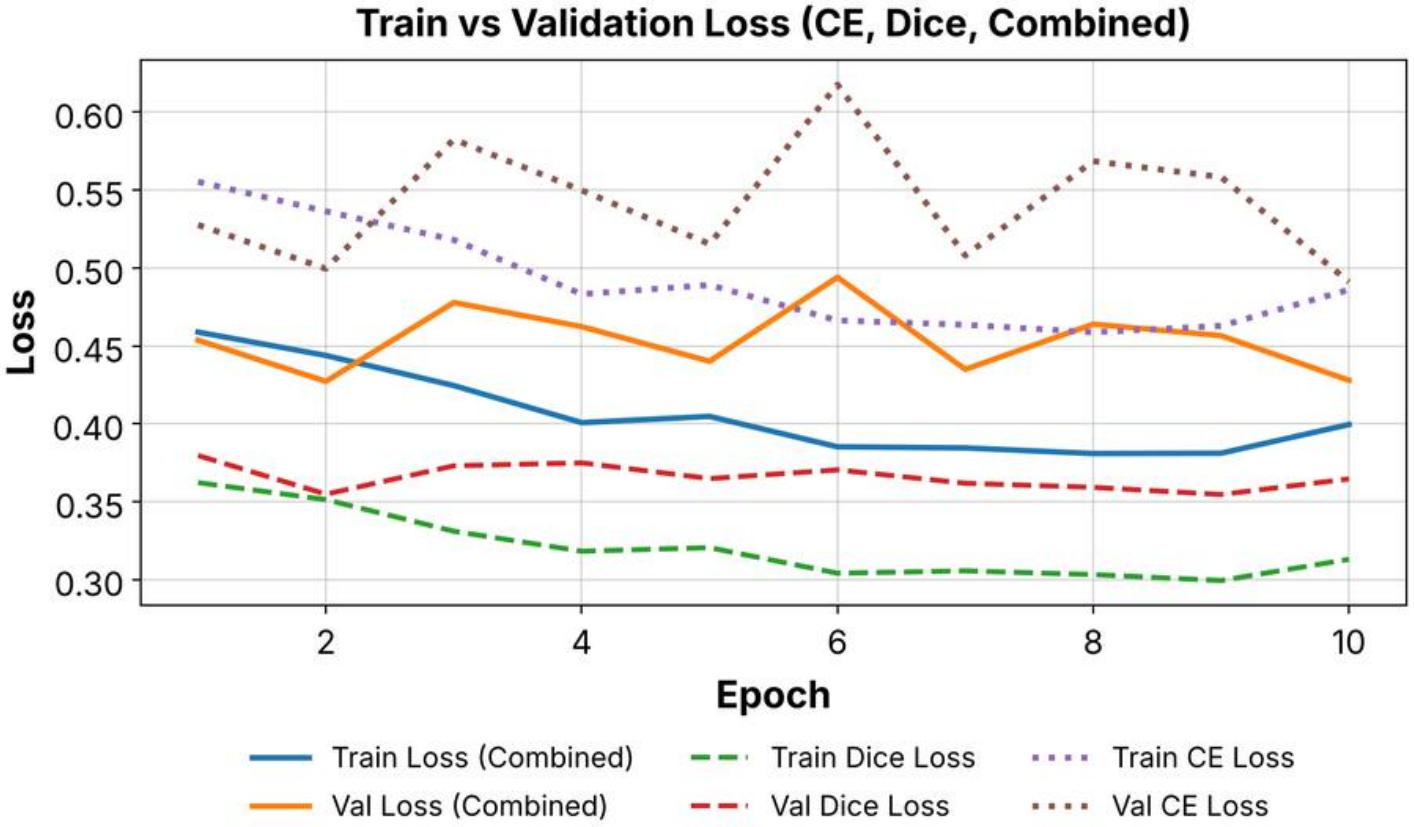
The same seed and model were also run inside 3_Seed_exp.ipynb (Section 2): the numbers on this page are identical to that seed's numbers there.

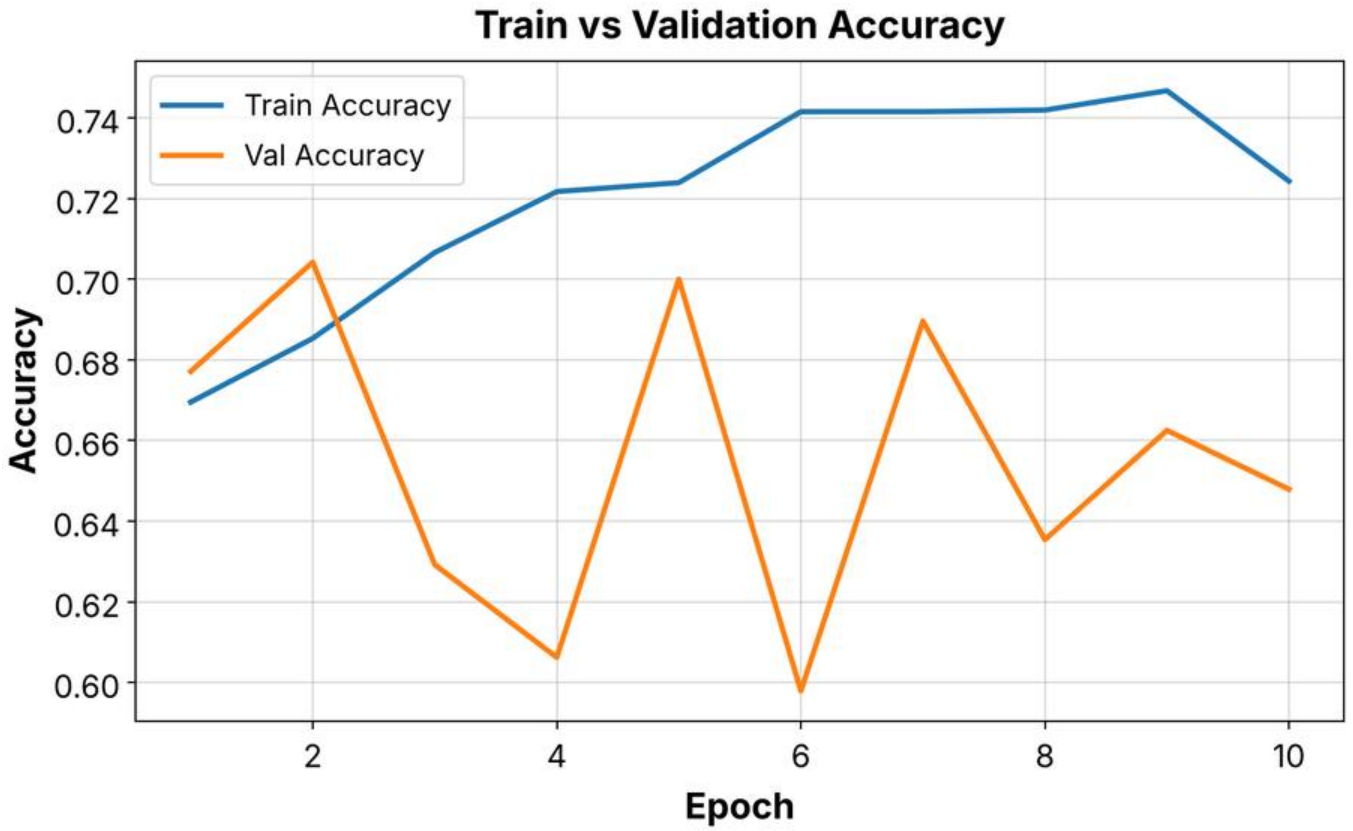
Confusion Matrix



3. EfficientNet-B0 + AdaBoost (single-seed notebook) — graphs

Training / Validation Graphs





4. MobileNetV2 — Direct (fine-tuned CNN) (single-seed notebook) — Seed 2622715460

Source notebook: (seeded) MobileNet and EfficientNet B0-Adaboost .ipynb, cell 4 (single-seed folder), fixed SEED = 2622715460

Accuracy: 92.44%

Precision: 92.57%

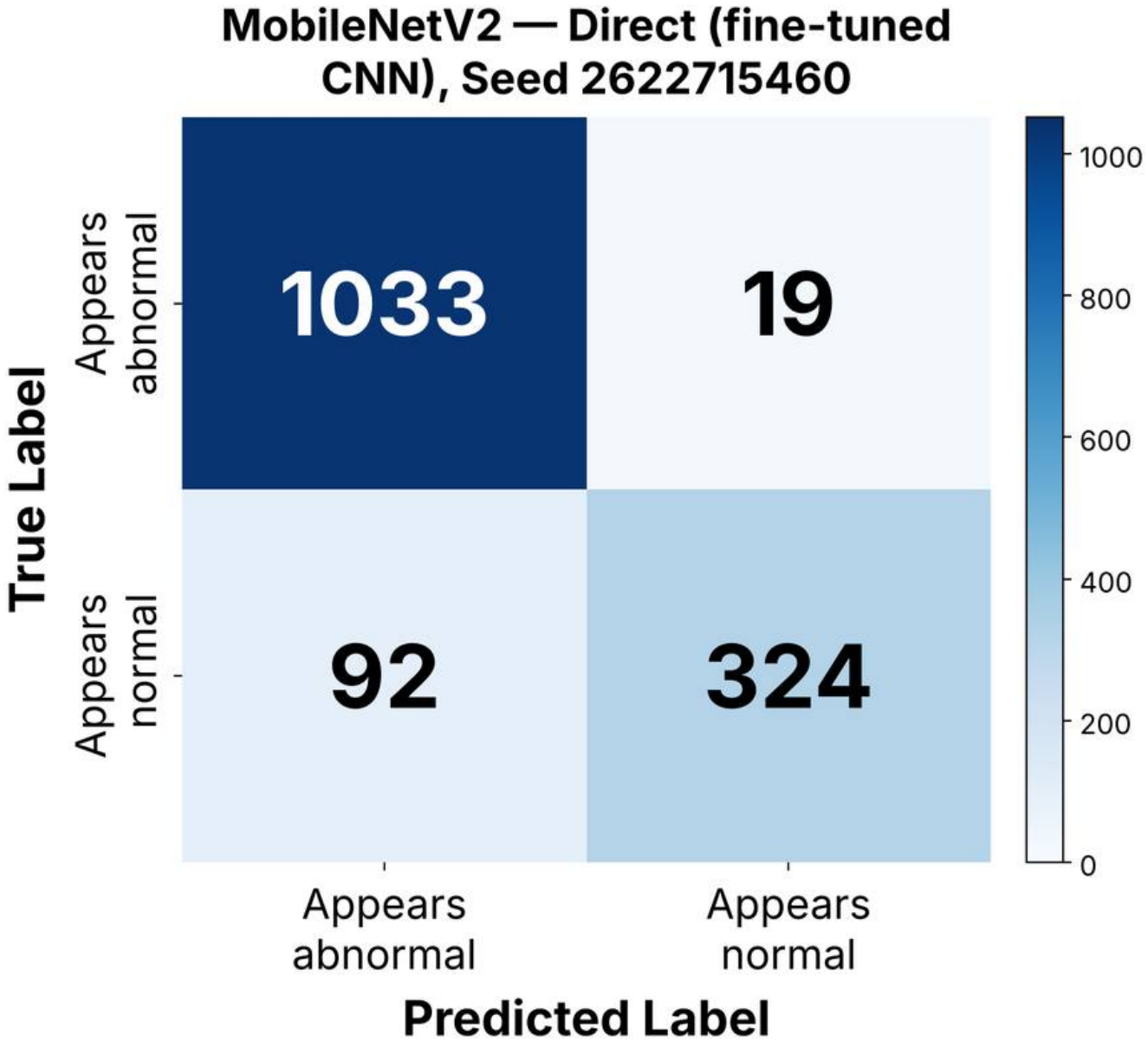
Recall: 92.44%

F1-score: 92.20%

Best validation accuracy: 72.08% **Test Dice loss:** 0.5267

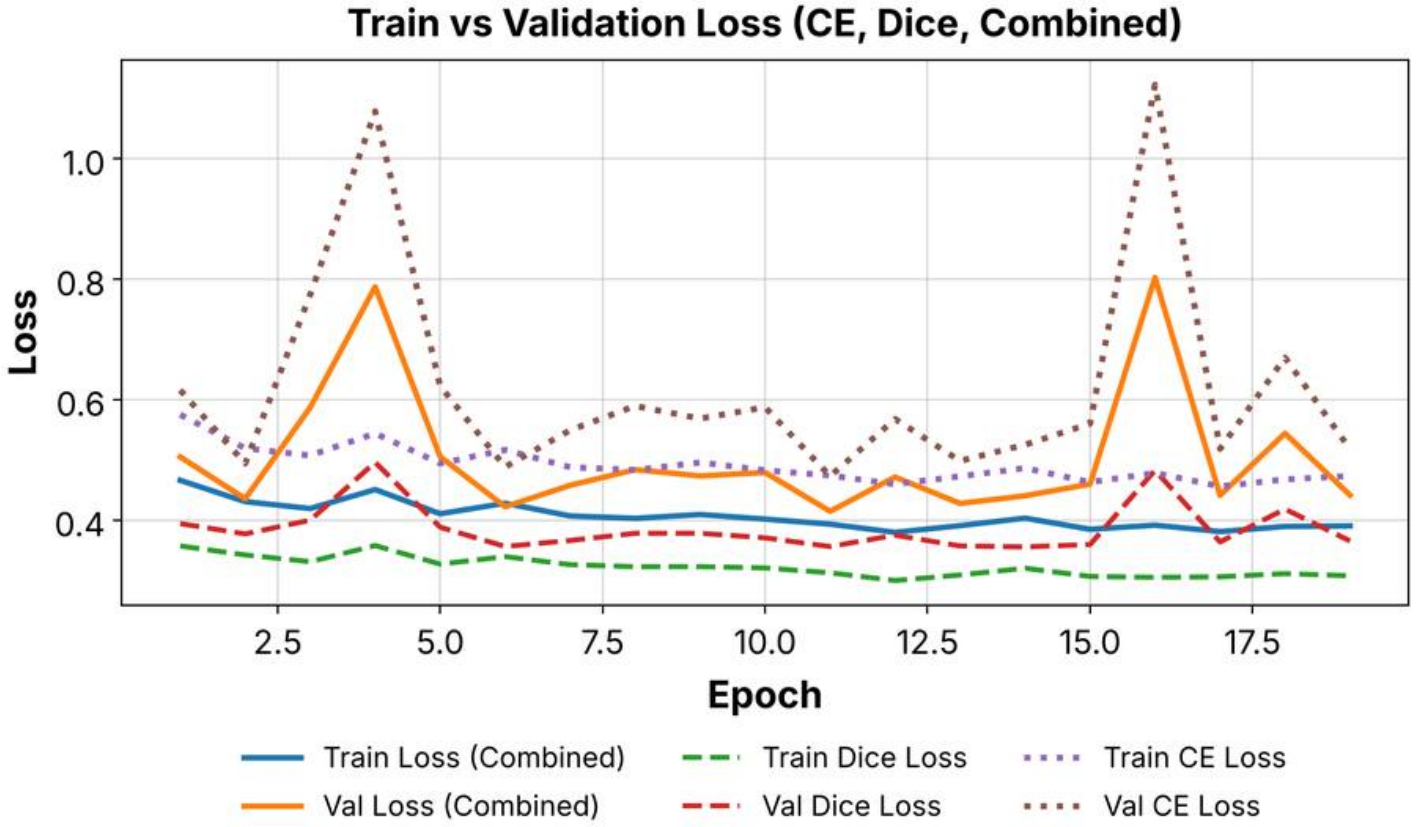
The same seed and model were also run inside 3_Seed_exp.ipynb (Section 2): the numbers on this page are identical to that seed's numbers there.

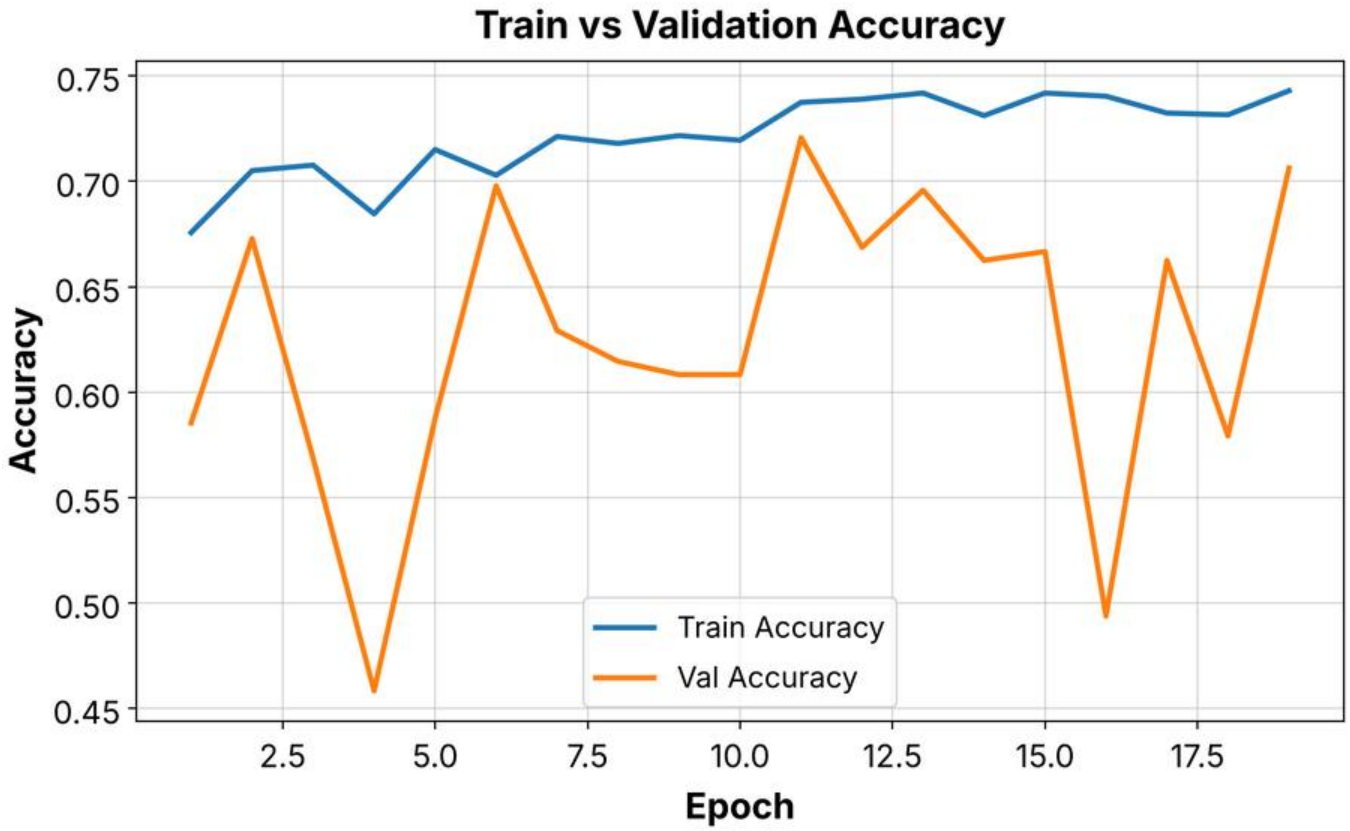
Confusion Matrix



4. MobileNetV2 — Direct (fine-tuned CNN) (single-seed notebook) — graphs

Training / Validation Graphs





5. EfficientNet-B0 — Direct (fine-tuned CNN) (single-seed notebook) — Seed 2622715460

Source notebook: (seeded) MobileNet and EfficientNet B0-Adaboost .ipynb, cell 8 (single-seed folder), fixed SEED = 2622715460

Accuracy: 89.03%

Precision: 88.91%

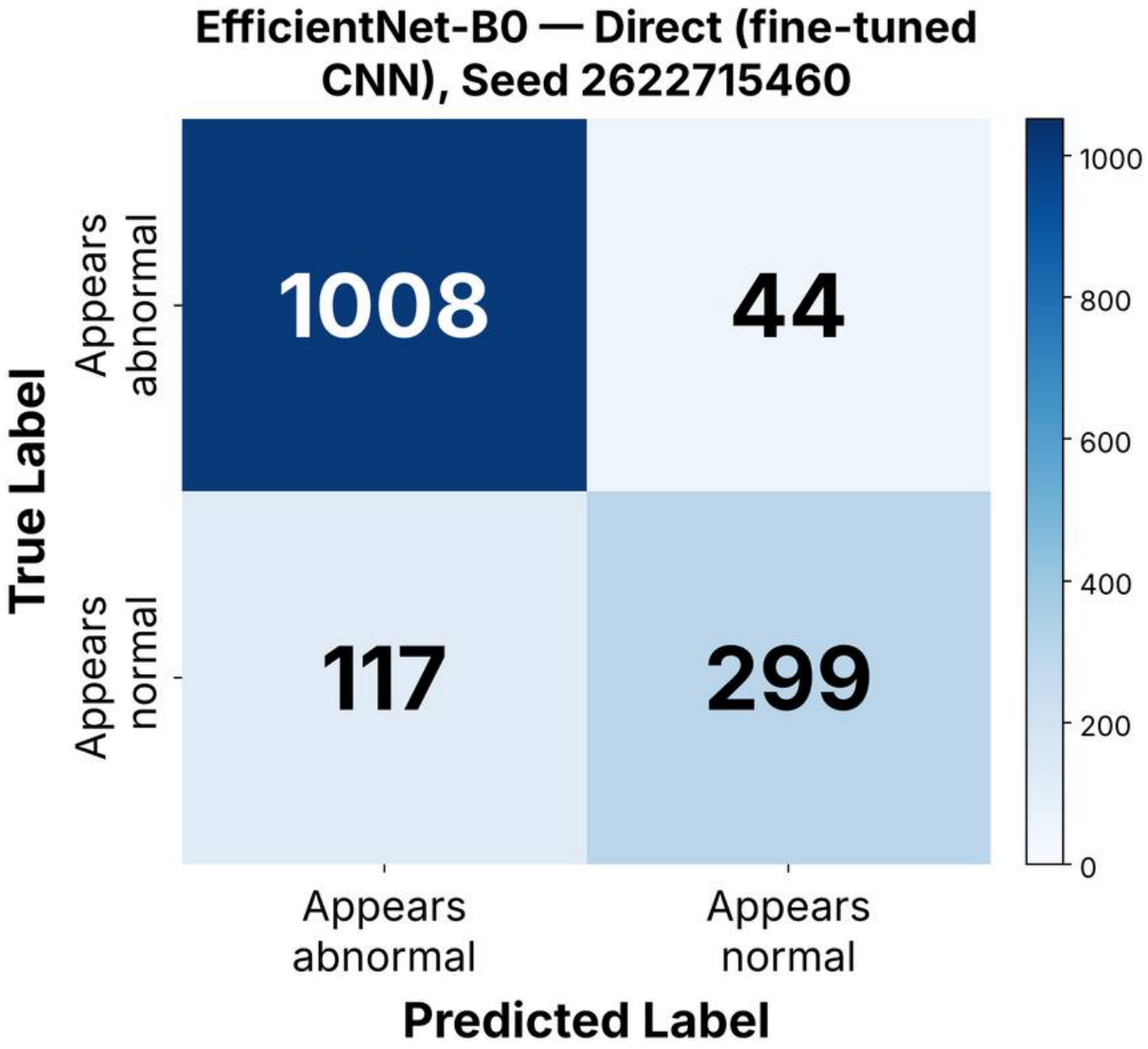
Recall: 89.03%

F1-score: 88.69%

Best validation accuracy: 70.42% **Test Dice loss:** 0.5193

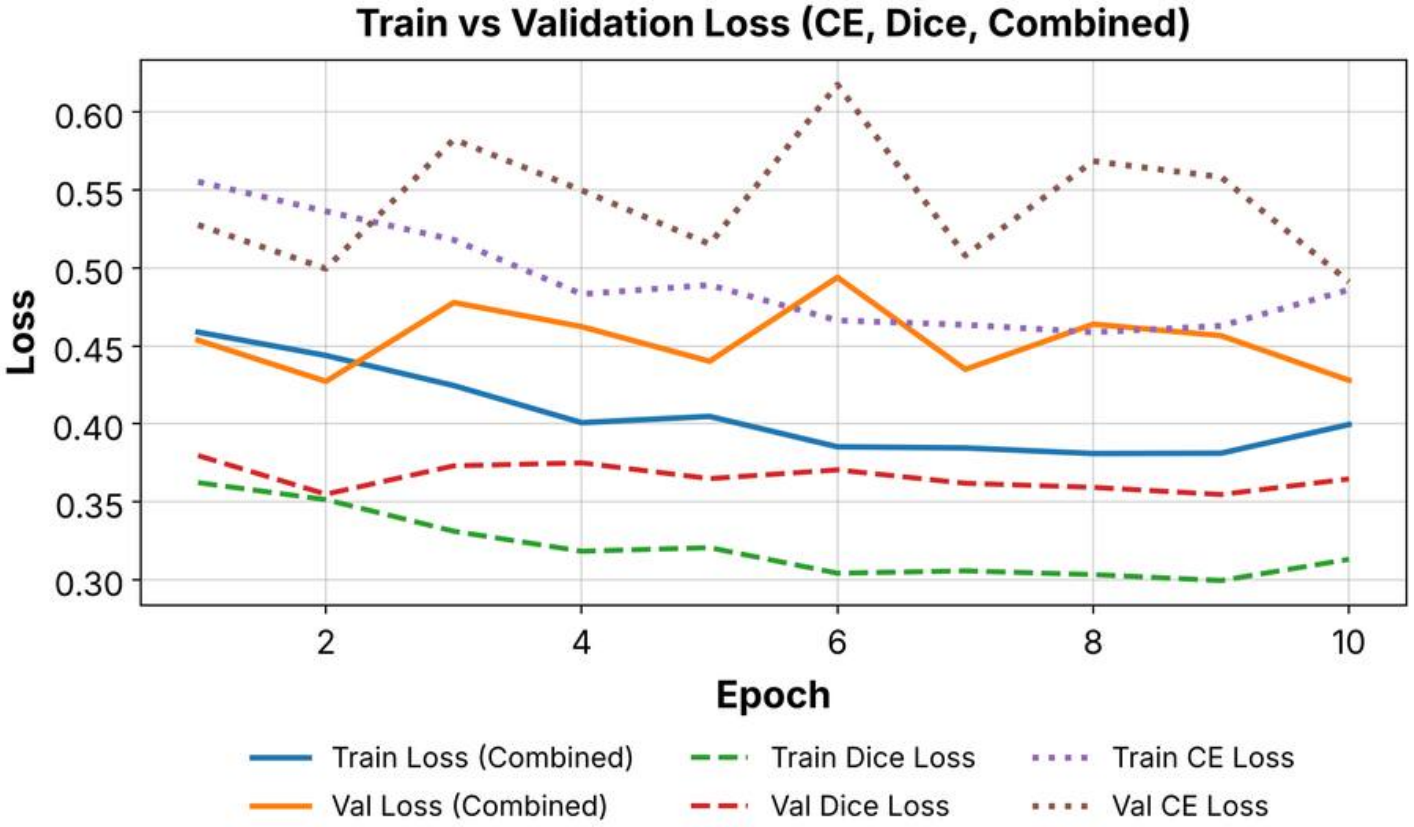
The same seed and model were also run inside 3_Seed_exp.ipynb (Section 2): the numbers on this page are identical to that seed's numbers there.

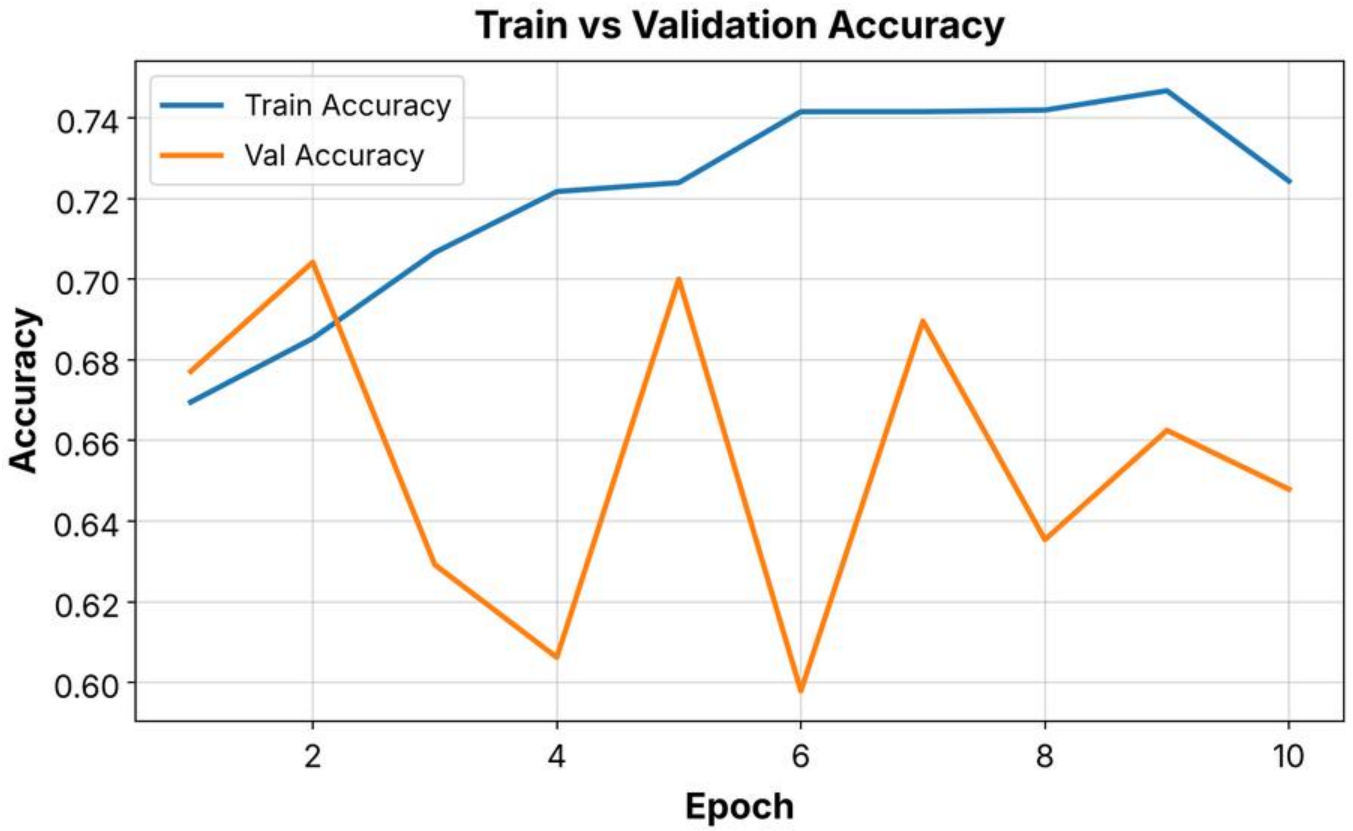
Confusion Matrix



5. EfficientNet-B0 — Direct (fine-tuned CNN) (single-seed notebook) — graphs

Training / Validation Graphs





6. EfficientNet-B3 — Direct (fine-tuned CNN) (single-seed notebook) — Seed 512

Source notebook: (seeded) EfficientNet B3 with Adaboost.ipynb, cell 4 (single-seed folder), fixed SEED = 512

Accuracy: 87.60%

Precision: 87.90%

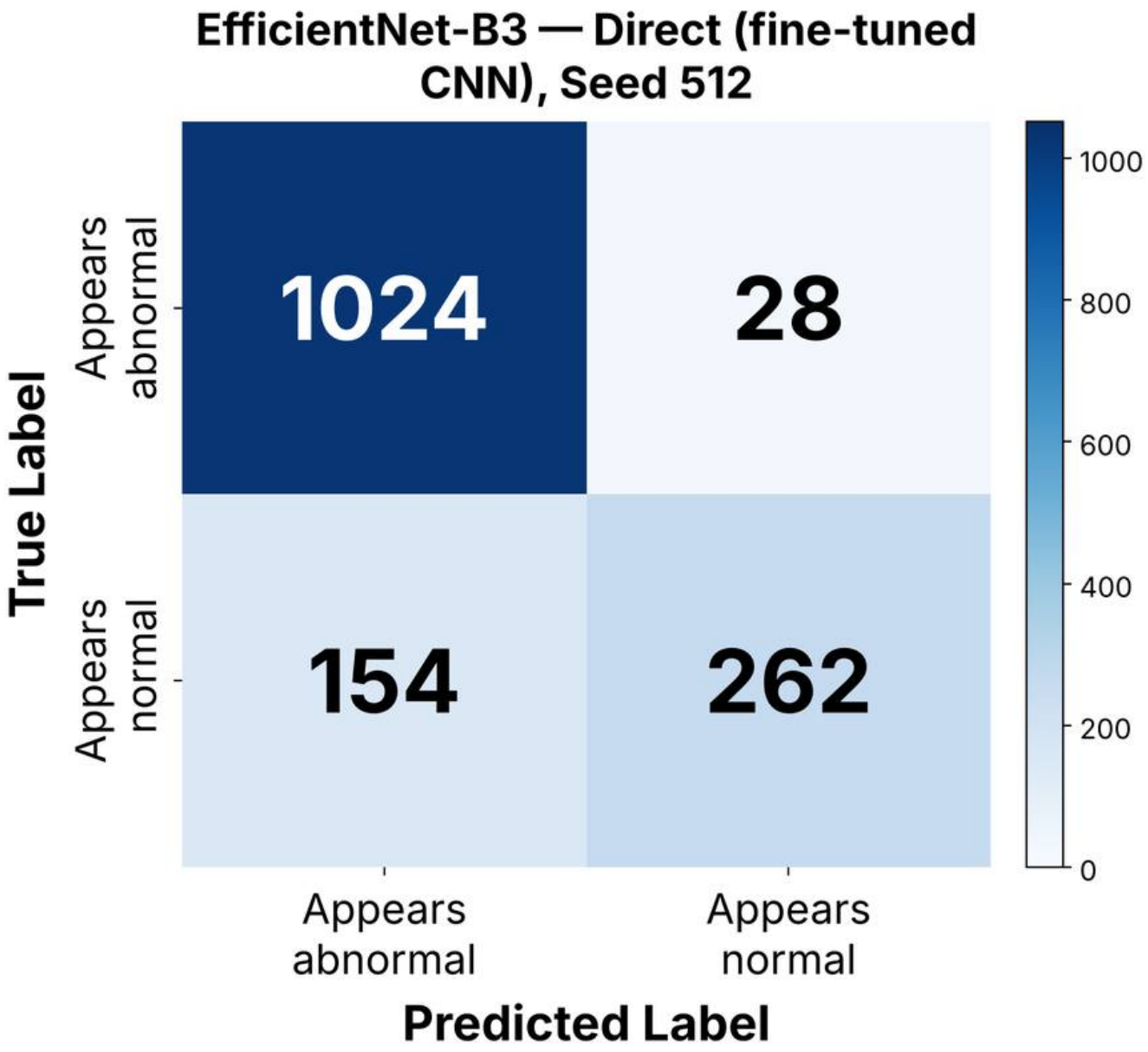
Recall: 87.60%

F1-score: 86.85%

Best validation accuracy: 75.62% **Test Dice loss:** 0.5218

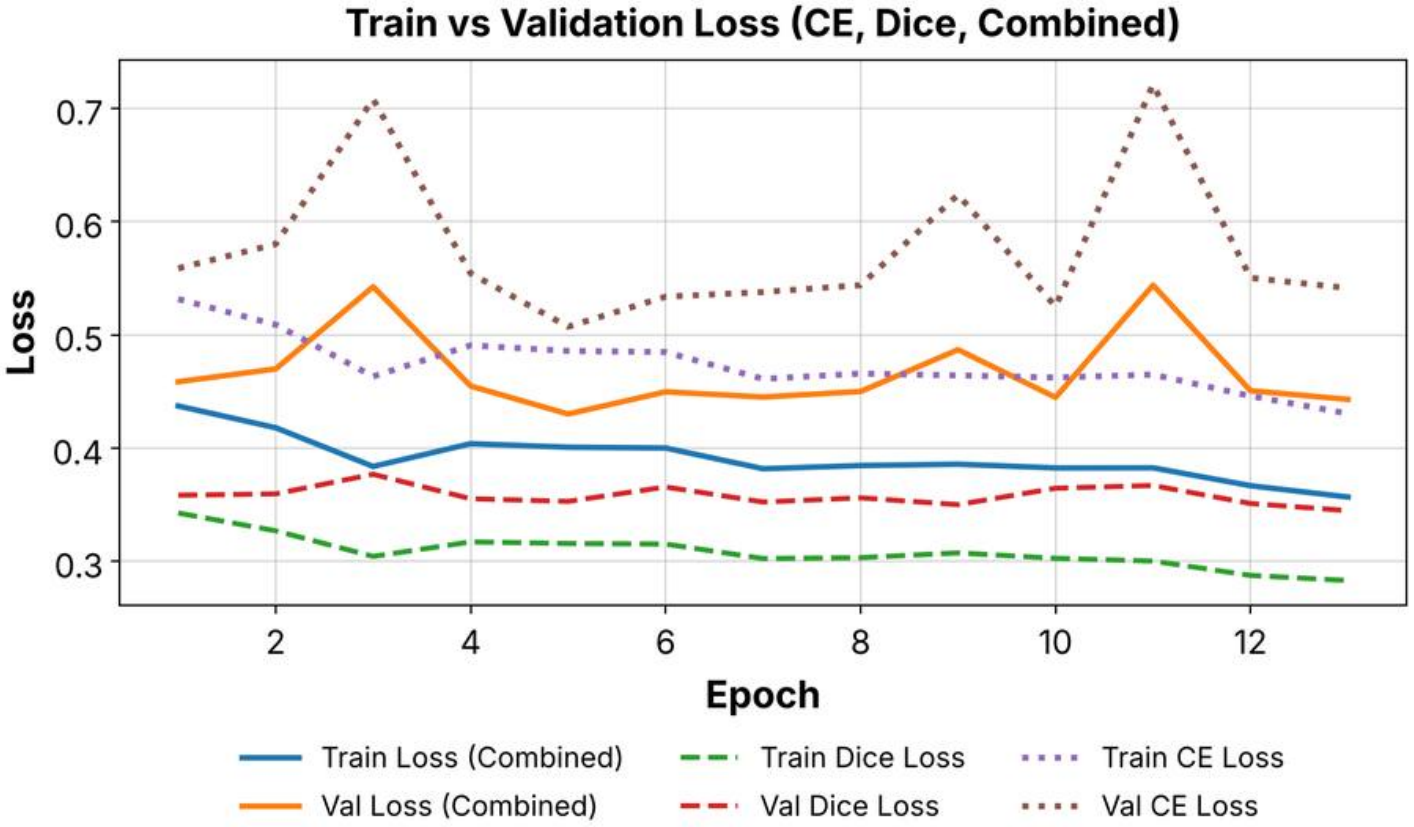
The same seed and model were also run inside 3_Seed_exp.ipynb (Section 2): the numbers on this page are identical to that seed's numbers there.

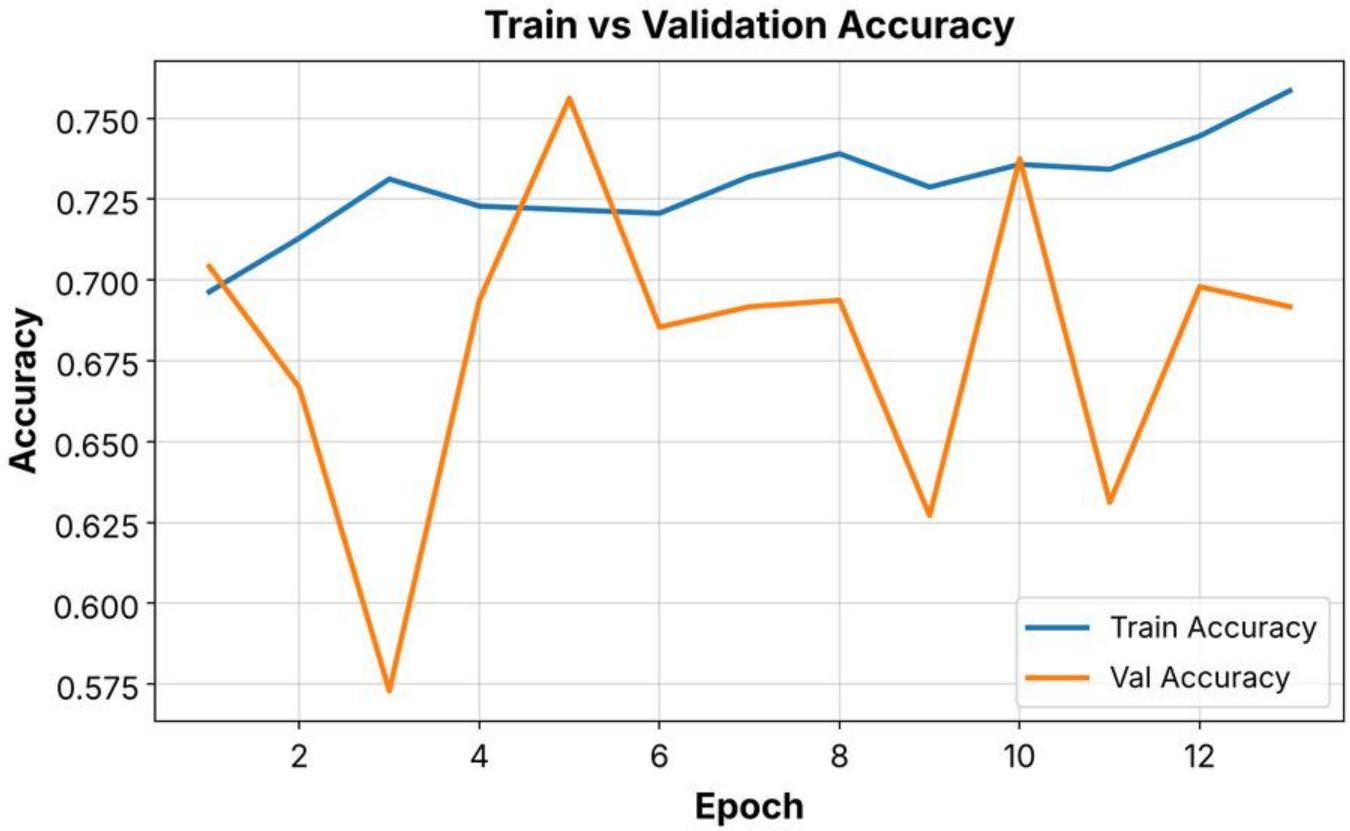
Confusion Matrix



6. EfficientNet-B3 — Direct (fine-tuned CNN) (single-seed notebook) — graphs

Training / Validation Graphs





Section 2 — 3-seed experiment

3_Seed_exp.ipynb trains the same three models with seeds 2622715460, 512 and 2. The main value is mean ± SD (sample SD, n = 3) over the three seeds; below it the individual result of each seed is listed in a small Seed | Value table, with the seed of highest test accuracy in bold (it is also the highest on precision, recall and F1-score). Rows are ranked by that best seed. Each row has one page per seed with results, confusion matrix and training graphs.

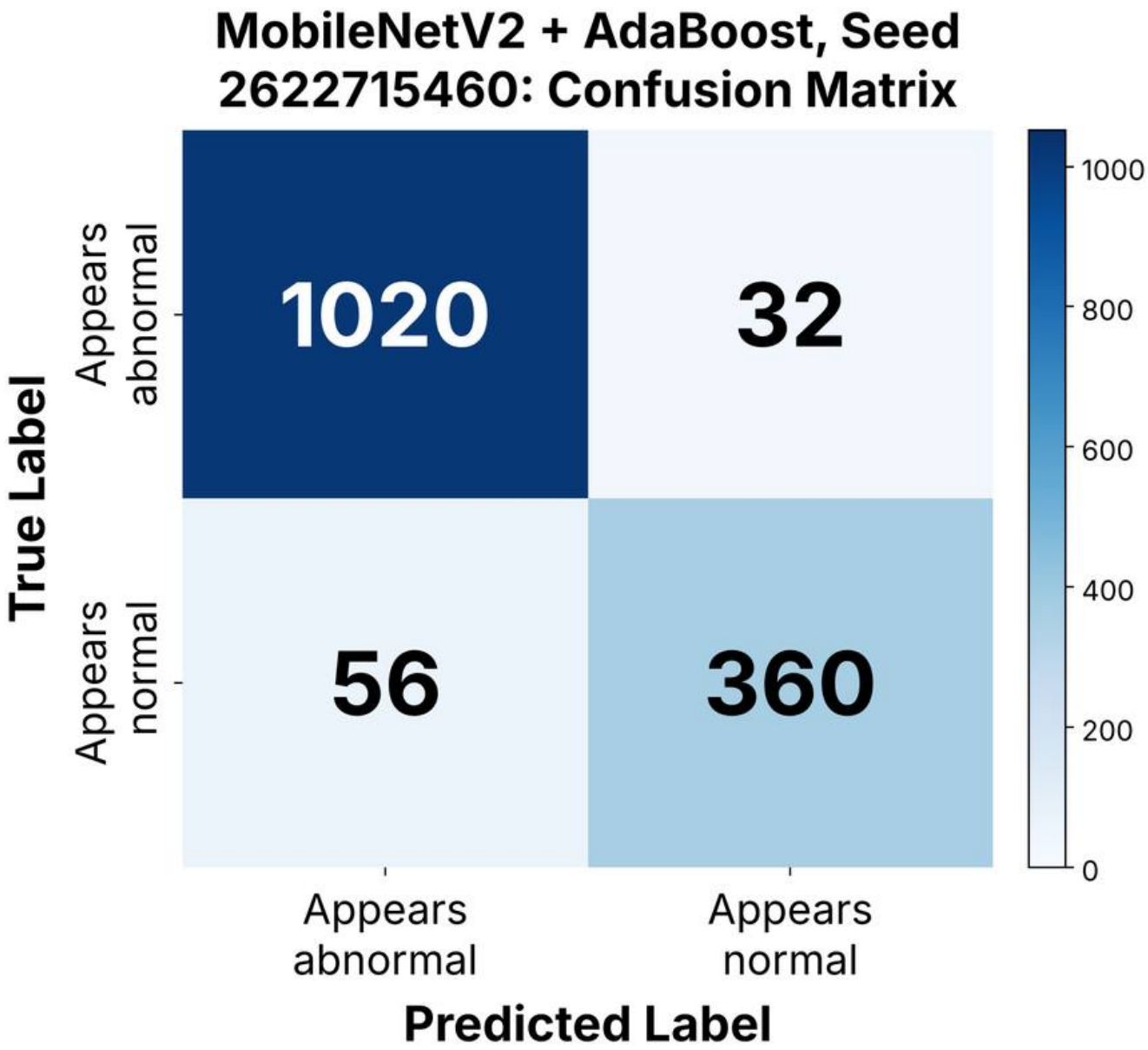
Rank	Model	Classifier	Test Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Source notebook (.ipynb)
7	MobileNetV2	AdaBoost	91.71 ± 2.07 **2622715460: 94.01** 512: 89.99 2: 91.14	91.67 ± 2.05 **2622715460: 93.96** 512: 90.00 2: 91.04	91.71 ± 2.07 **2622715460: 94.01** 512: 89.99 2: 91.14	91.52 ± 2.21 **2622715460: 93.95** 512: 89.63 2: 90.97	3_Seed_exp.ipynb (cell 4)
8	EfficientNet-B3	AdaBoost	91.71 ± 0.88 2622715460: 91.35 **512: 92.71** 2: 91.08	91.62 ± 0.89 2622715460: 91.24 **512: 92.64** 2: 90.99	91.71 ± 0.88 2622715460: 91.35 **512: 92.71** 2: 91.08	91.59 ± 0.93 2622715460: 91.25 **512: 92.64** 2: 90.88	3_Seed_exp.ipynb (cell 8)
9	EfficientNet-B0	AdaBoost	92.08 ± 0.49 **2622715460: 92.64** 512: 91.83 2: 91.76	92.05 ± 0.50 **2622715460: 92.62** 512: 91.73 2: 91.79	92.08 ± 0.49 **2622715460: 92.64** 512: 91.83 2: 91.76	91.99 ± 0.43 **2622715460: 92.49** 512: 91.72 2: 91.77	3_Seed_exp.ipynb (cell 12)
10	MobileNetV2	Direct (fine-tuned CNN)	88.96 ± 3.32 **2622715460: 92.44** 512: 85.83 2: 88.62	89.13 ± 3.41 **2622715460: 92.57** 512: 85.76 2: 89.07	88.96 ± 3.32 **2622715460: 92.44** 512: 85.83 2: 88.62	88.39 ± 3.61 **2622715460: 92.20** 512: 85.02 2: 87.94	3_Seed_exp.ipynb (cell 4)
11	EfficientNet-B3	Direct (fine-tuned CNN)	89.12 ± 1.33 **2622715460: 90.05** 512: 87.60 2: 89.71	89.28 ± 1.21 **2622715460: 90.10** 512: 87.90 2: 89.84	89.12 ± 1.33 **2622715460: 90.05** 512: 87.60 2: 89.71	88.60 ± 1.53 **2622715460: 89.69** 512: 86.85 2: 89.27	3_Seed_exp.ipynb (cell 8)
12	EfficientNet-B0	Direct (fine-tuned CNN)	89.01 ± 0.58 2622715460: 89.03 **512: 89.58** 2: 88.42	89.04 ± 0.44 2622715460: 88.91 **512: 89.52** 2: 88.68	89.01 ± 0.58 2622715460: 89.03 **512: 89.58** 2: 88.42	88.57 ± 0.73 2622715460: 88.69 **512: 89.23** 2: 87.78	3_Seed_exp.ipynb (cell 12)

7. MobileNetV2 + AdaBoost — Seed 2622715460 (seed 1 of 3)

Source: 3_Seed_exp.ipynb, cell 4. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 4, seed 2622715460* (same numbers for that seed).

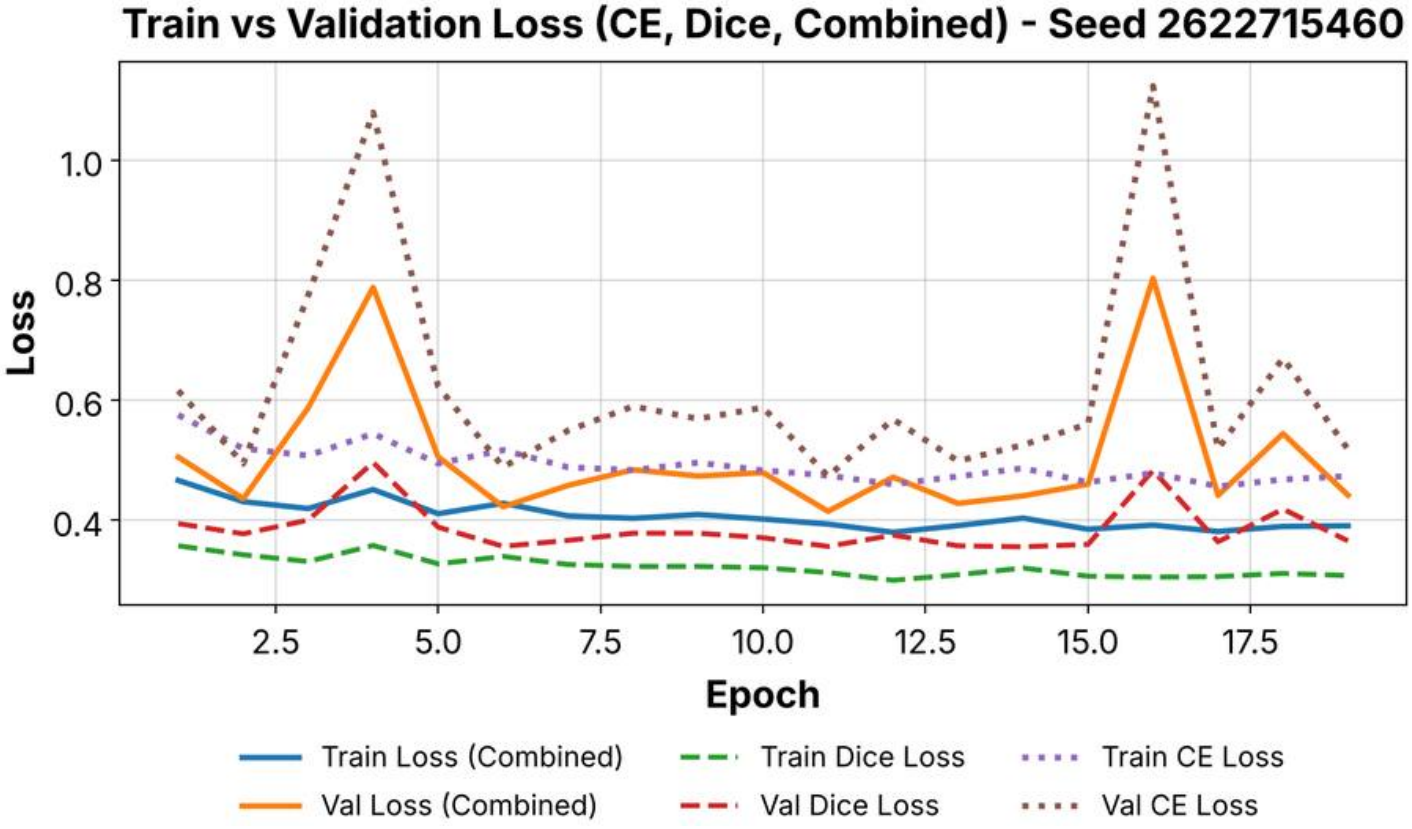
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**94.01%**	89.99%	91.14%	91.71 ± 2.07 %
Precision (weighted)	**93.96%**	90.00%	91.04%	91.67 ± 2.05 %
Recall (weighted)	**94.01%**	89.99%	91.14%	91.71 ± 2.07 %
F1-score (weighted)	**93.95%**	89.63%	90.97%	91.52 ± 2.21 %
AdaBoost validation acc.	67.29%	65.62%	63.54%	65.49 ± 1.88 %
Best validation acc. (CNN)	72.08%	71.25%	77.50%	73.61 ± 3.39 %

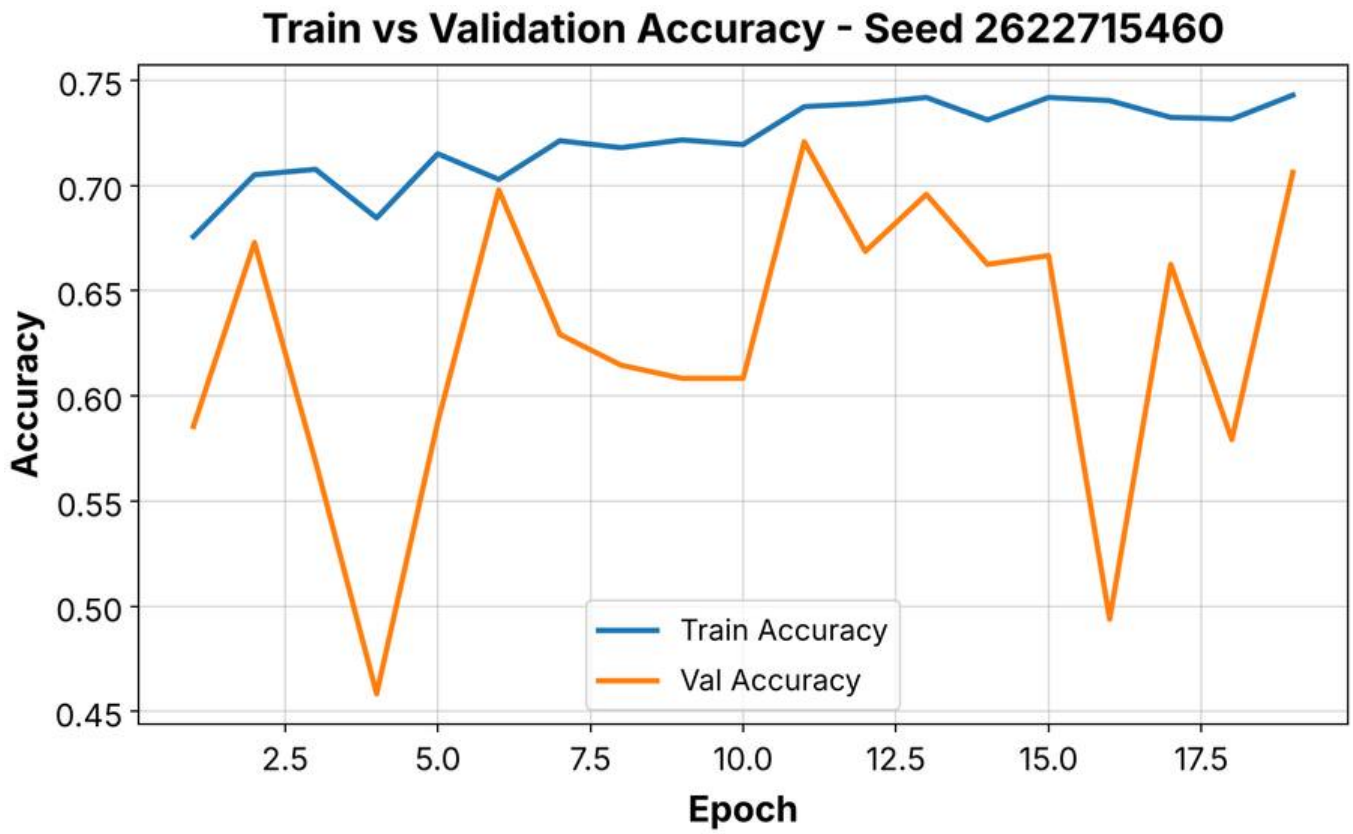
Confusion Matrix



7. MobileNetV2 + AdaBoost, Seed 2622715460 — graphs

Training / Validation Graphs



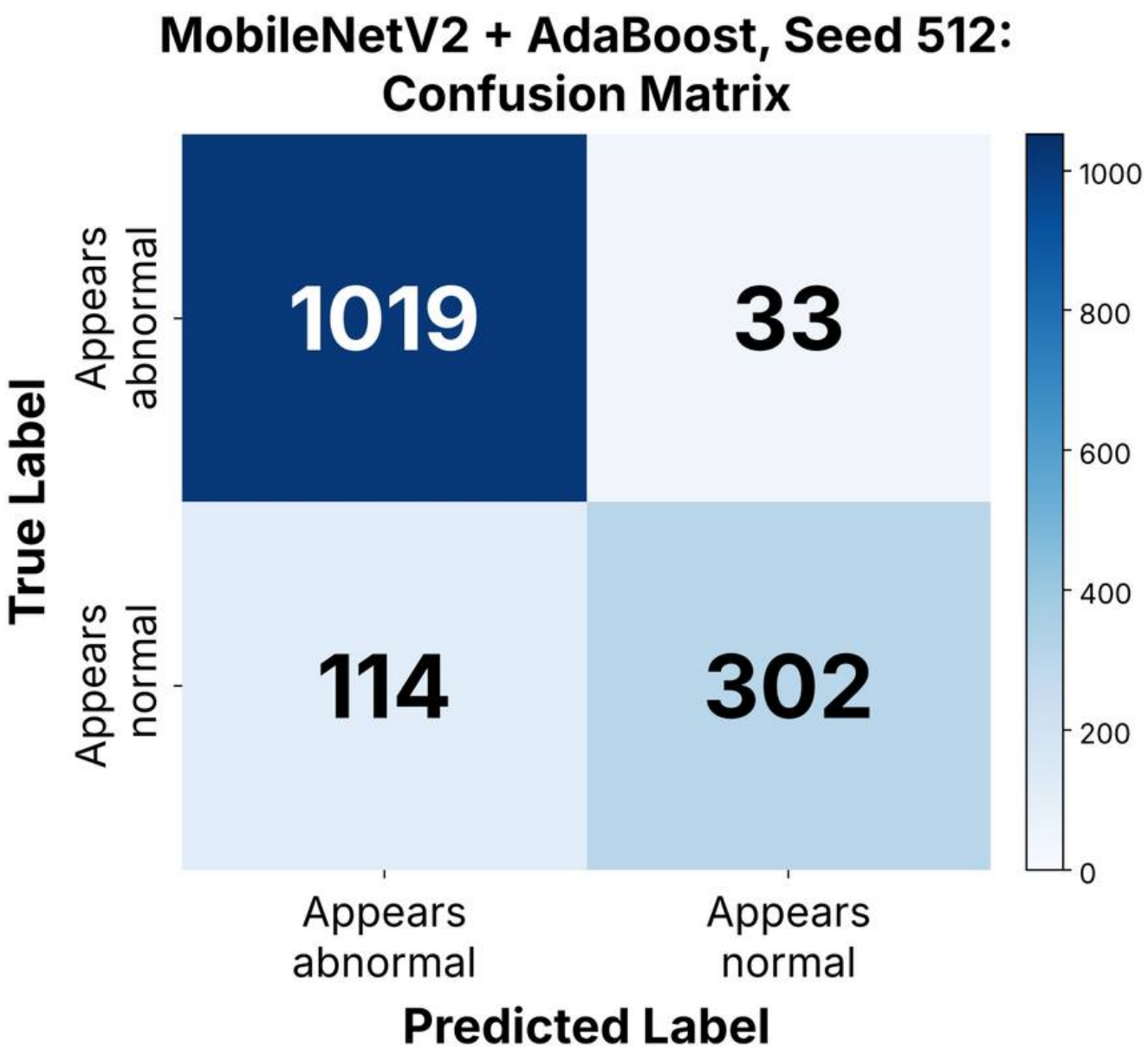


7. MobileNetV2 + AdaBoost — Seed 512 (seed 2 of 3)

Source: 3_Seed_exp.ipynb, cell 4. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 4, seed 2622715460* (same numbers for that seed).

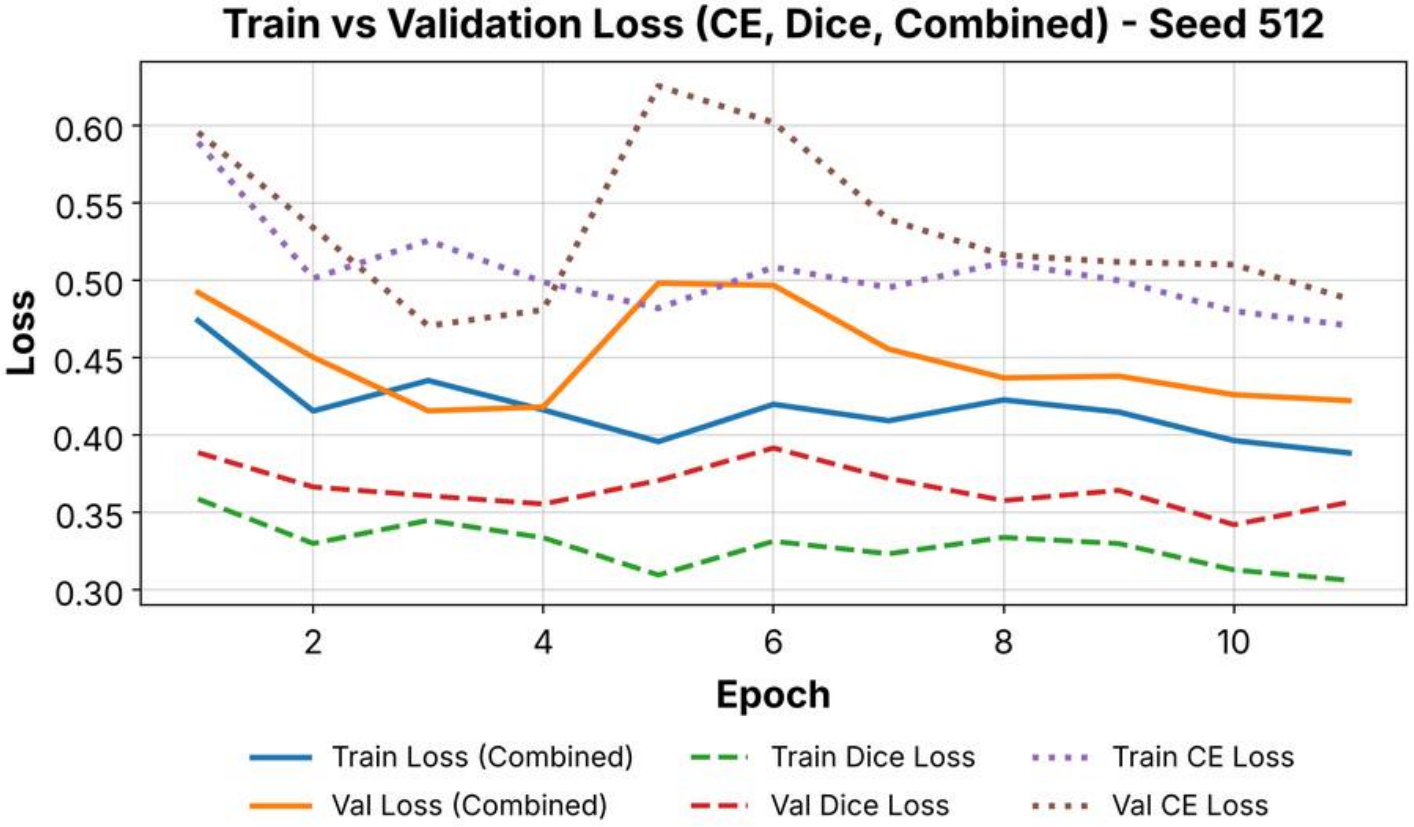
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	94.01%	**89.99%**	91.14%	91.71 ± 2.07 %
Precision (weighted)	93.96%	**90.00%**	91.04%	91.67 ± 2.05 %
Recall (weighted)	94.01%	**89.99%**	91.14%	91.71 ± 2.07 %
F1-score (weighted)	93.95%	**89.63%**	90.97%	91.52 ± 2.21 %
AdaBoost validation acc.	67.29%	65.62%	63.54%	65.49 ± 1.88 %
Best validation acc. (CNN)	72.08%	71.25%	77.50%	73.61 ± 3.39 %

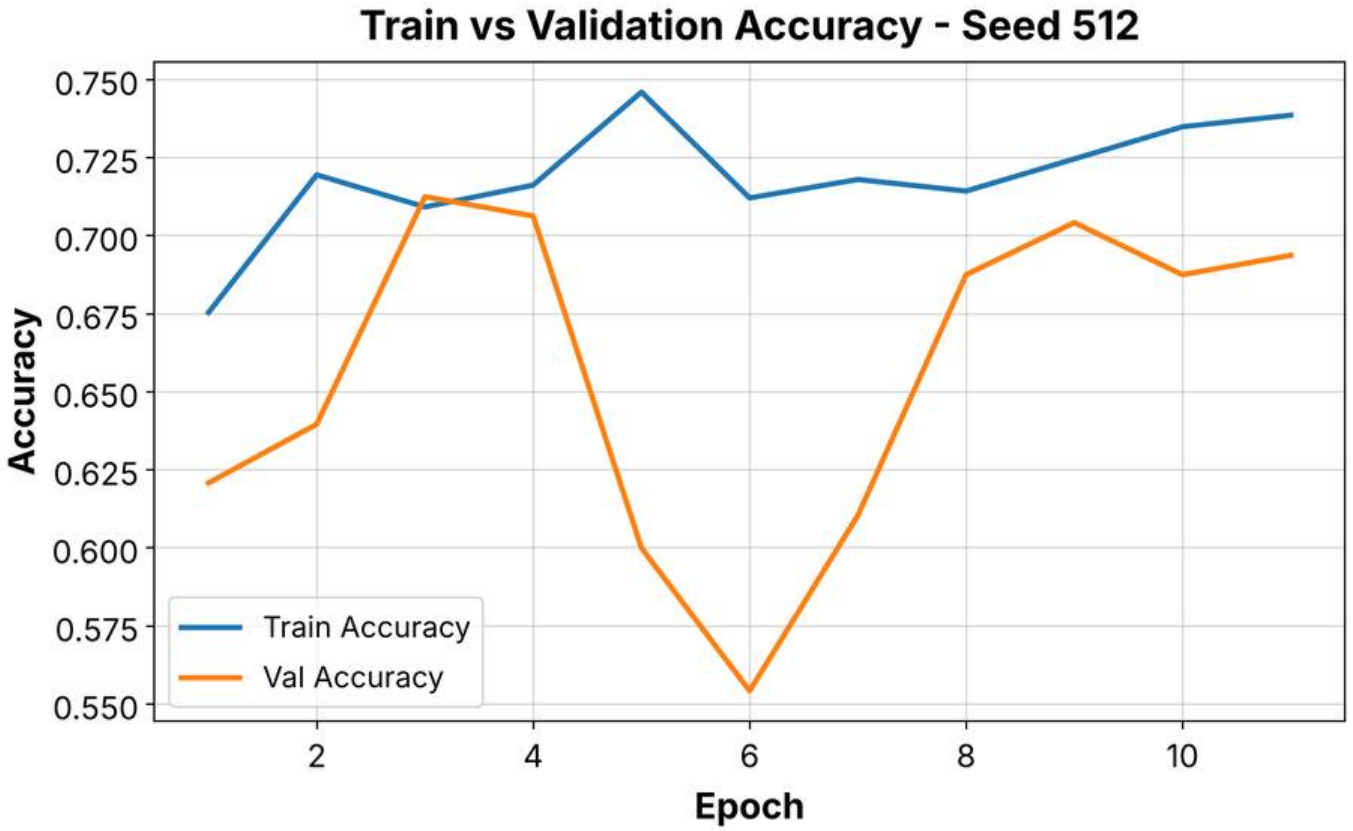
Confusion Matrix



7. MobileNetV2 + AdaBoost, Seed 512 — graphs

Training / Validation Graphs



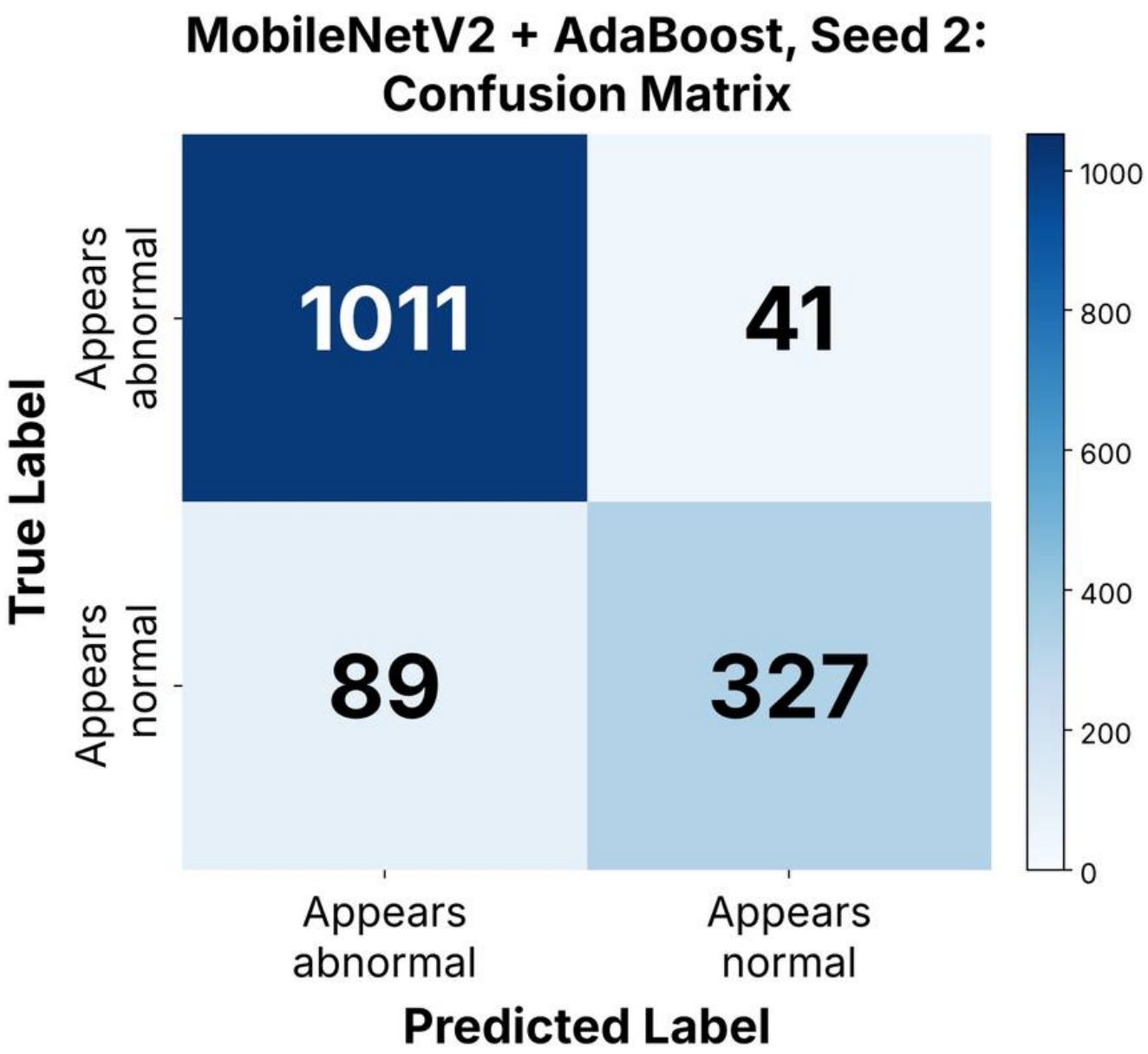


7. MobileNetV2 + AdaBoost — Seed 2 (seed 3 of 3)

Source: 3_Seed_exp.ipynb, cell 4. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 4, seed 2622715460* (same numbers for that seed).

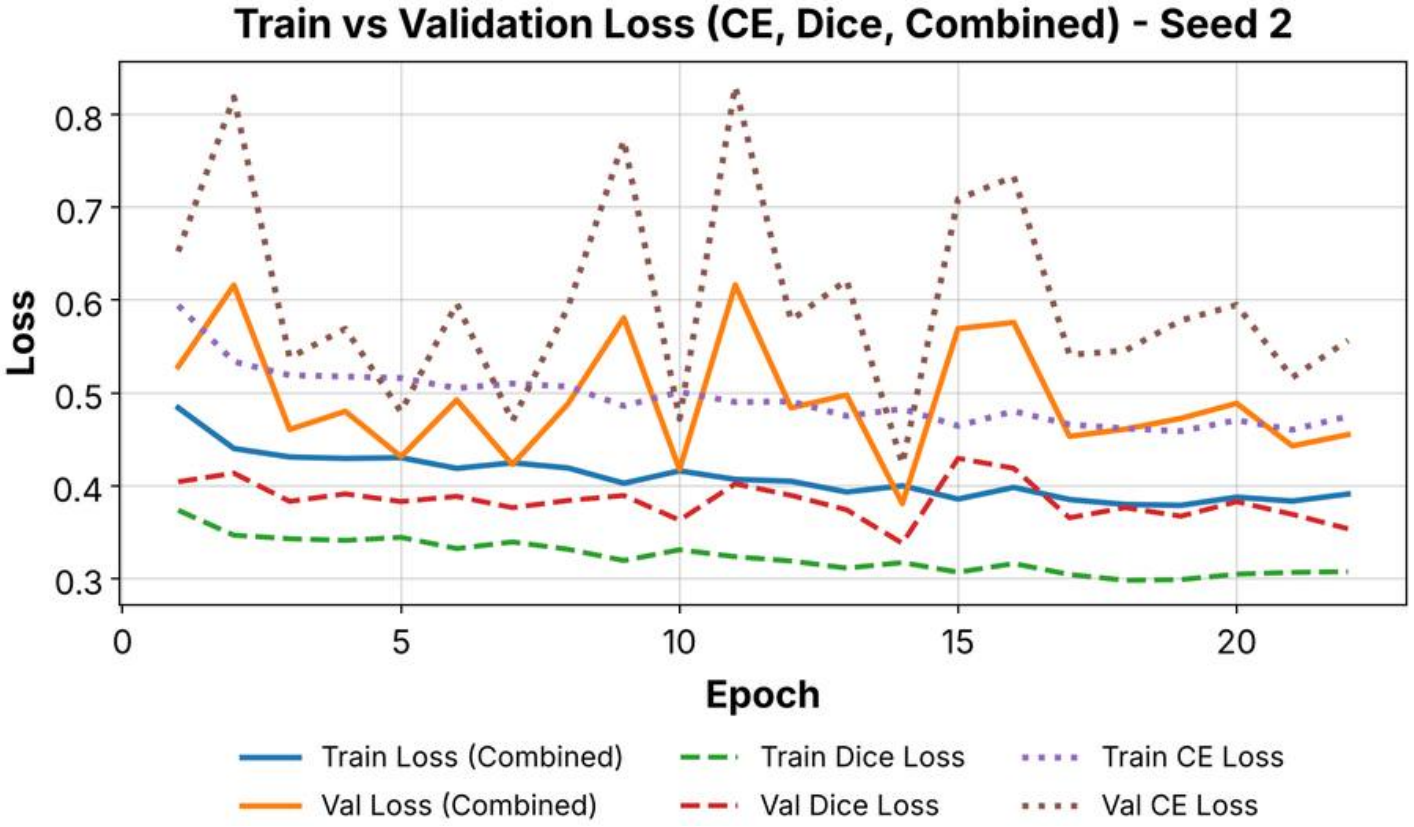
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	94.01%	89.99%	**91.14%**	91.71 ± 2.07 %
Precision (weighted)	93.96%	90.00%	**91.04%**	91.67 ± 2.05 %
Recall (weighted)	94.01%	89.99%	**91.14%**	91.71 ± 2.07 %
F1-score (weighted)	93.95%	89.63%	**90.97%**	91.52 ± 2.21 %
AdaBoost validation acc.	67.29%	65.62%	63.54%	65.49 ± 1.88 %
Best validation acc. (CNN)	72.08%	71.25%	77.50%	73.61 ± 3.39 %

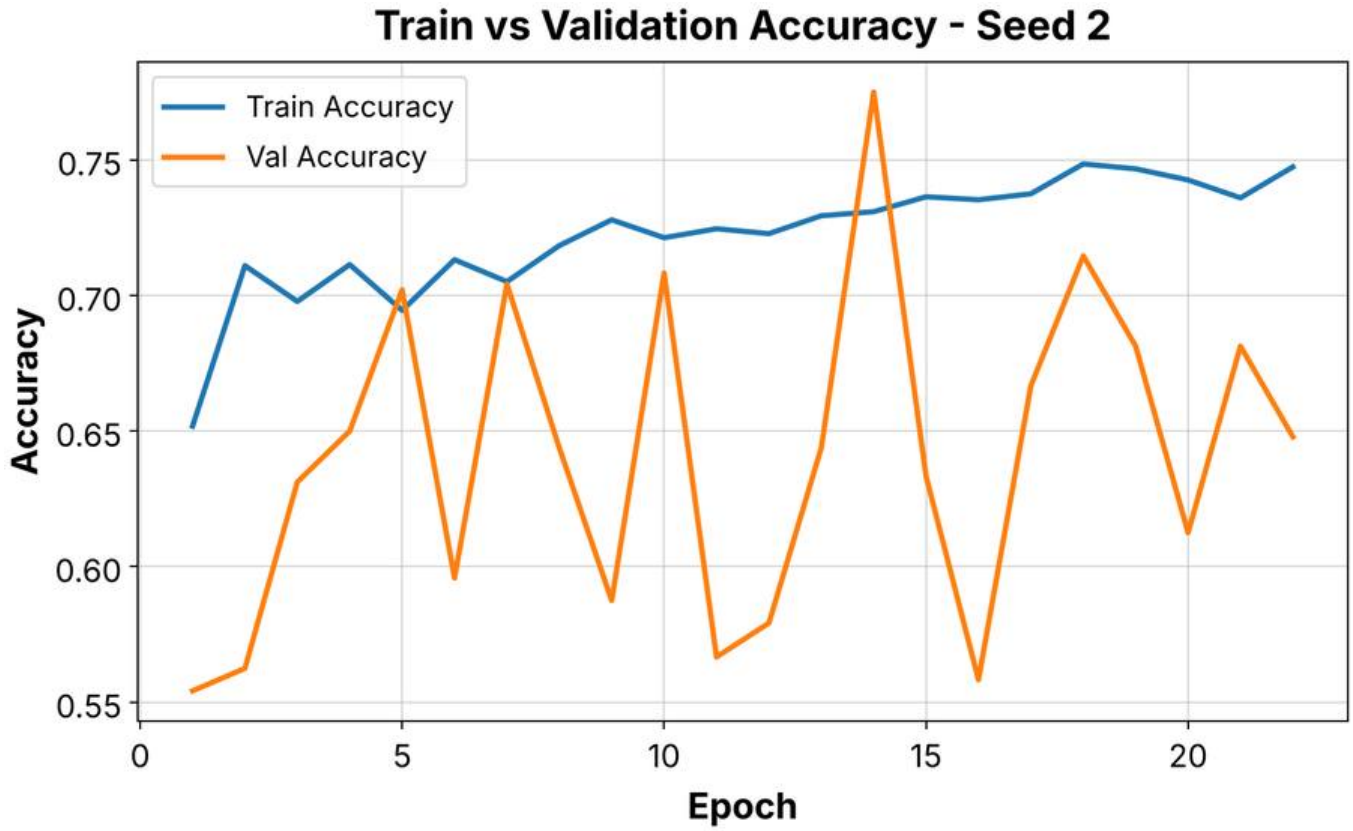
Confusion Matrix



7. MobileNetV2 + AdaBoost, Seed 2 — graphs

Training / Validation Graphs



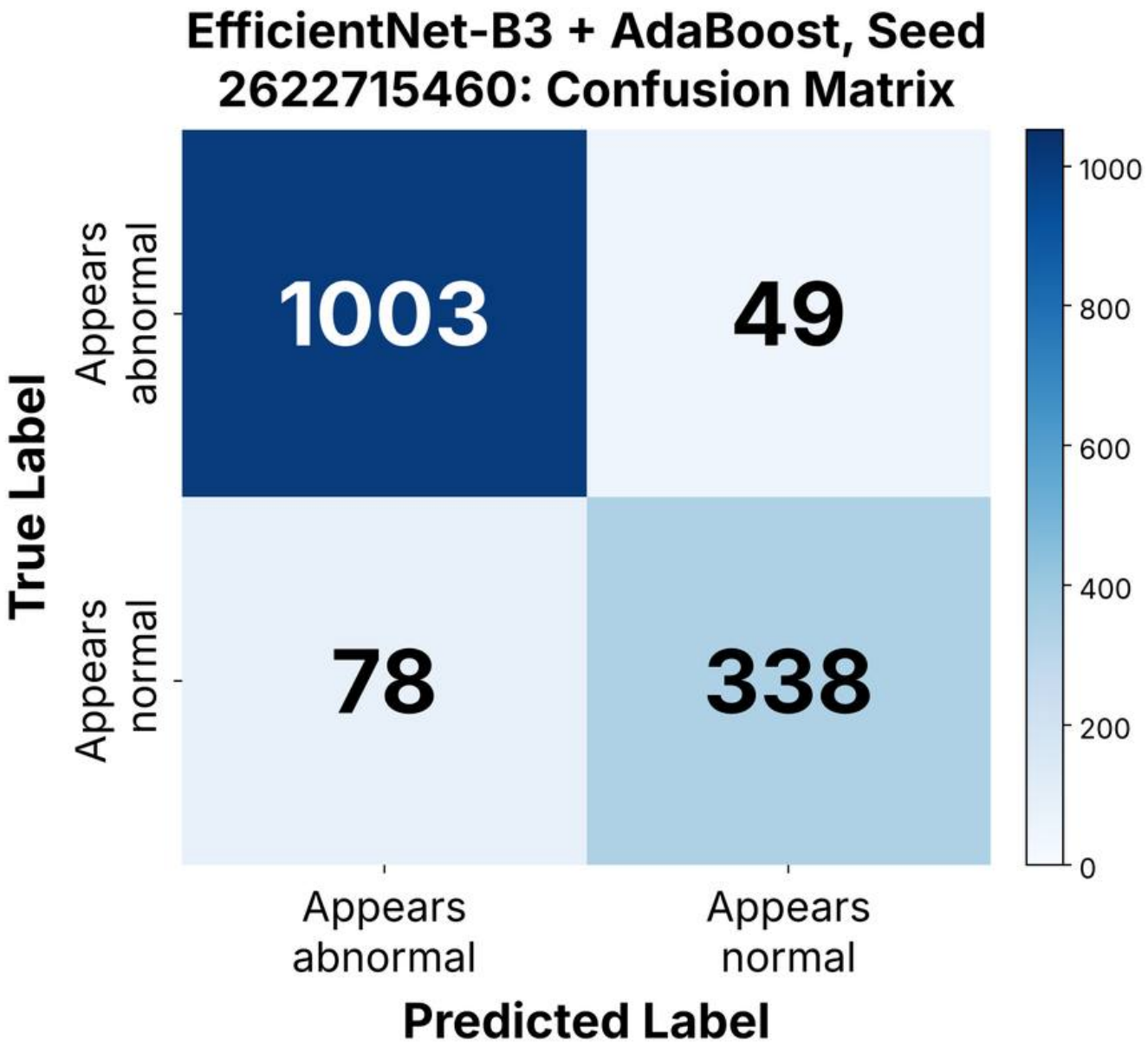


8. EfficientNet-B3 + AdaBoost — Seed 2622715460 (seed 1 of 3)

Source: 3_Seed_exp.ipynb, cell 8. Re-run independently in *(seeded) EfficientNet B3 with Adaboost.ipynb, cell 4, seed 512* (same numbers for that seed).

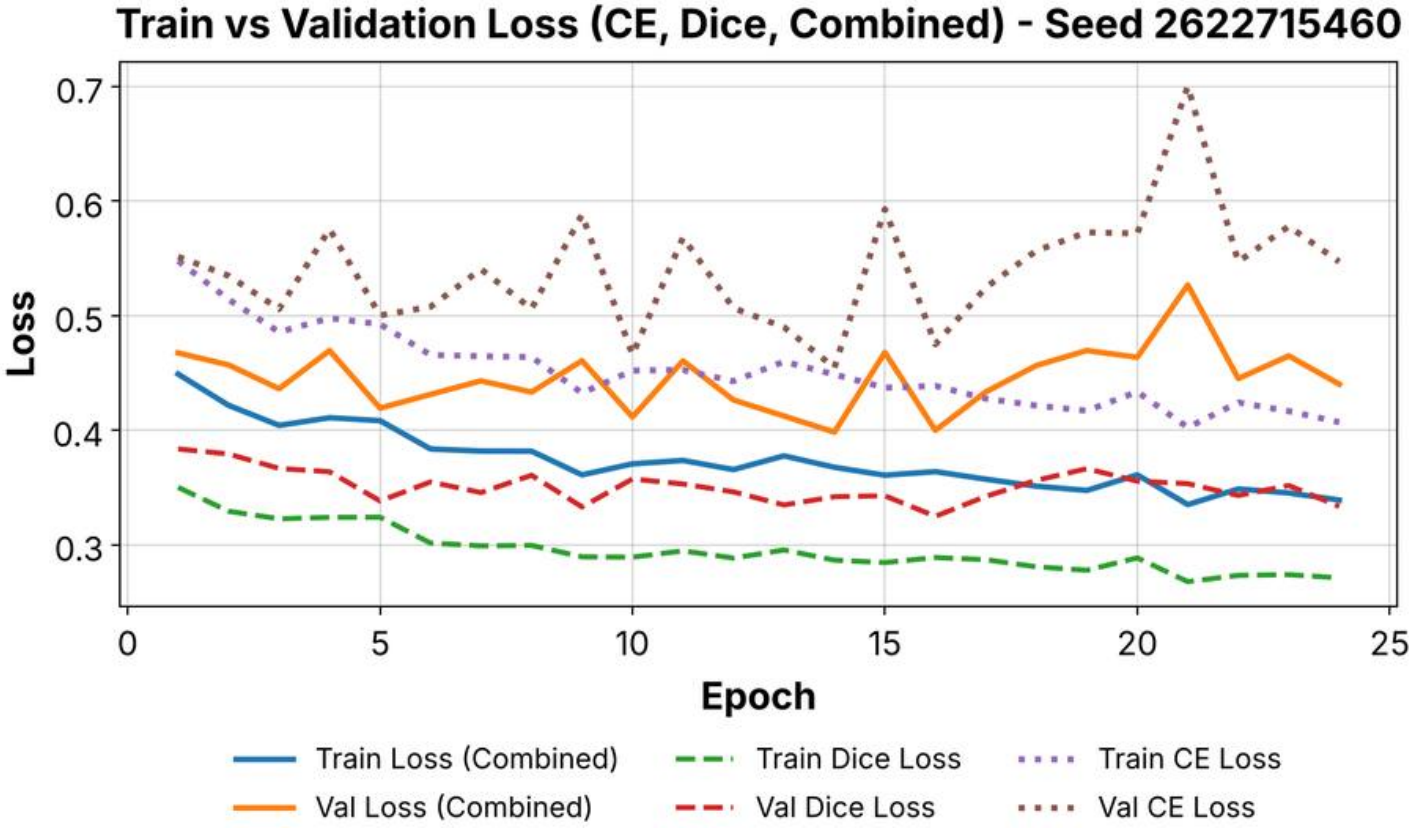
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**91.35%**	92.71%	91.08%	91.71 ± 0.88 %
Precision (weighted)	**91.24%**	92.64%	90.99%	91.62 ± 0.89 %
Recall (weighted)	**91.35%**	92.71%	91.08%	91.71 ± 0.88 %
F1-score (weighted)	**91.25%**	92.64%	90.88%	91.59 ± 0.93 %
AdaBoost validation acc.	69.37%	69.17%	68.33%	68.96 ± 0.55 %
Best validation acc. (CNN)	76.46%	75.62%	72.50%	74.86 ± 2.09 %

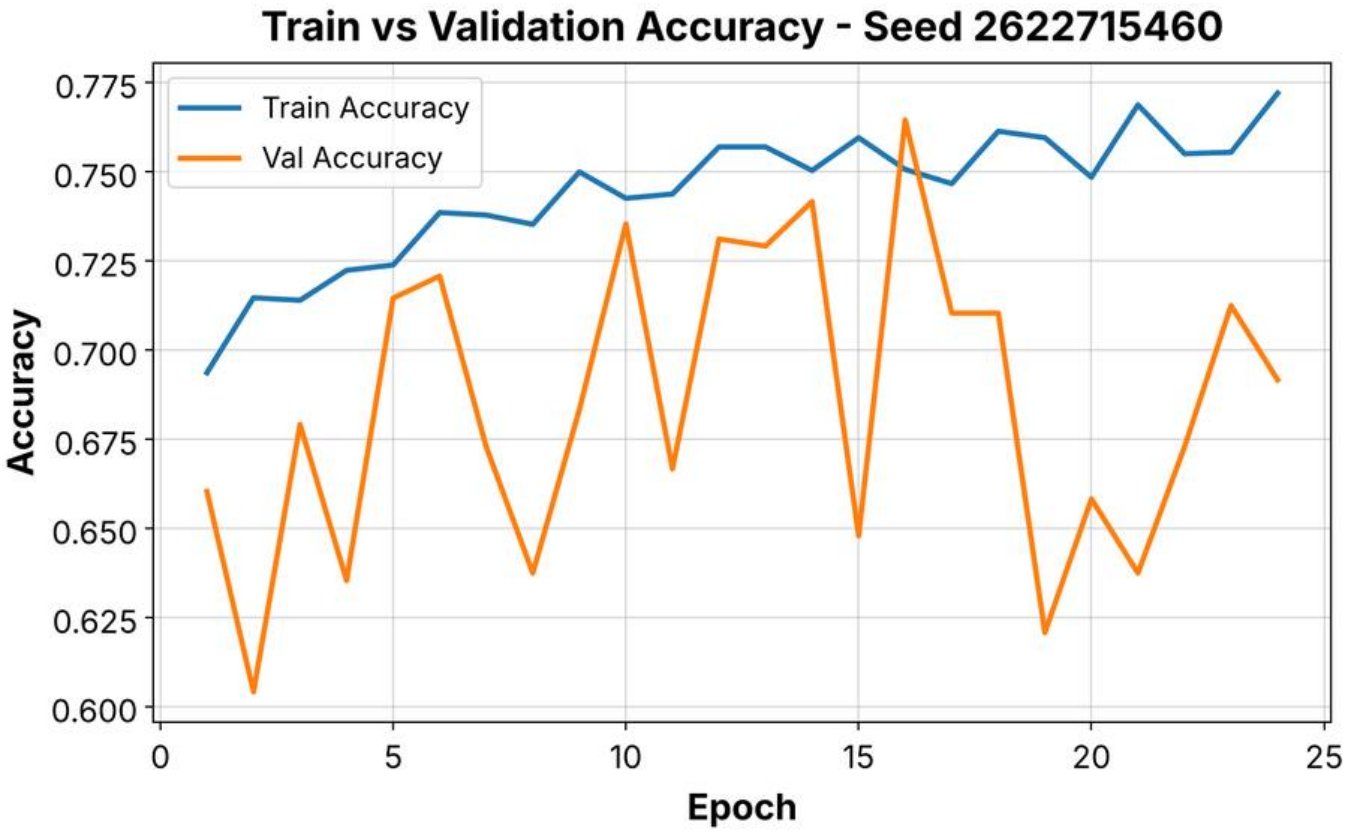
Confusion Matrix



8. EfficientNet-B3 + AdaBoost, Seed 2622715460 — graphs

Training / Validation Graphs



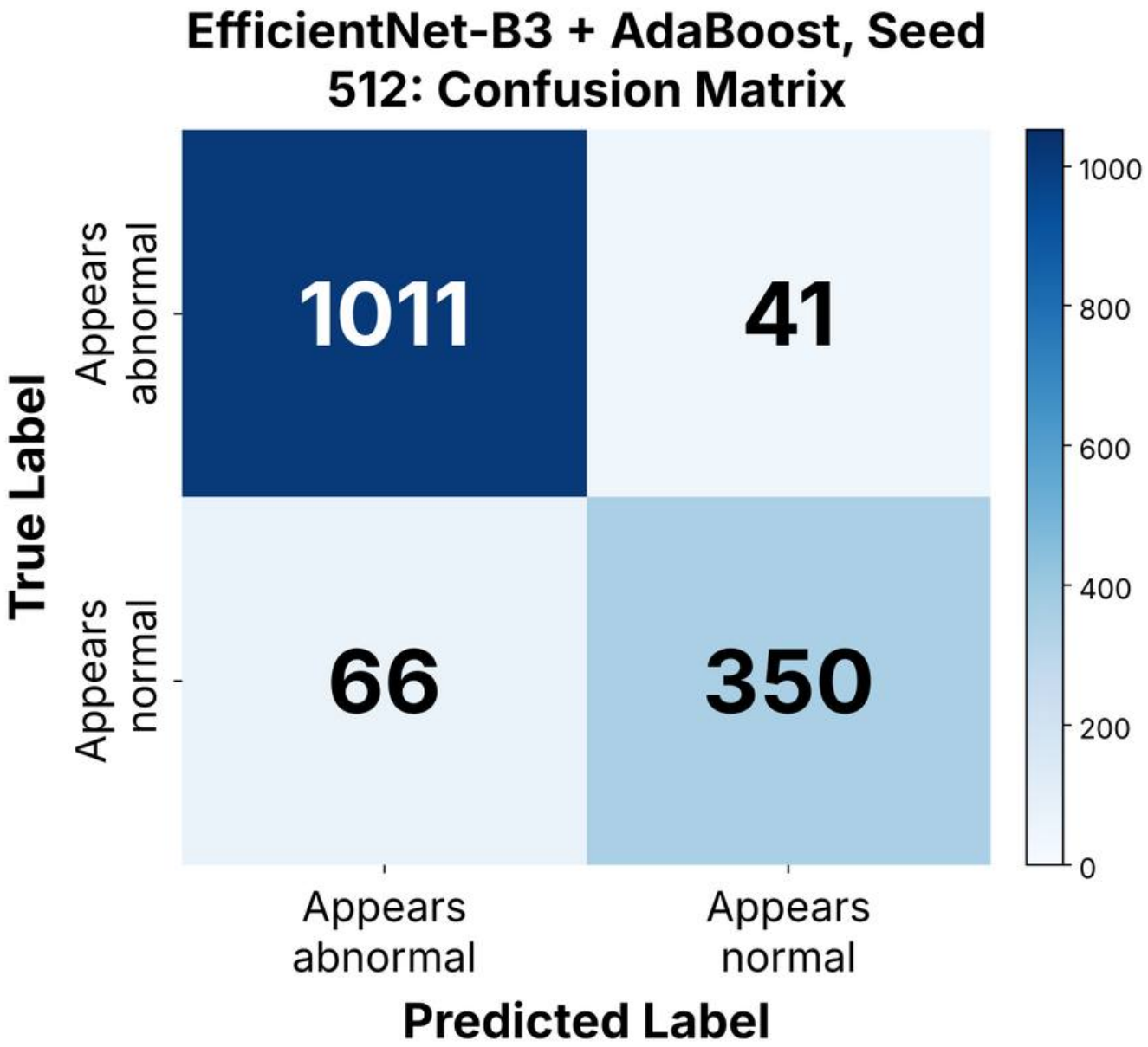


8. EfficientNet-B3 + AdaBoost — Seed 512 (seed 2 of 3)

Source: 3_Seed_exp.ipynb, cell 8. Re-run independently in *(seeded) EfficientNet B3 with Adaboost.ipynb, cell 4, seed 512* (same numbers for that seed).

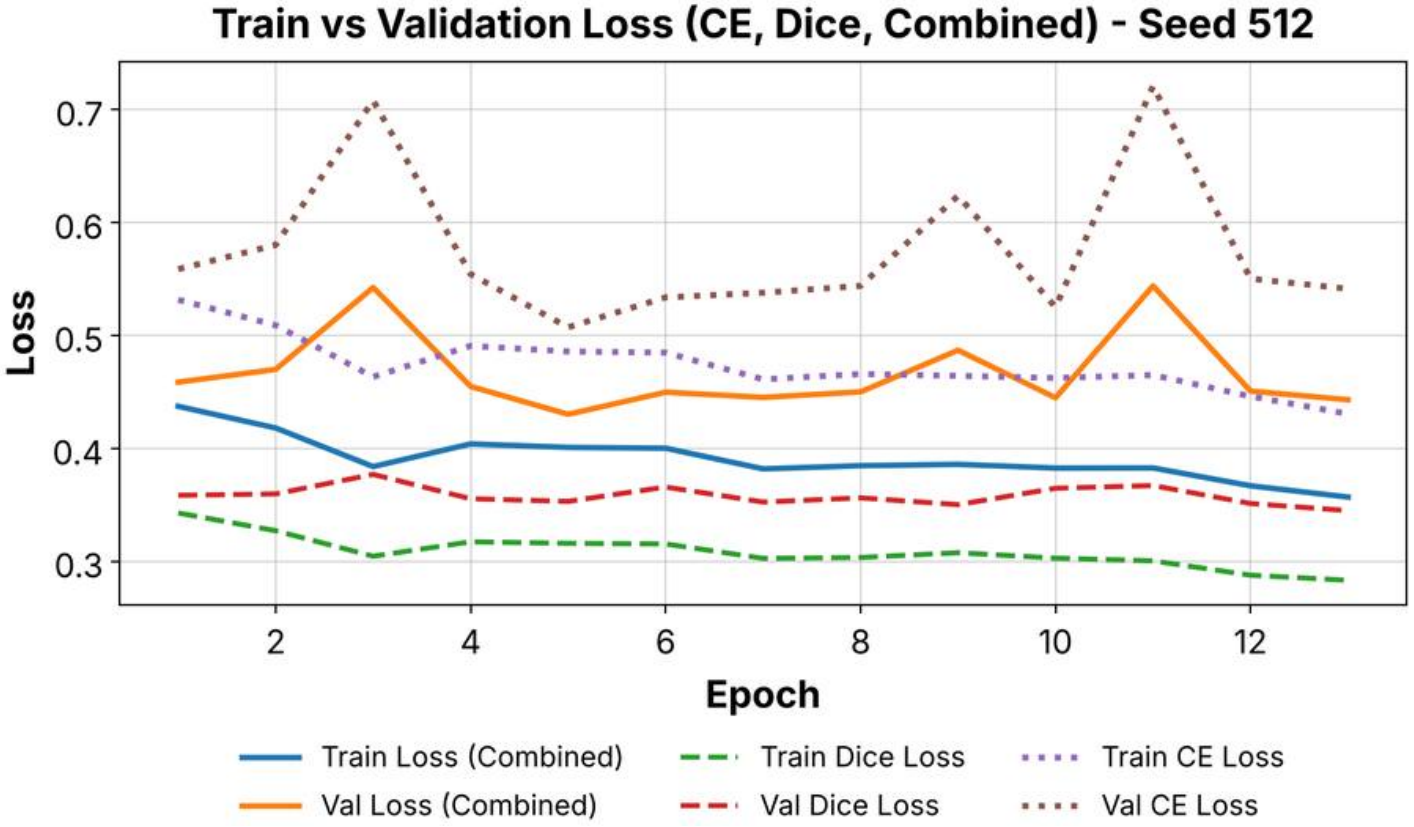
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	91.35%	**92.71%**	91.08%	91.71 ± 0.88 %
Precision (weighted)	91.24%	**92.64%**	90.99%	91.62 ± 0.89 %
Recall (weighted)	91.35%	**92.71%**	91.08%	91.71 ± 0.88 %
F1-score (weighted)	91.25%	**92.64%**	90.88%	91.59 ± 0.93 %
AdaBoost validation acc.	69.37%	69.17%	68.33%	68.96 ± 0.55 %
Best validation acc. (CNN)	76.46%	75.62%	72.50%	74.86 ± 2.09 %

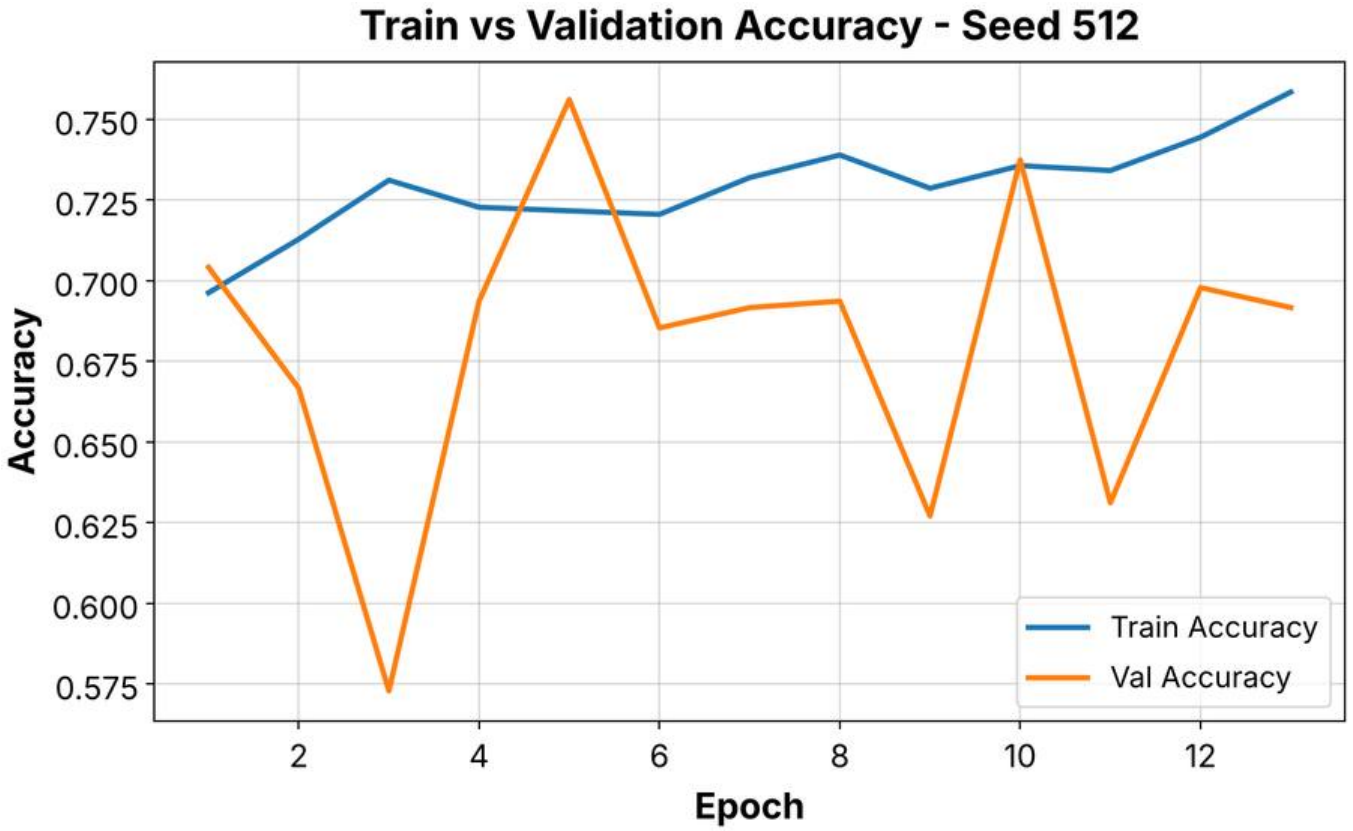
Confusion Matrix



8. EfficientNet-B3 + AdaBoost, Seed 512 — graphs

Training / Validation Graphs



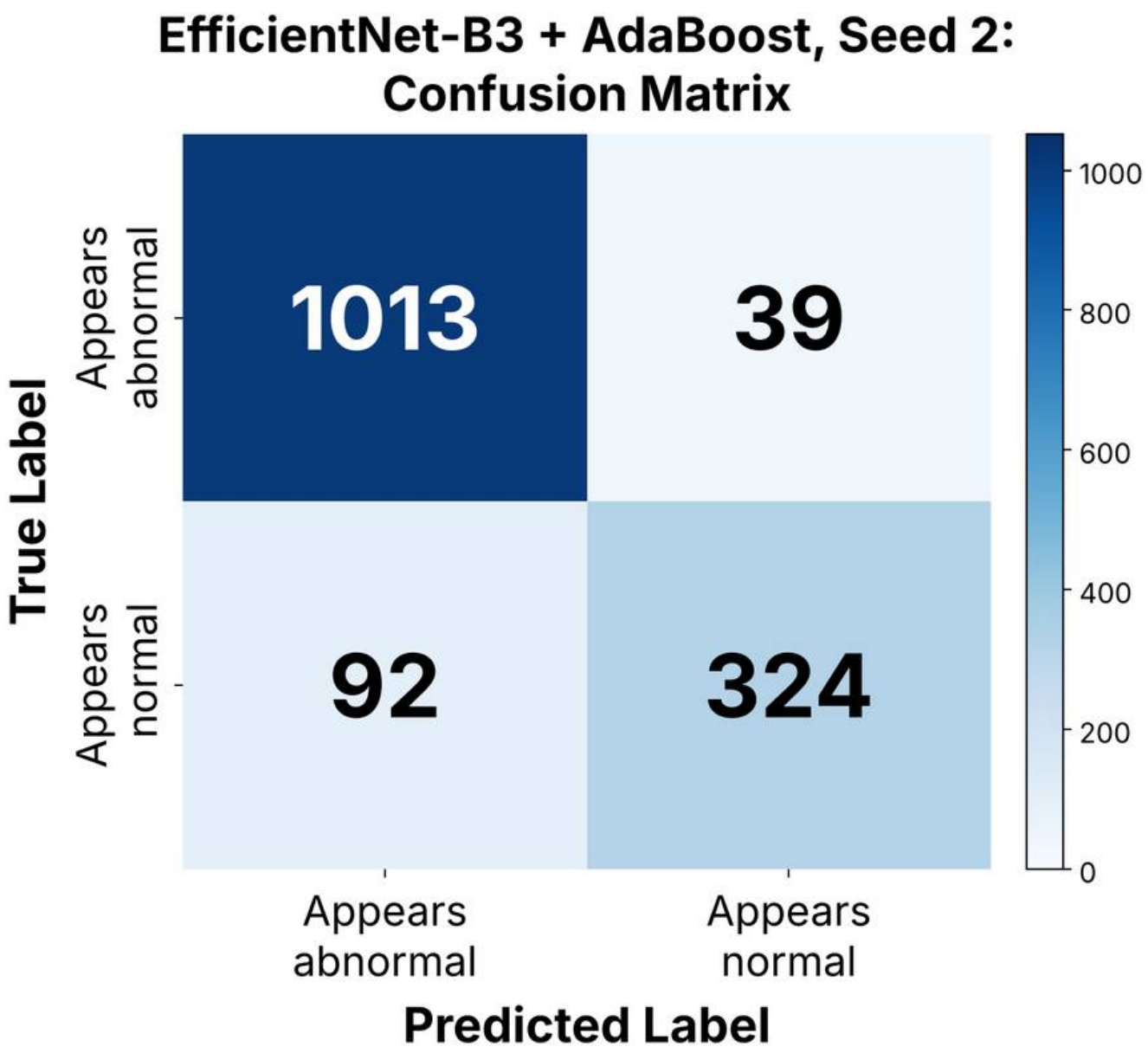


8. EfficientNet-B3 + AdaBoost — Seed 2 (seed 3 of 3)

Source: 3_Seed_exp.ipynb, cell 8. Re-run independently in *(seeded) EfficientNet B3 with Adaboost.ipynb, cell 4, seed 512* (same numbers for that seed).

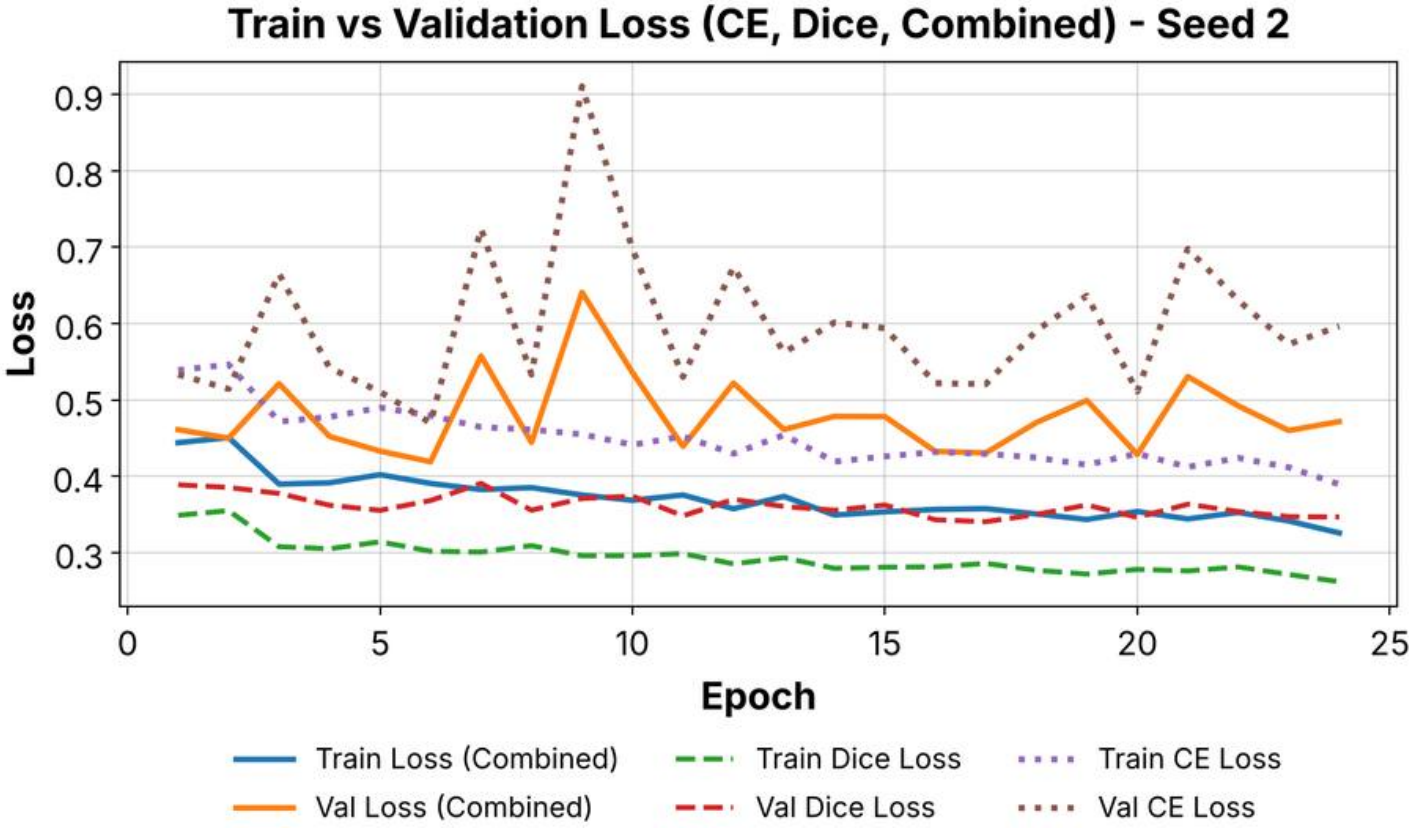
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	91.35%	92.71%	**91.08%**	91.71 ± 0.88 %
Precision (weighted)	91.24%	92.64%	**90.99%**	91.62 ± 0.89 %
Recall (weighted)	91.35%	92.71%	**91.08%**	91.71 ± 0.88 %
F1-score (weighted)	91.25%	92.64%	**90.88%**	91.59 ± 0.93 %
AdaBoost validation acc.	69.37%	69.17%	68.33%	68.96 ± 0.55 %
Best validation acc. (CNN)	76.46%	75.62%	72.50%	74.86 ± 2.09 %

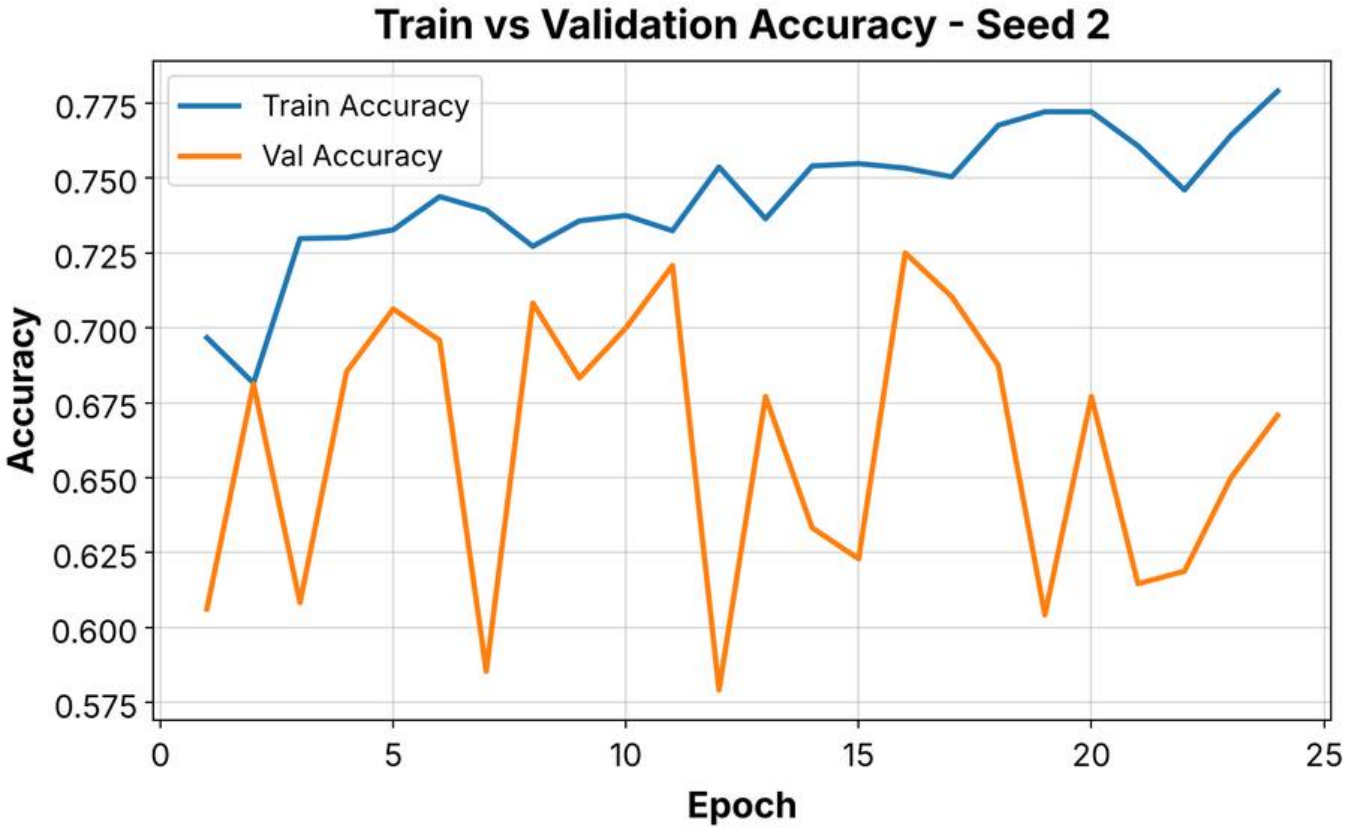
Confusion Matrix



8. EfficientNet-B3 + AdaBoost, Seed 2 — graphs

Training / Validation Graphs



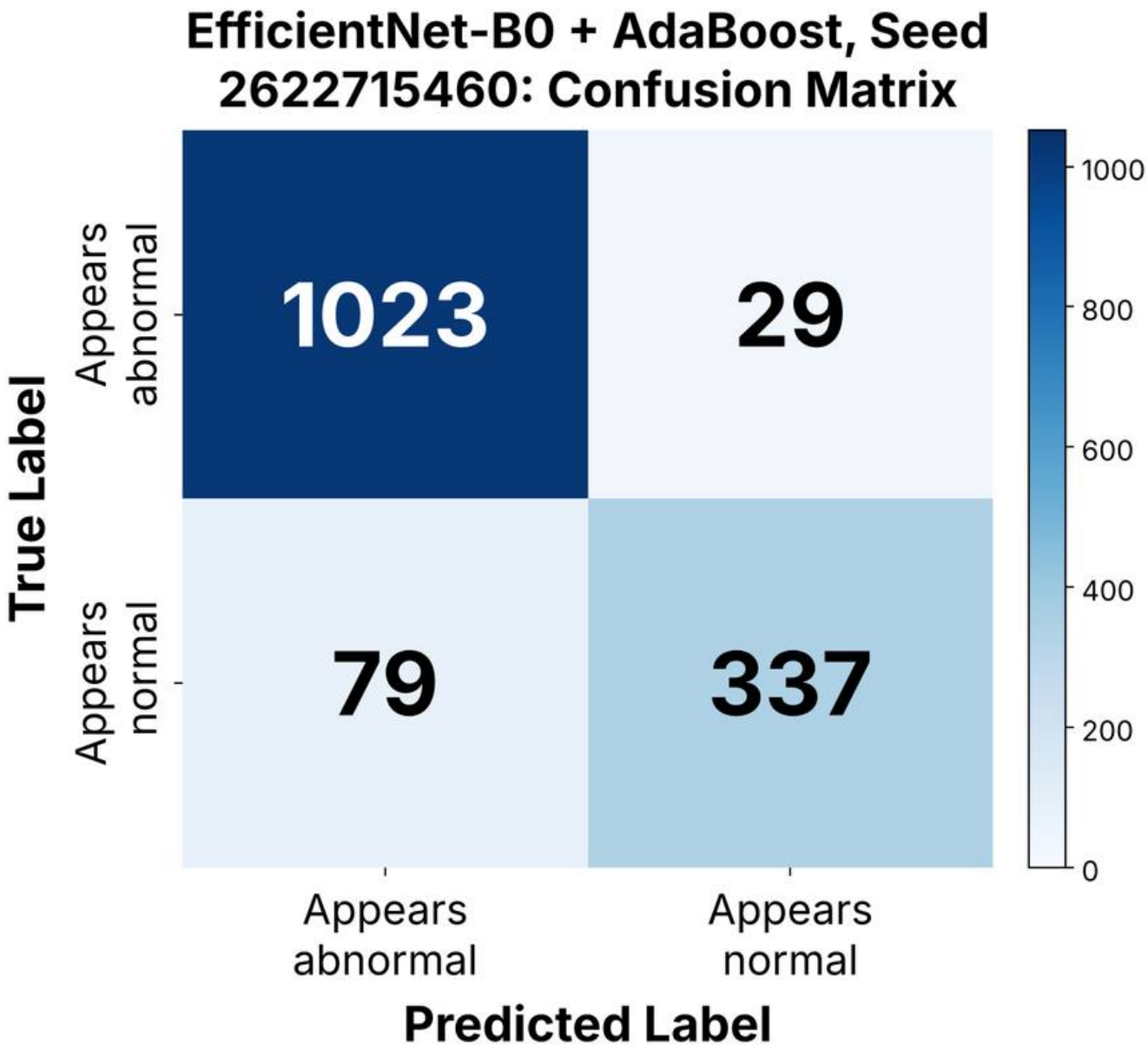


9. EfficientNet-B0 + AdaBoost — Seed 2622715460 (seed 1 of 3)

Source: 3_Seed_exp.ipynb, cell 12. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 8, seed 2622715460* (same numbers for that seed).

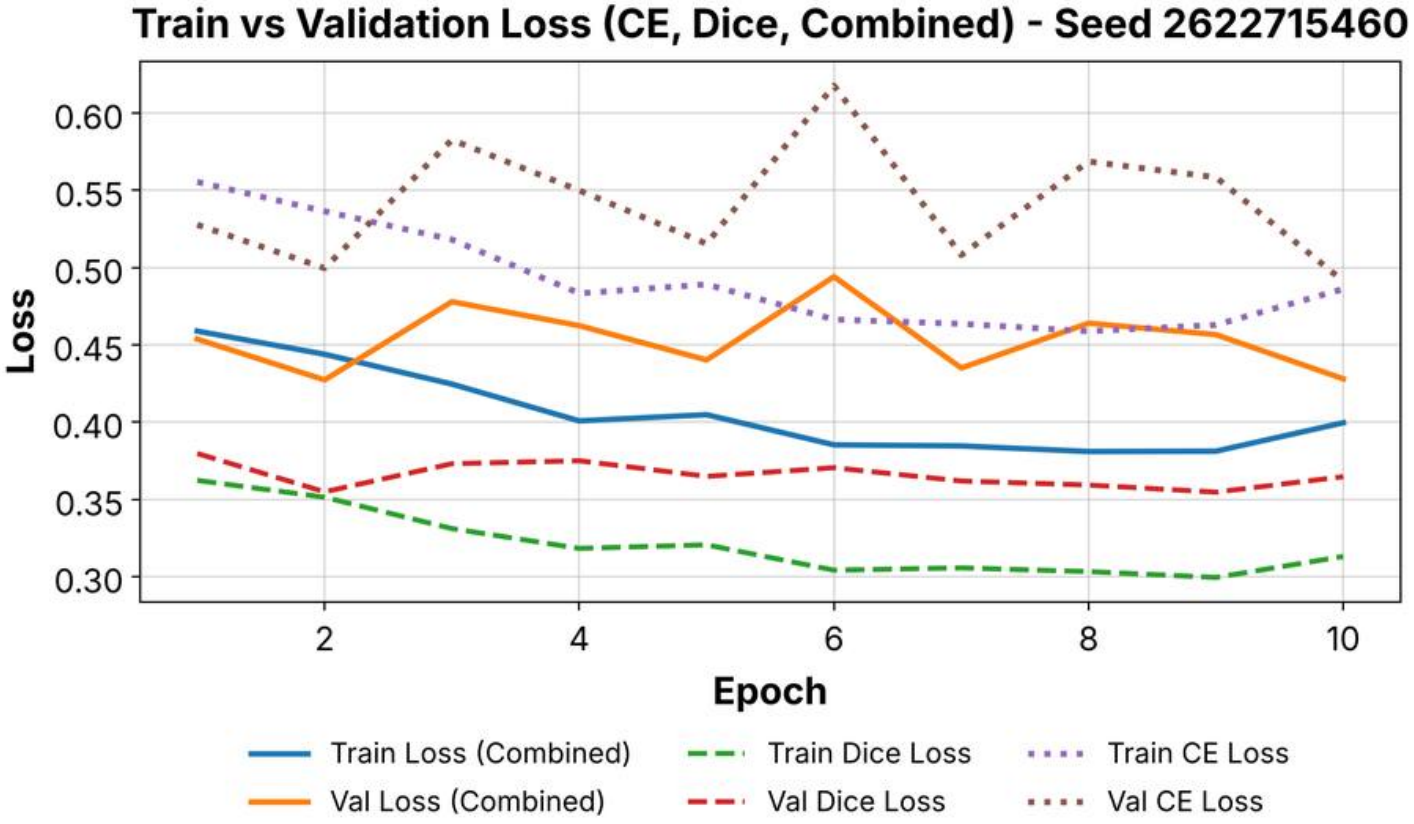
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**92.64%**	91.83%	91.76%	92.08 ± 0.49 %
Precision (weighted)	**92.62%**	91.73%	91.79%	92.05 ± 0.50 %
Recall (weighted)	**92.64%**	91.83%	91.76%	92.08 ± 0.49 %
F1-score (weighted)	**92.49%**	91.72%	91.77%	91.99 ± 0.43 %
AdaBoost validation acc.	71.46%	67.50%	60.00%	66.32 ± 5.82 %
Best validation acc. (CNN)	70.42%	75.62%	75.62%	73.89 ± 3.01 %

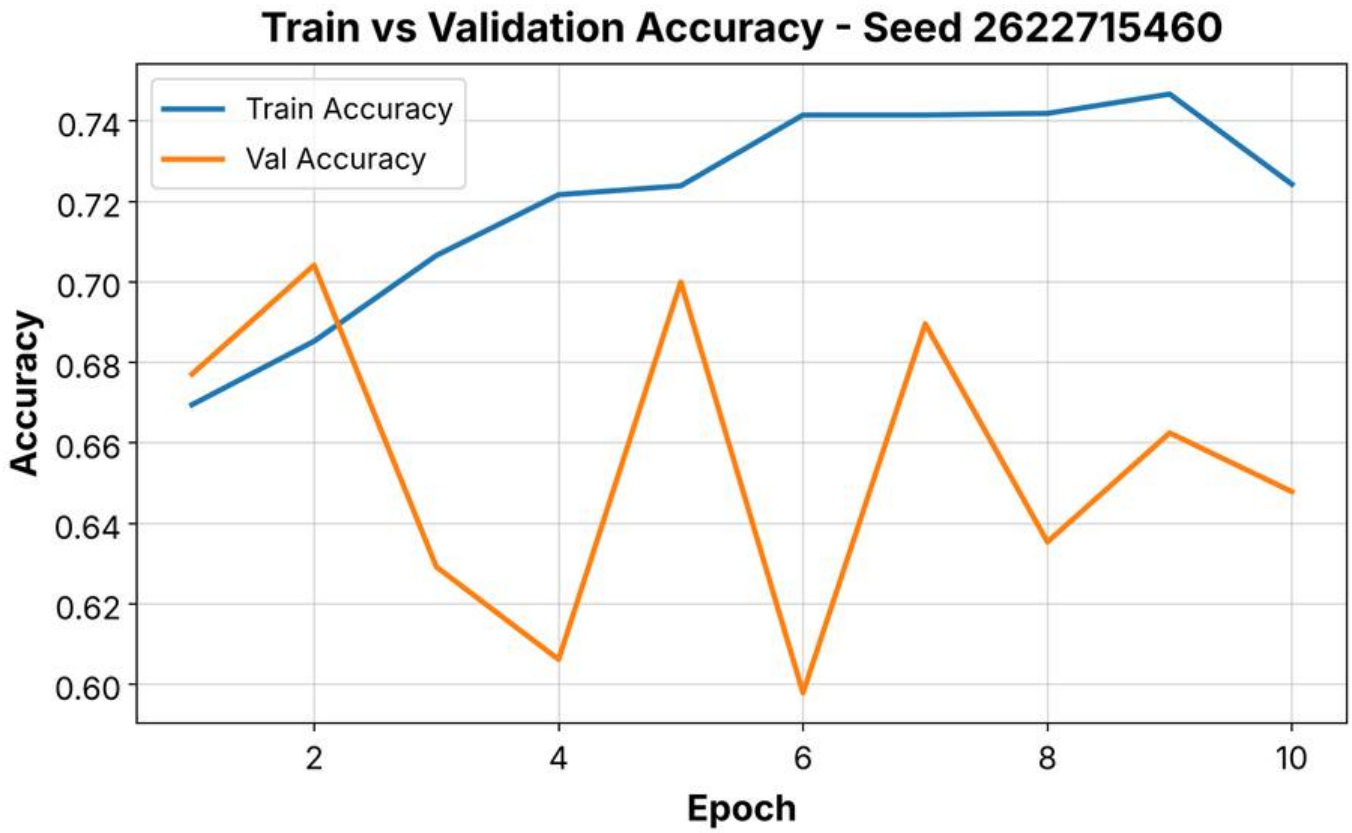
Confusion Matrix



9. EfficientNet-B0 + AdaBoost, Seed 2622715460 — graphs

Training / Validation Graphs



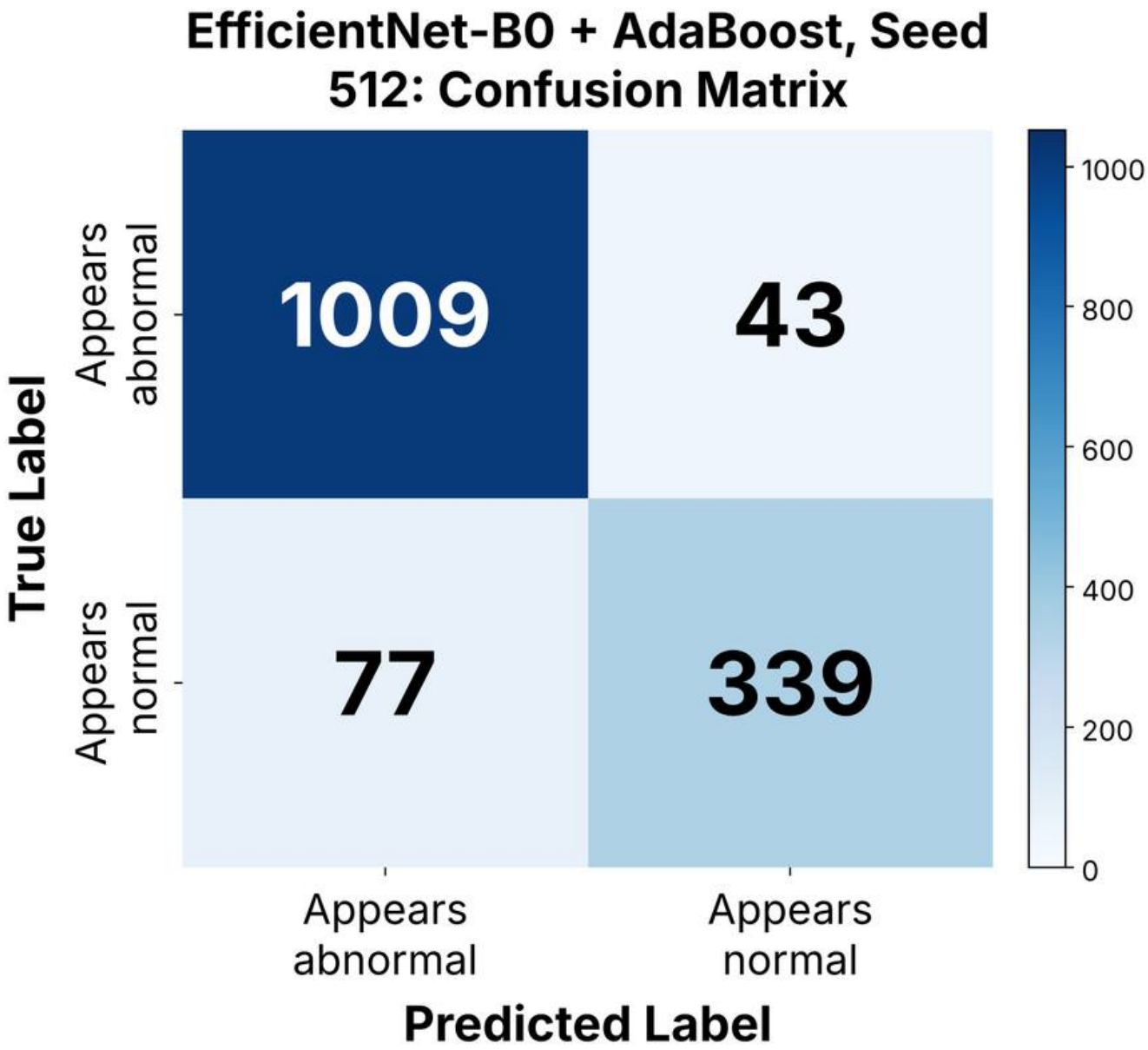


9. EfficientNet-B0 + AdaBoost — Seed 512 (seed 2 of 3)

Source: 3_Seed_exp.ipynb, cell 12. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 8, seed 2622715460* (same numbers for that seed).

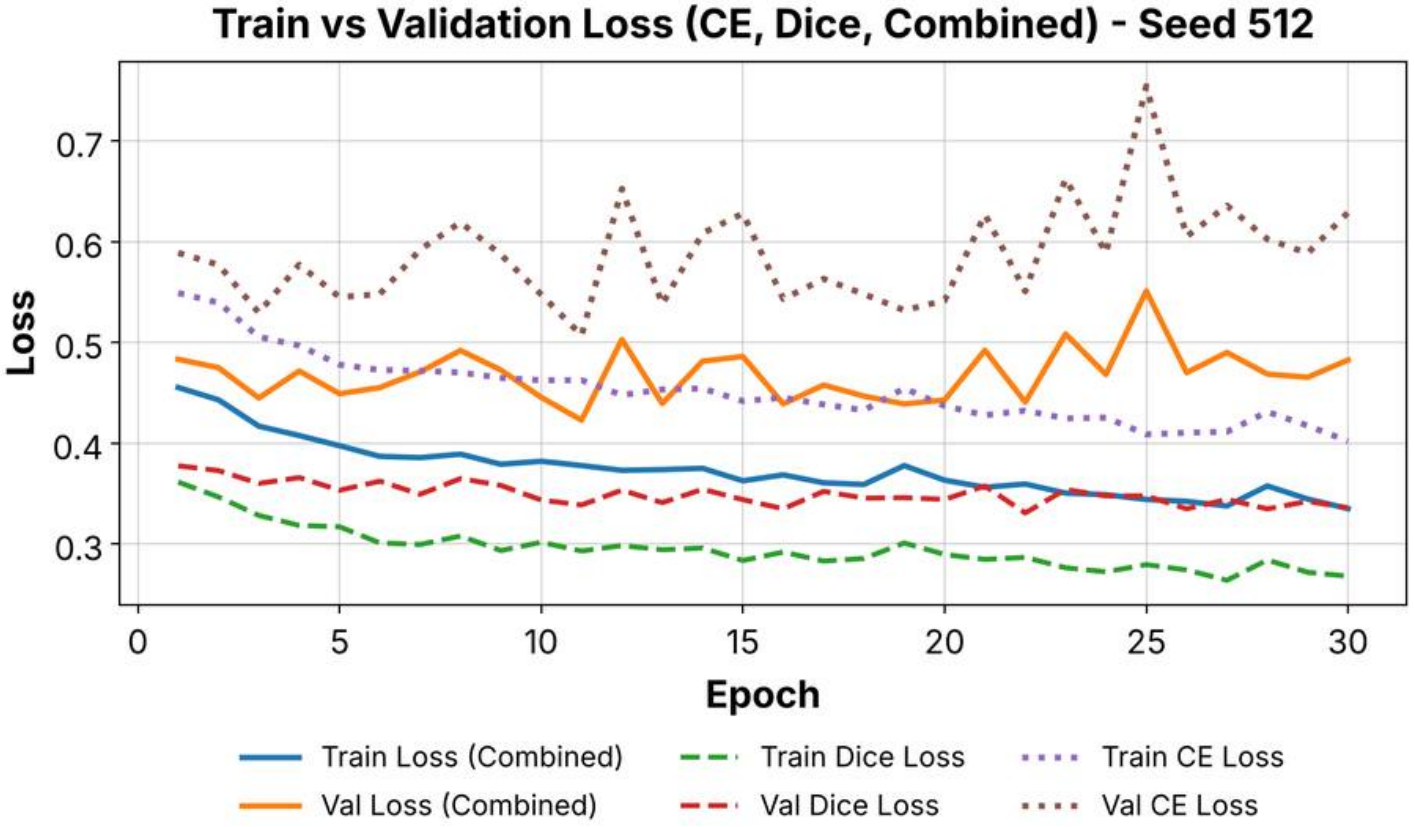
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	92.64%	**91.83%**	91.76%	92.08 ± 0.49 %
Precision (weighted)	92.62%	**91.73%**	91.79%	92.05 ± 0.50 %
Recall (weighted)	92.64%	**91.83%**	91.76%	92.08 ± 0.49 %
F1-score (weighted)	92.49%	**91.72%**	91.77%	91.99 ± 0.43 %
AdaBoost validation acc.	71.46%	67.50%	60.00%	66.32 ± 5.82 %
Best validation acc. (CNN)	70.42%	75.62%	75.62%	73.89 ± 3.01 %

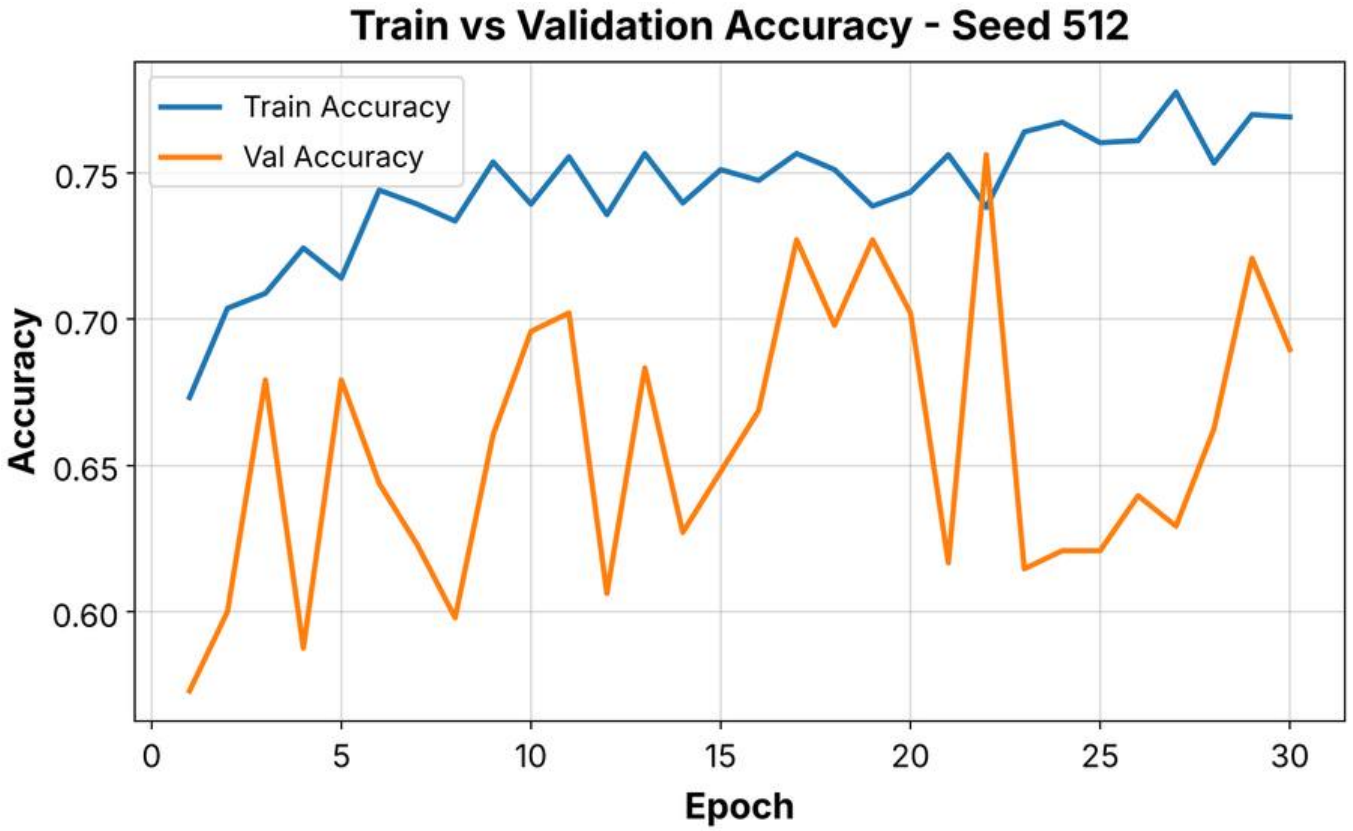
Confusion Matrix



9. EfficientNet-B0 + AdaBoost, Seed 512 — graphs

Training / Validation Graphs



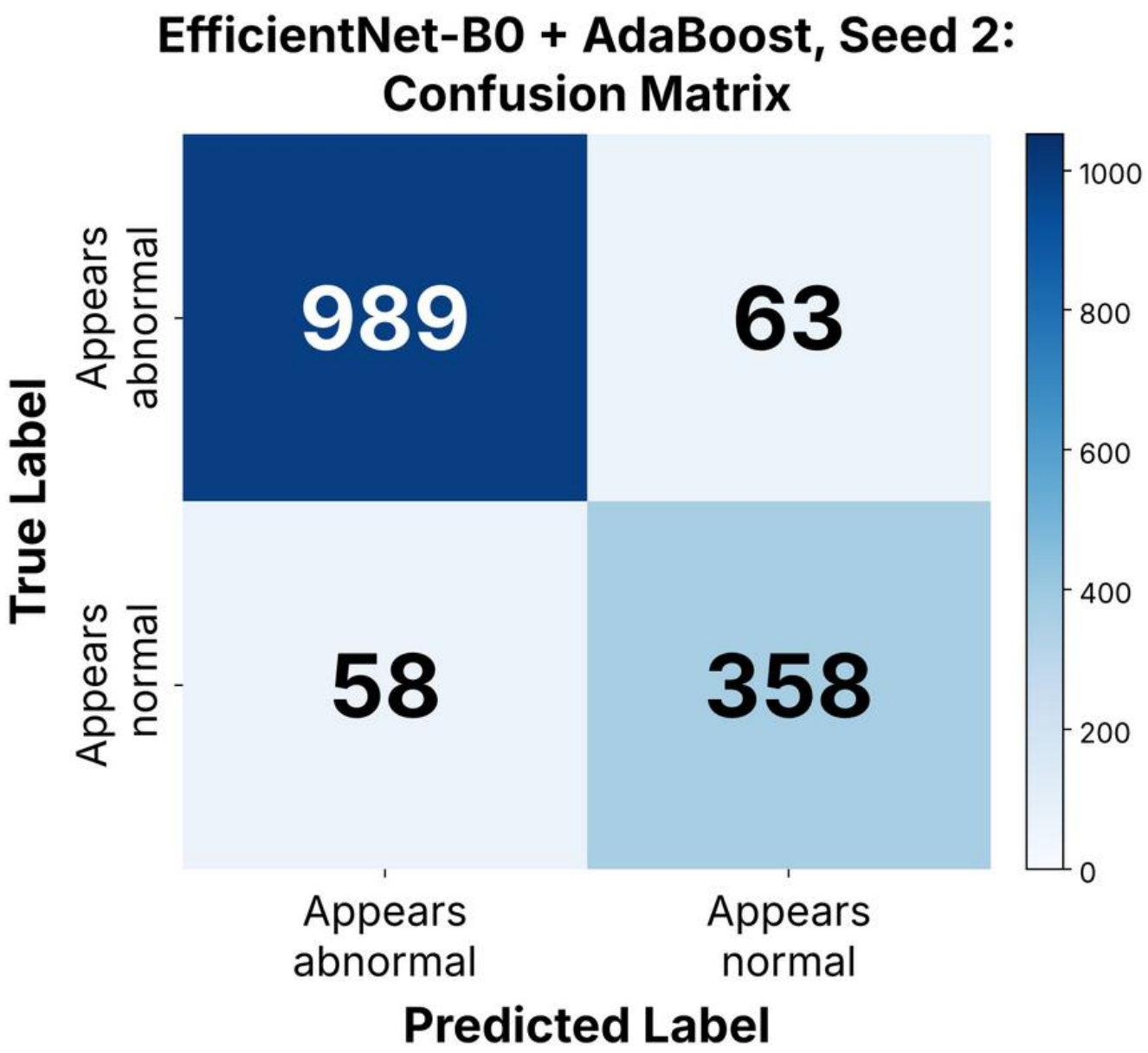


9. EfficientNet-B0 + AdaBoost — Seed 2 (seed 3 of 3)

Source: 3_Seed_exp.ipynb, cell 12. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 8, seed 2622715460* (same numbers for that seed).

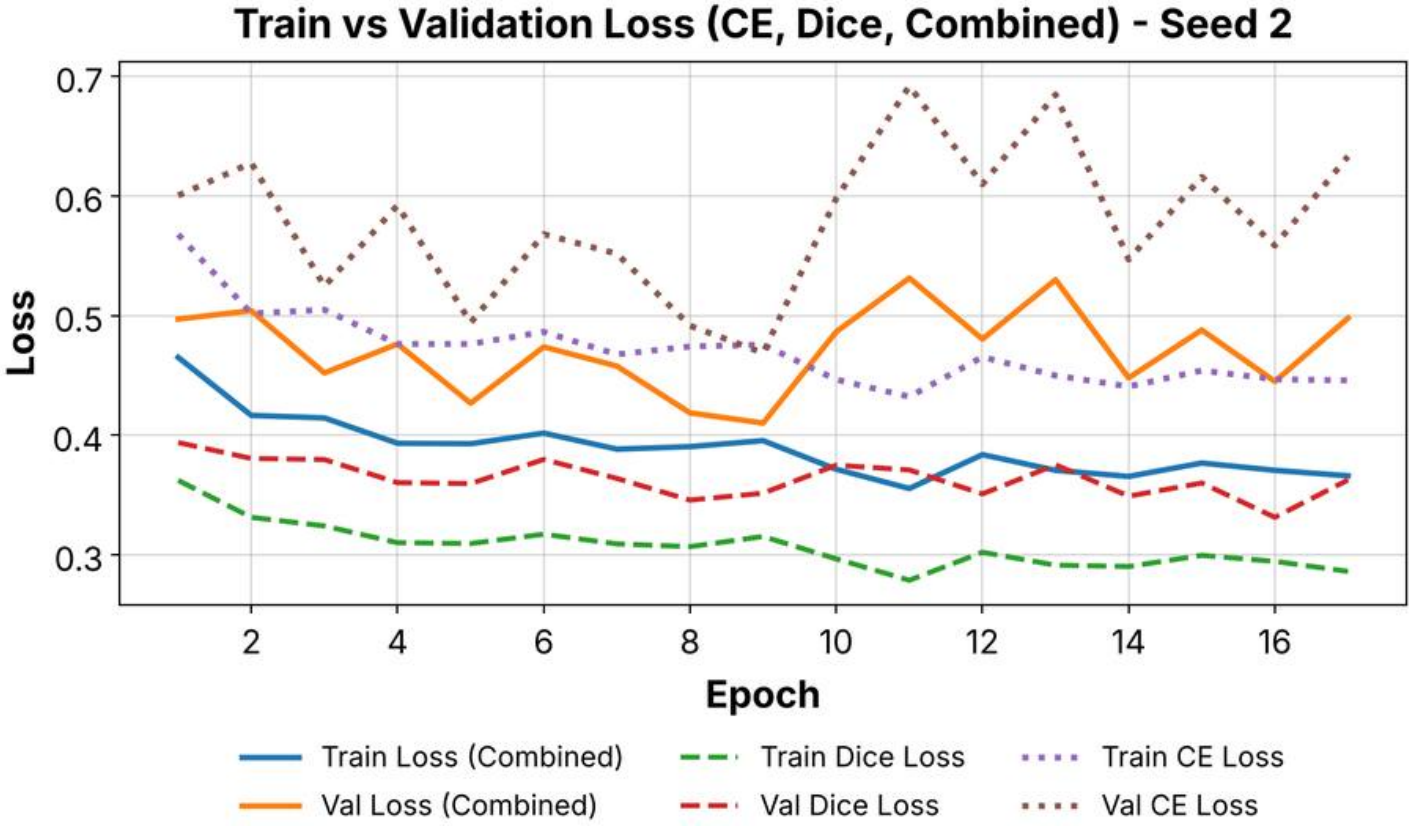
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	92.64%	91.83%	**91.76%**	92.08 ± 0.49 %
Precision (weighted)	92.62%	91.73%	**91.79%**	92.05 ± 0.50 %
Recall (weighted)	92.64%	91.83%	**91.76%**	92.08 ± 0.49 %
F1-score (weighted)	92.49%	91.72%	**91.77%**	91.99 ± 0.43 %
AdaBoost validation acc.	71.46%	67.50%	60.00%	66.32 ± 5.82 %
Best validation acc. (CNN)	70.42%	75.62%	75.62%	73.89 ± 3.01 %

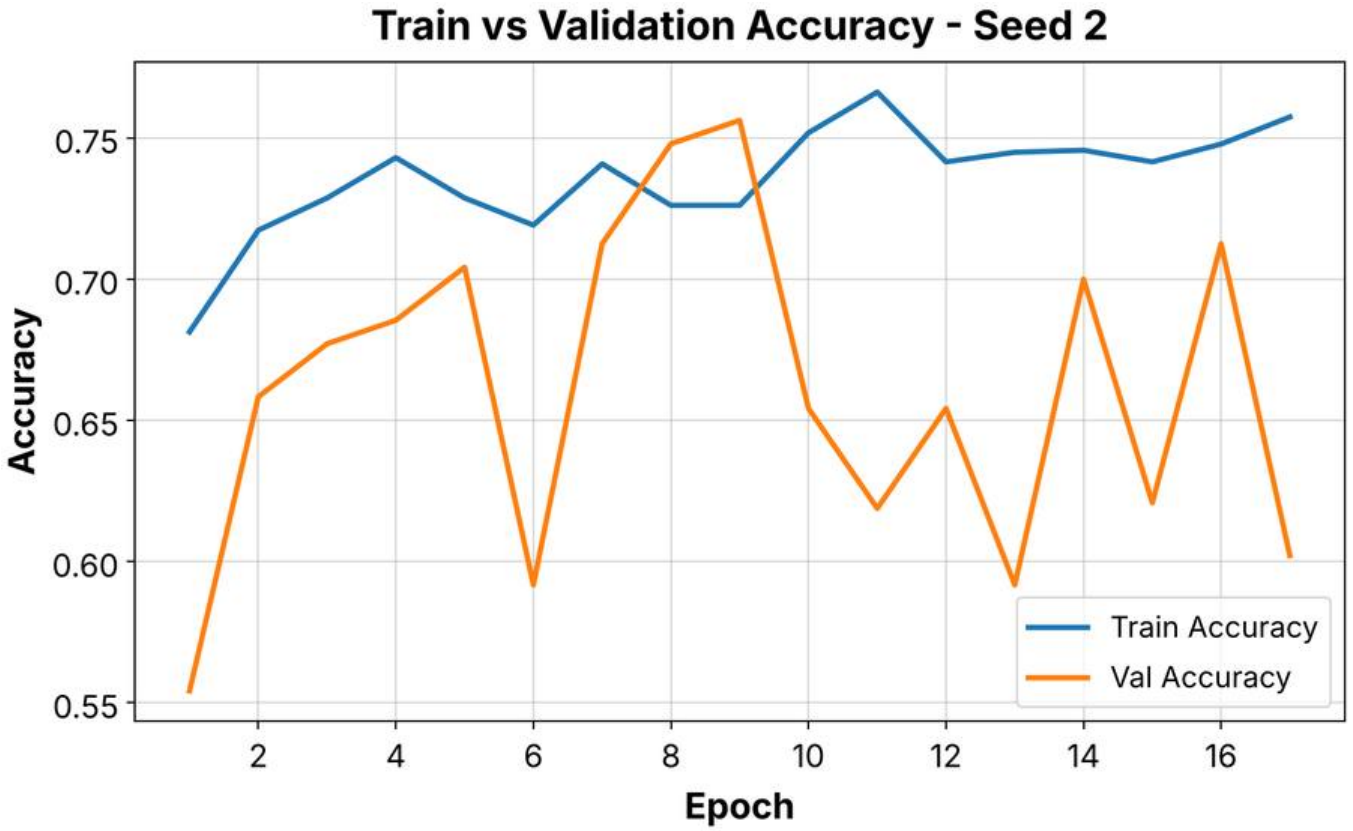
Confusion Matrix



9. EfficientNet-B0 + AdaBoost, Seed 2 — graphs

Training / Validation Graphs



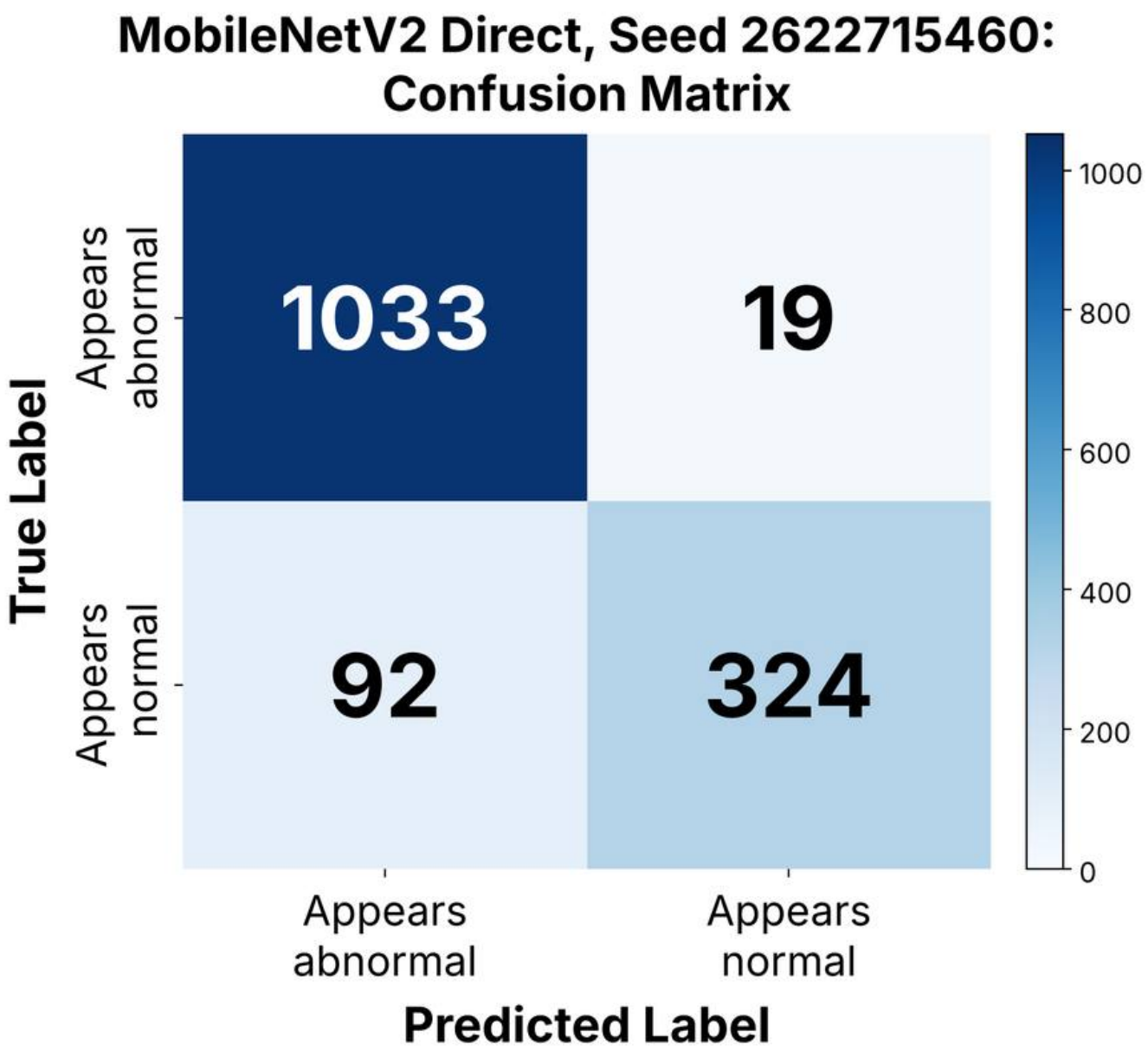


10. MobileNetV2 — Direct (fine-tuned CNN) — Seed 2622715460 (seed 1 of 3)

Source: 3_Seed_exp.ipynb, cell 4. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 4, seed 2622715460* (same numbers for that seed).

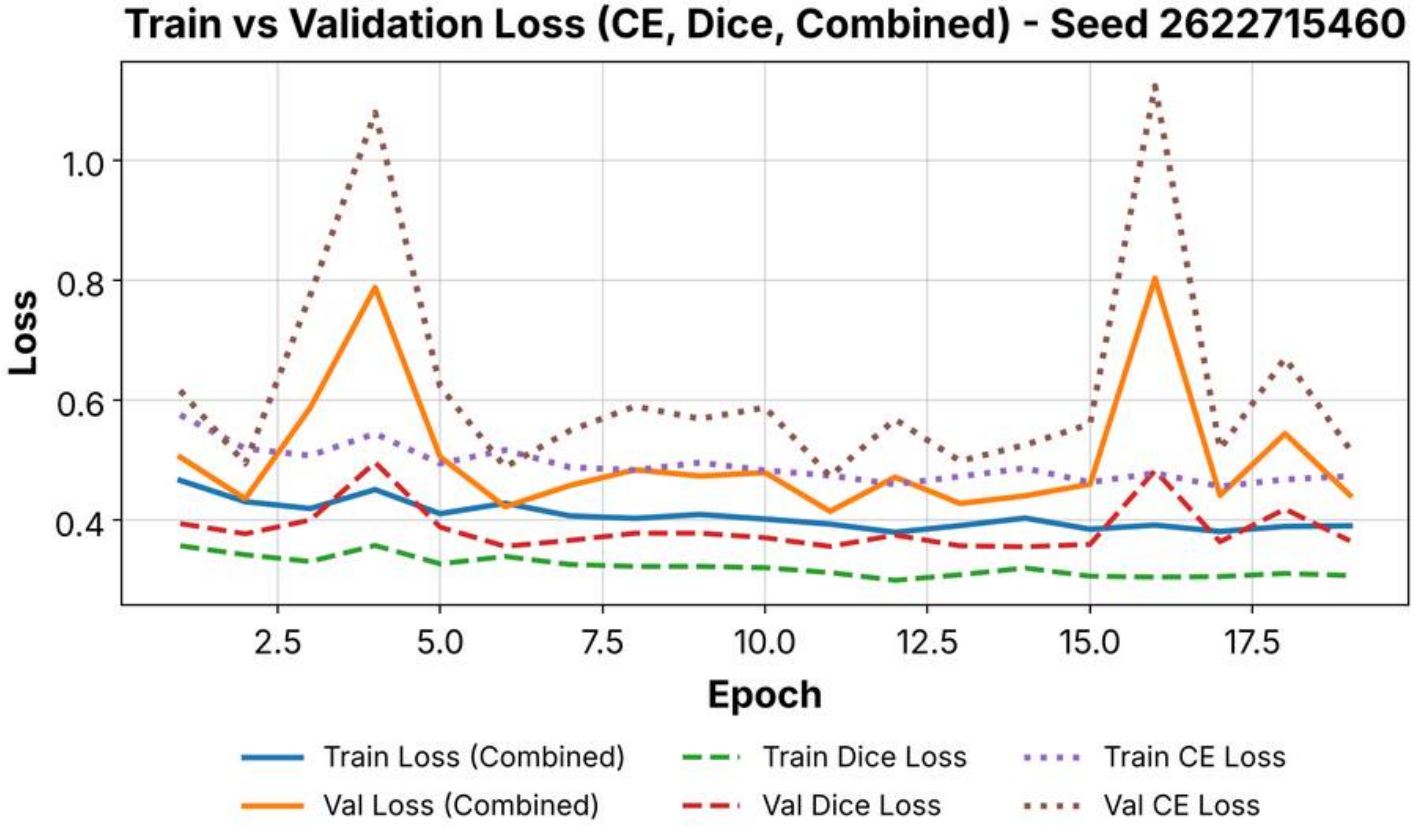
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**92.44%**	85.83%	88.62%	88.96 ± 3.32 %
Precision (weighted)	**92.57%**	85.76%	89.07%	89.13 ± 3.41 %
Recall (weighted)	**92.44%**	85.83%	88.62%	88.96 ± 3.32 %
F1-score (weighted)	**92.20%**	85.02%	87.94%	88.39 ± 3.61 %
Test Dice loss	0.5267	0.5283	0.5245	0.5265 ± 0.0019
Best validation acc. (CNN)	72.08%	71.25%	77.50%	73.61 ± 3.39 %

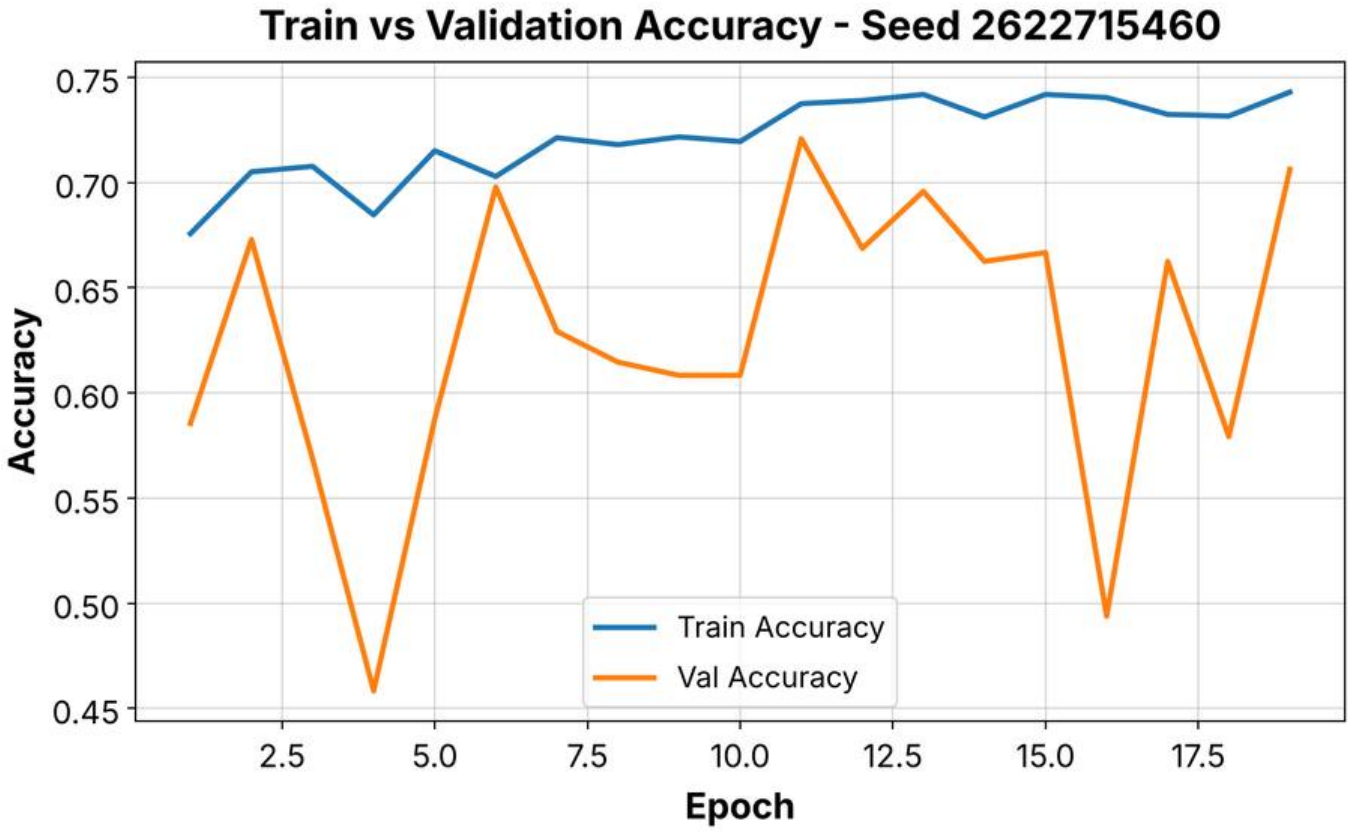
Confusion Matrix



10. MobileNetV2 — Direct (fine-tuned CNN), Seed 2622715460 — graphs

Training / Validation Graphs



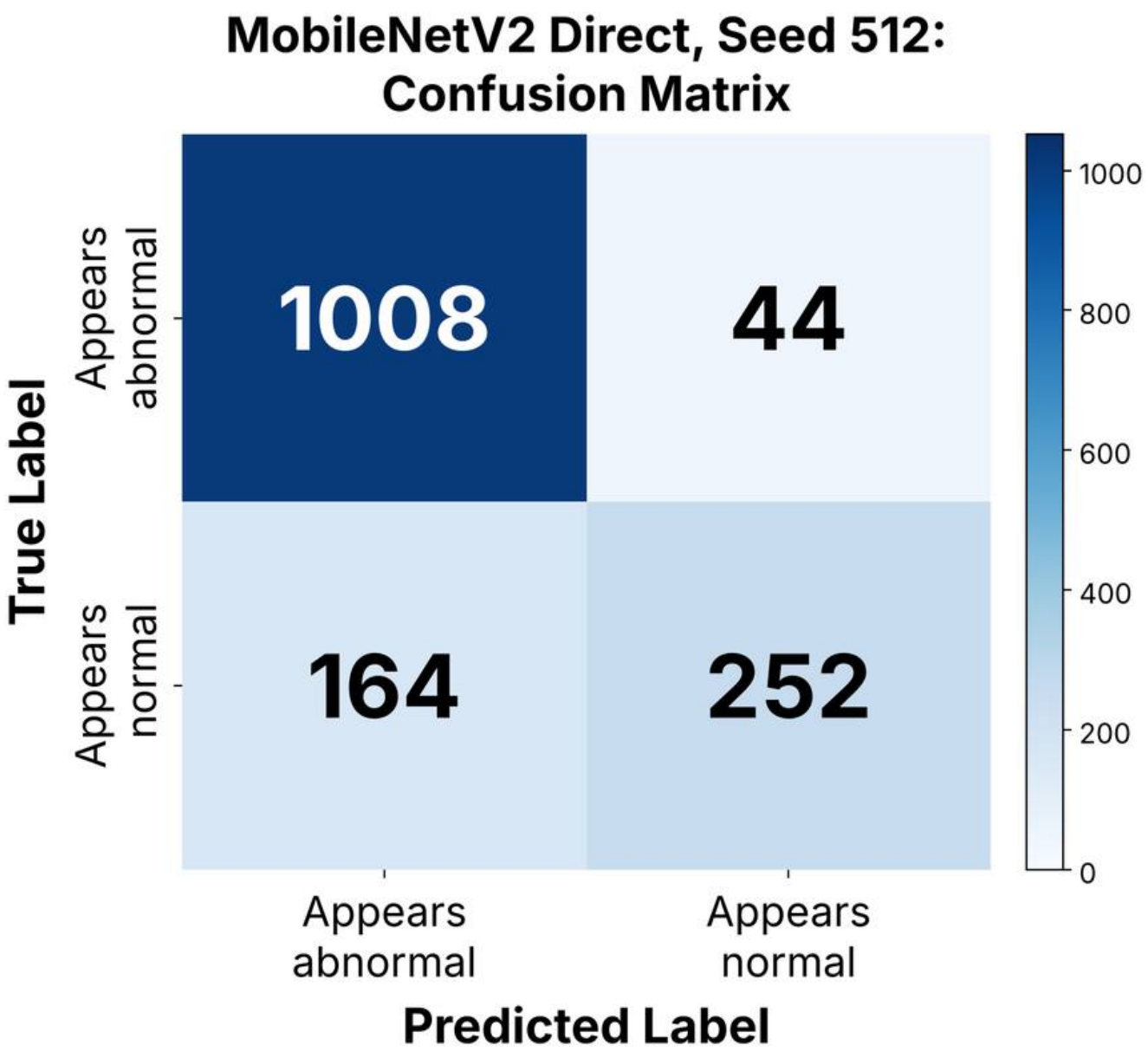


10. MobileNetV2 — Direct (fine-tuned CNN) — Seed 512 (seed 2 of 3)

Source: 3_Seed_exp.ipynb, cell 4. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 4, seed 2622715460* (same numbers for that seed).

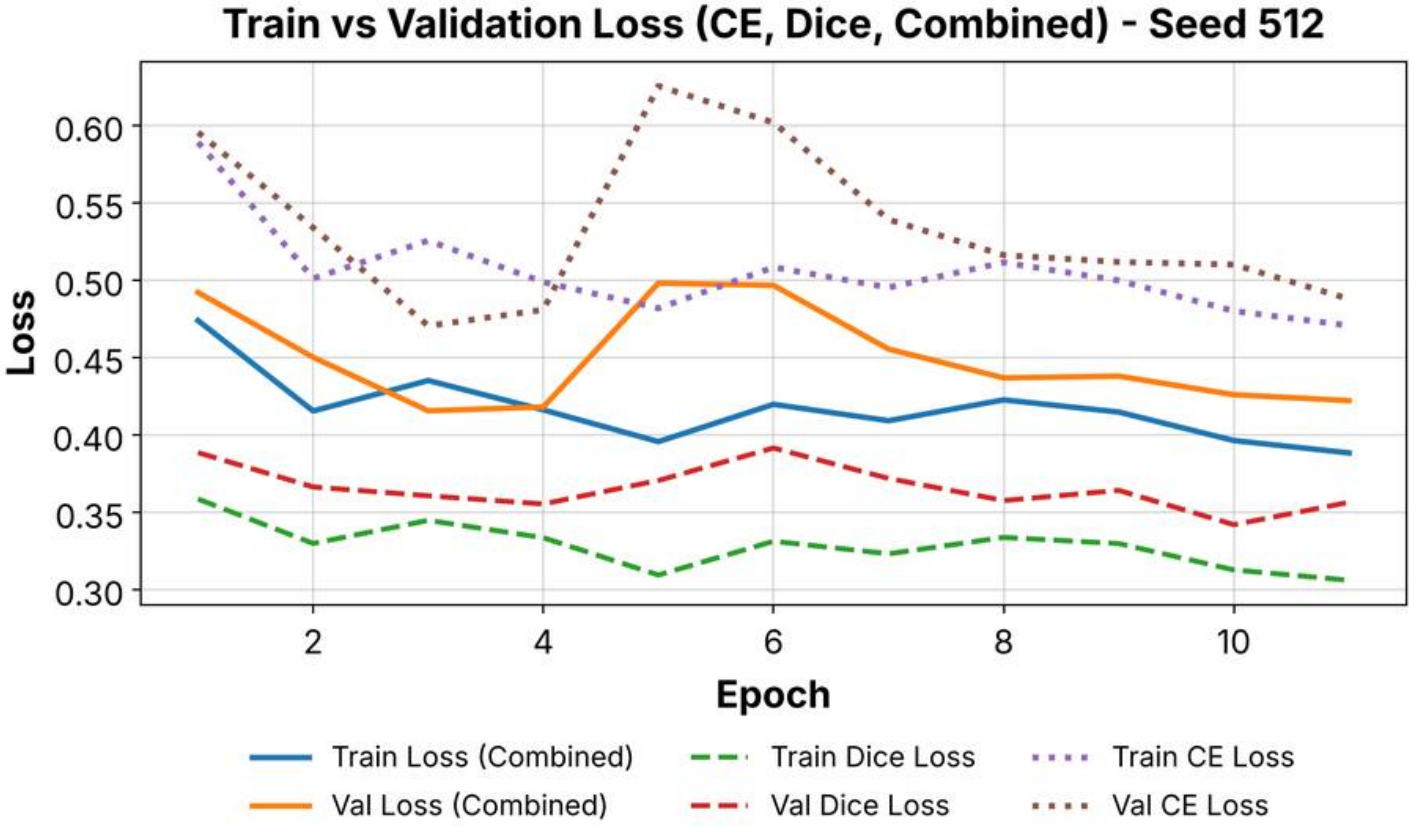
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	92.44%	**85.83%**	88.62%	88.96 ± 3.32 %
Precision (weighted)	92.57%	**85.76%**	89.07%	89.13 ± 3.41 %
Recall (weighted)	92.44%	**85.83%**	88.62%	88.96 ± 3.32 %
F1-score (weighted)	92.20%	**85.02%**	87.94%	88.39 ± 3.61 %
Test Dice loss	0.5267	0.5283	0.5245	0.5265 ± 0.0019
Best validation acc. (CNN)	72.08%	71.25%	77.50%	73.61 ± 3.39 %

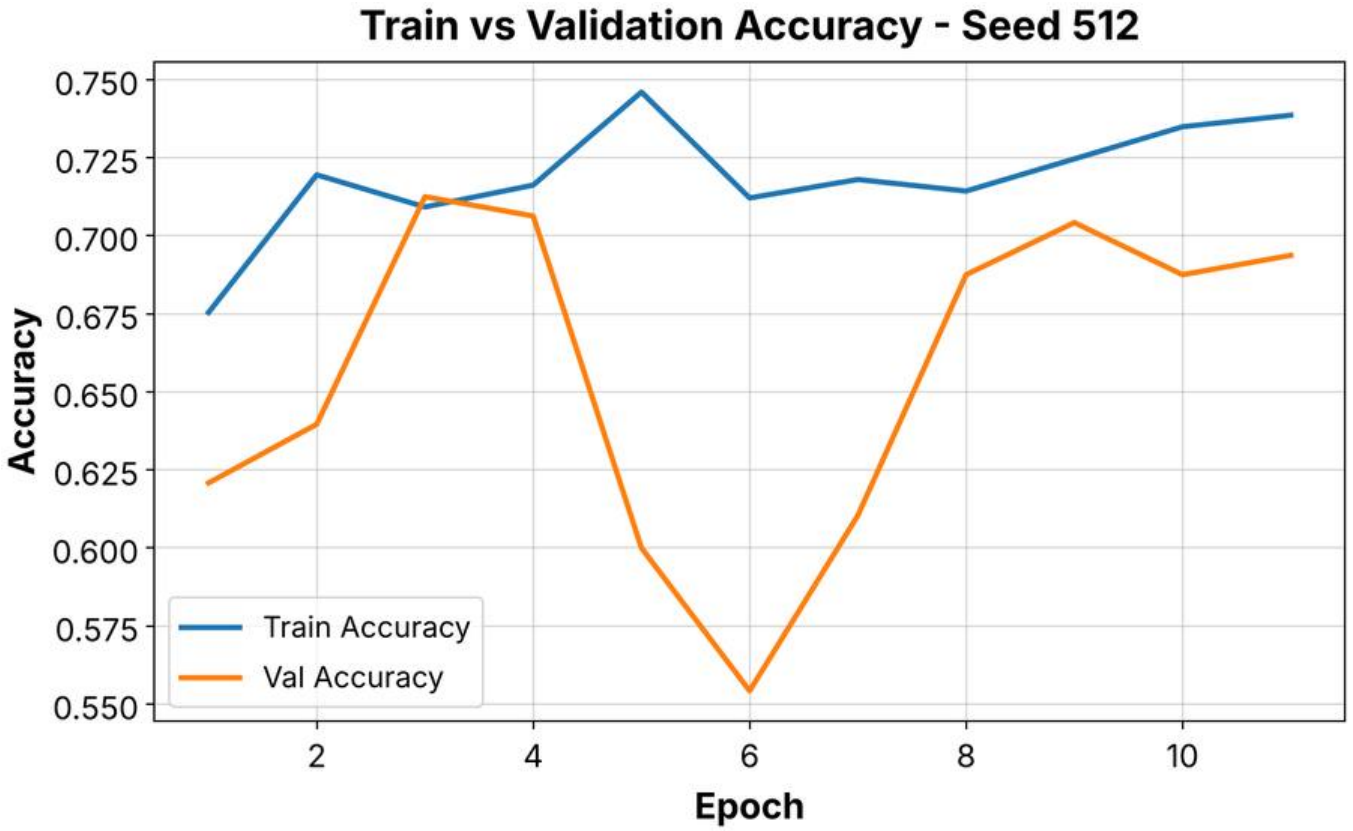
Confusion Matrix



10. MobileNetV2 — Direct (fine-tuned CNN), Seed 512 — graphs

Training / Validation Graphs



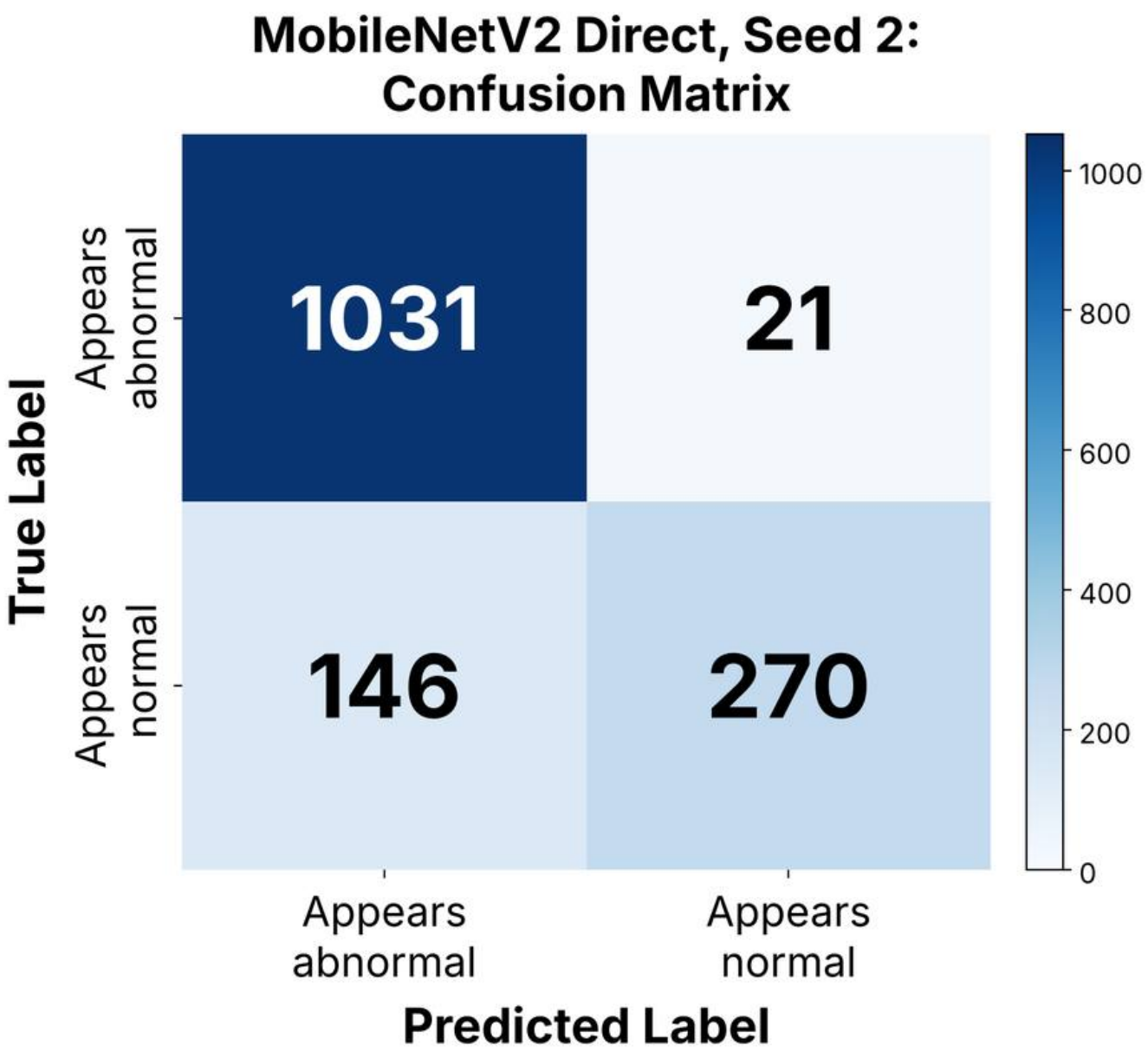


10. MobileNetV2 — Direct (fine-tuned CNN) — Seed 2 (seed 3 of 3)

Source: 3_Seed_exp.ipynb, cell 4. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 4, seed 2622715460* (same numbers for that seed).

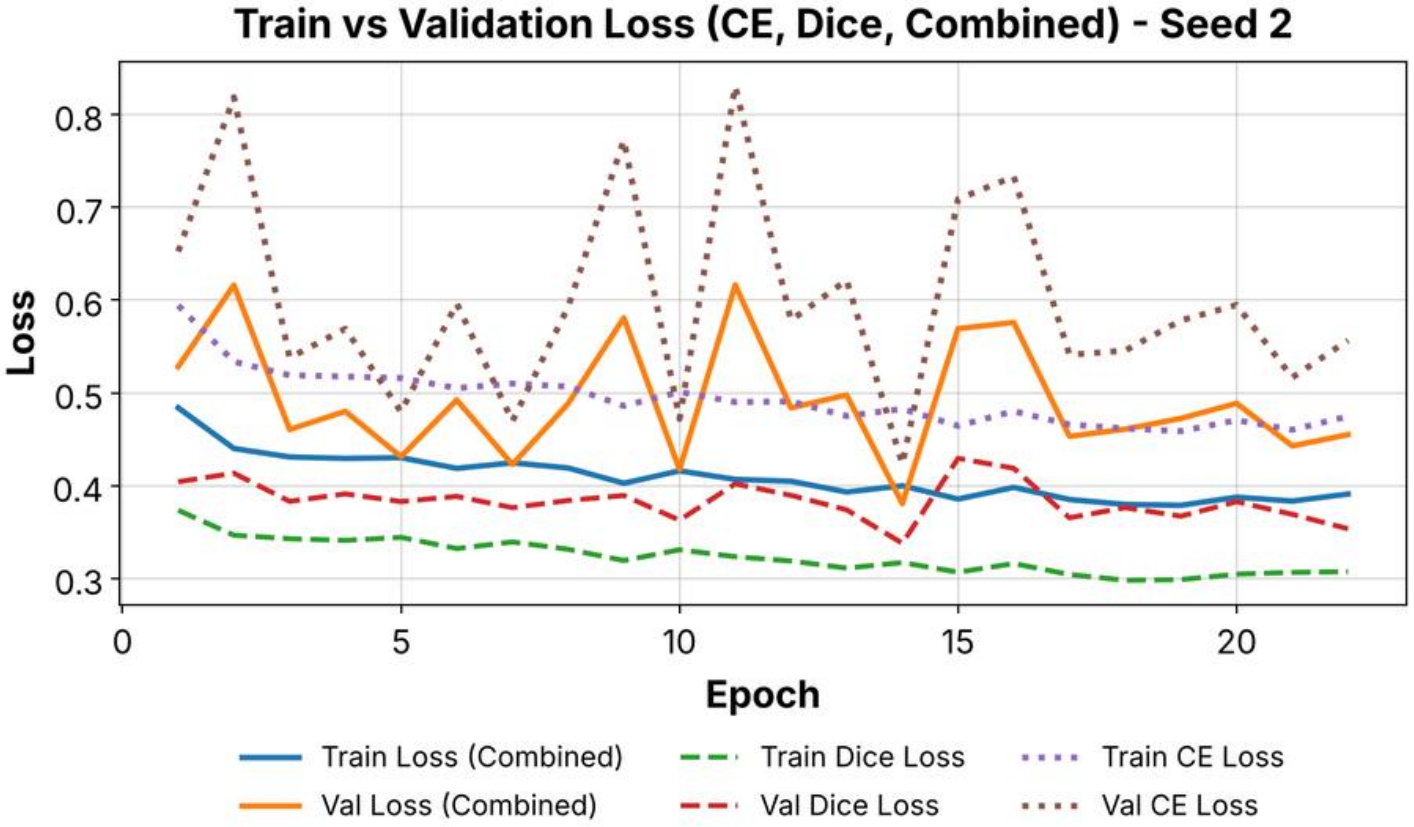
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	92.44%	85.83%	**88.62%**	88.96 ± 3.32 %
Precision (weighted)	92.57%	85.76%	**89.07%**	89.13 ± 3.41 %
Recall (weighted)	92.44%	85.83%	**88.62%**	88.96 ± 3.32 %
F1-score (weighted)	92.20%	85.02%	**87.94%**	88.39 ± 3.61 %
Test Dice loss	0.5267	0.5283	0.5245	0.5265 ± 0.0019
Best validation acc. (CNN)	72.08%	71.25%	77.50%	73.61 ± 3.39 %

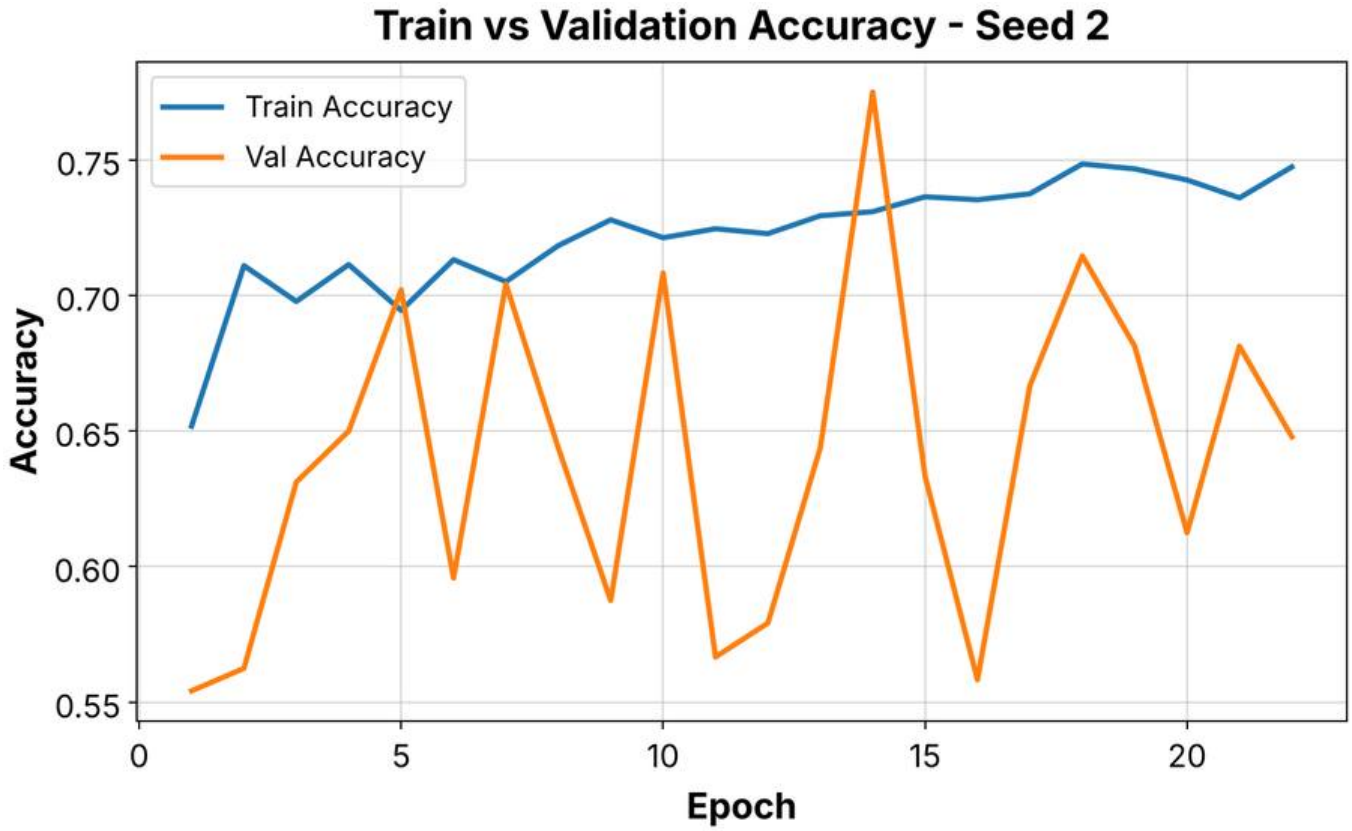
Confusion Matrix



10. MobileNetV2 — Direct (fine-tuned CNN), Seed 2 — graphs

Training / Validation Graphs



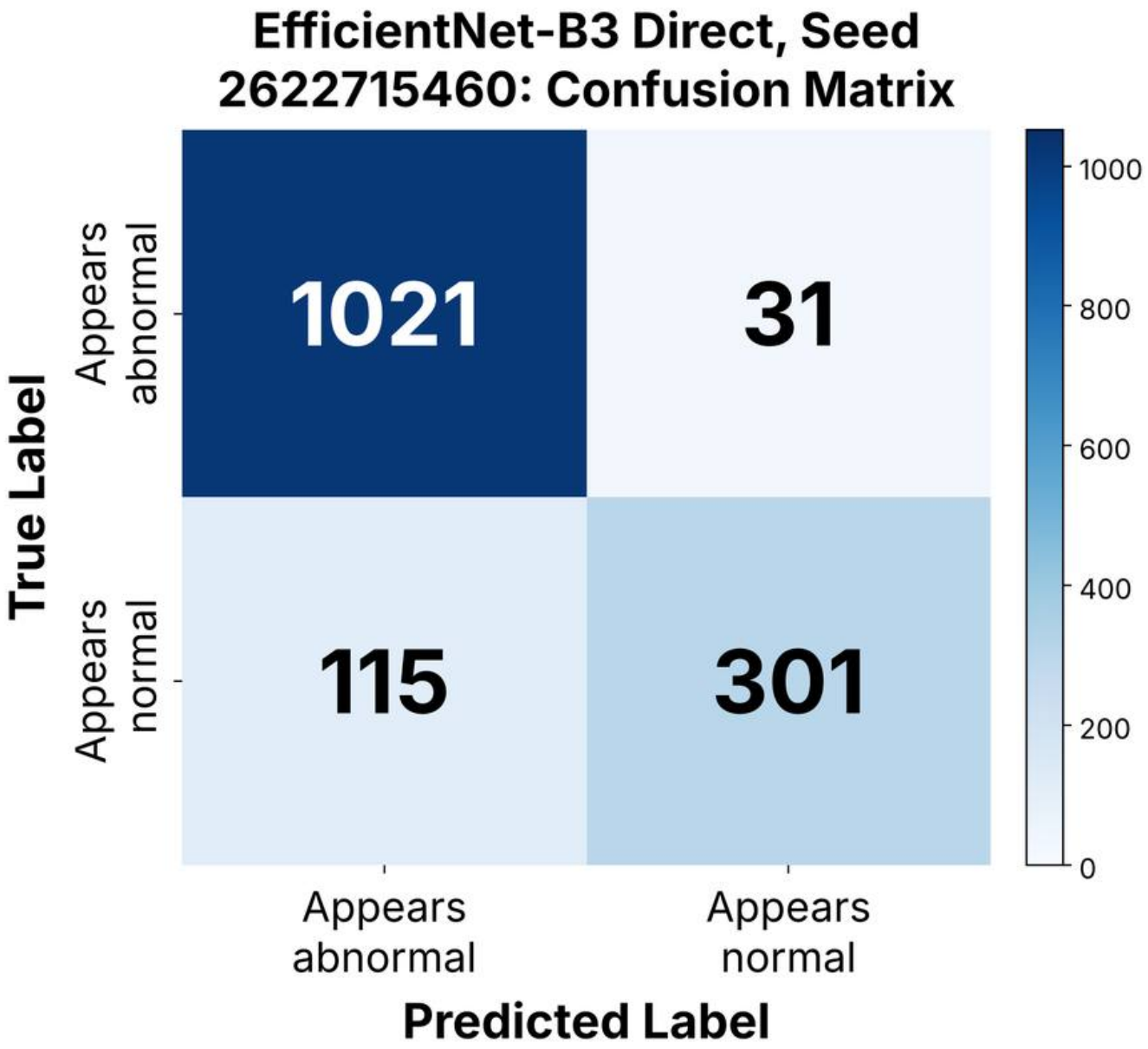


11. EfficientNet-B3 — Direct (fine-tuned CNN) — Seed 2622715460 (seed 1 of 3)

Source: 3_Seed_exp.ipynb, cell 8. Re-run independently in *(seeded) EfficientNet B3 with Adaboost.ipynb, cell 4, seed 512* (same numbers for that seed).

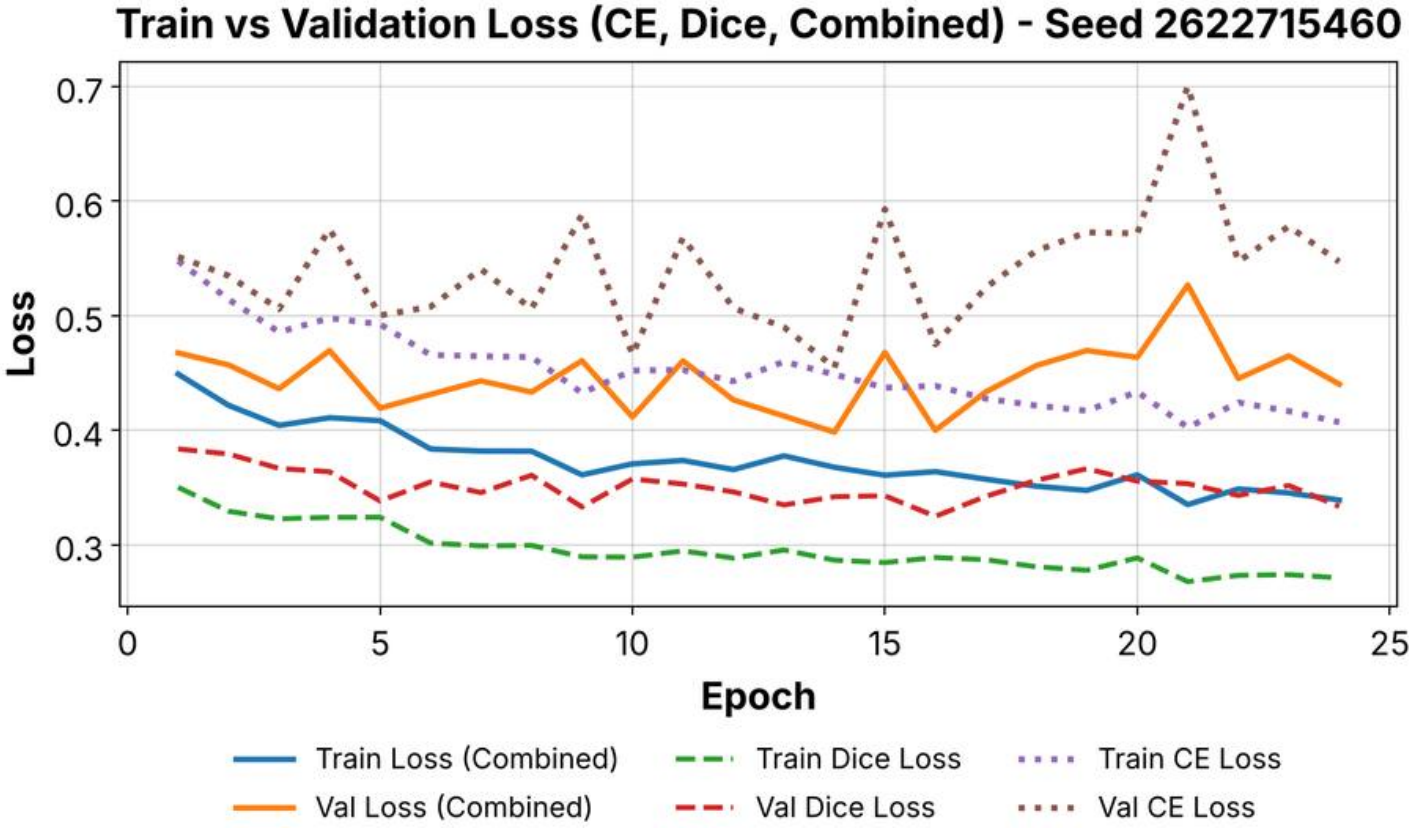
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**90.05%**	87.60%	89.71%	89.12 ± 1.33 %
Precision (weighted)	**90.10%**	87.90%	89.84%	89.28 ± 1.21 %
Recall (weighted)	**90.05%**	87.60%	89.71%	89.12 ± 1.33 %
F1-score (weighted)	**89.69%**	86.85%	89.27%	88.60 ± 1.53 %
Test Dice loss	0.5196	0.5218	0.5247	0.5220 ± 0.0025
Best validation acc. (CNN)	76.46%	75.62%	72.50%	74.86 ± 2.09 %

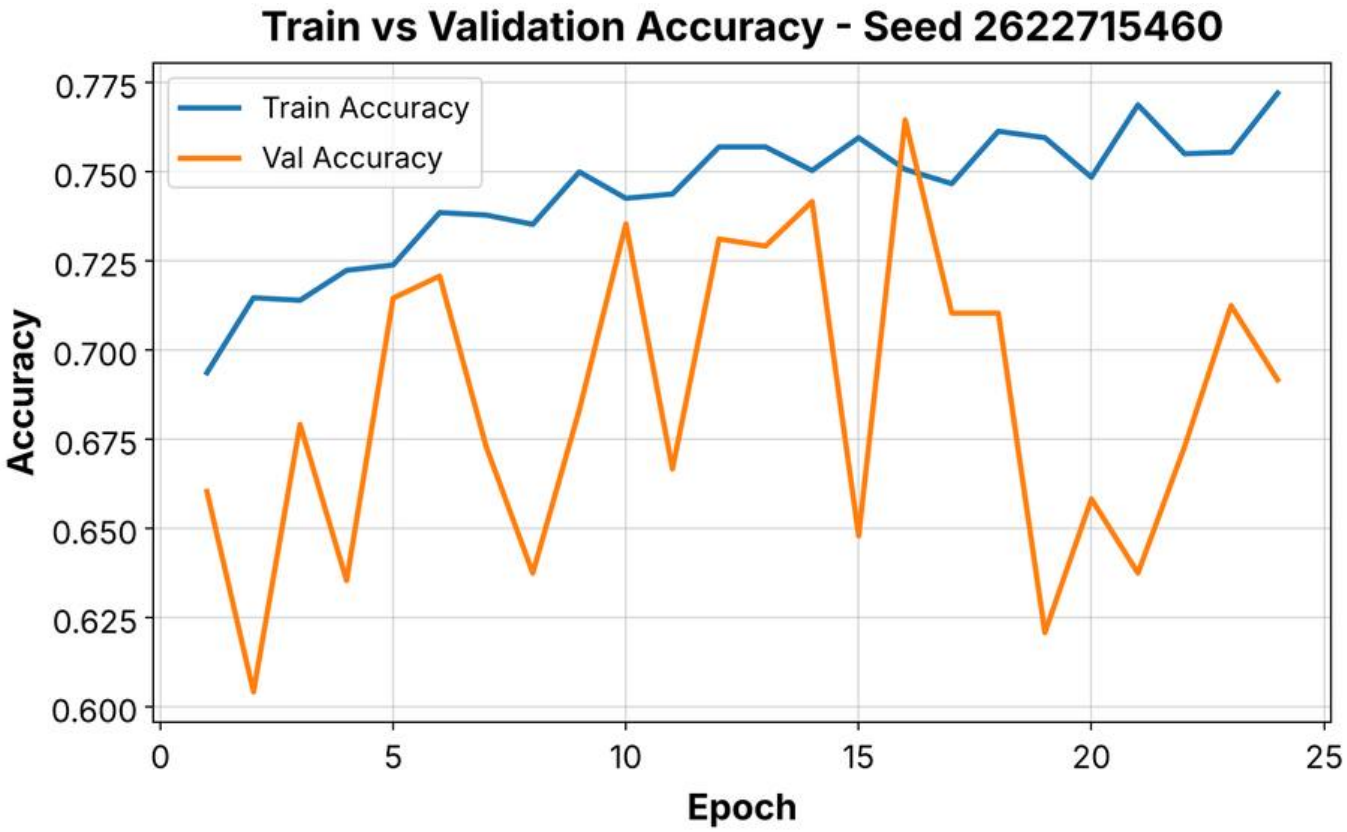
Confusion Matrix



11. EfficientNet-B3 — Direct (fine-tuned CNN), Seed 2622715460 — graphs

Training / Validation Graphs



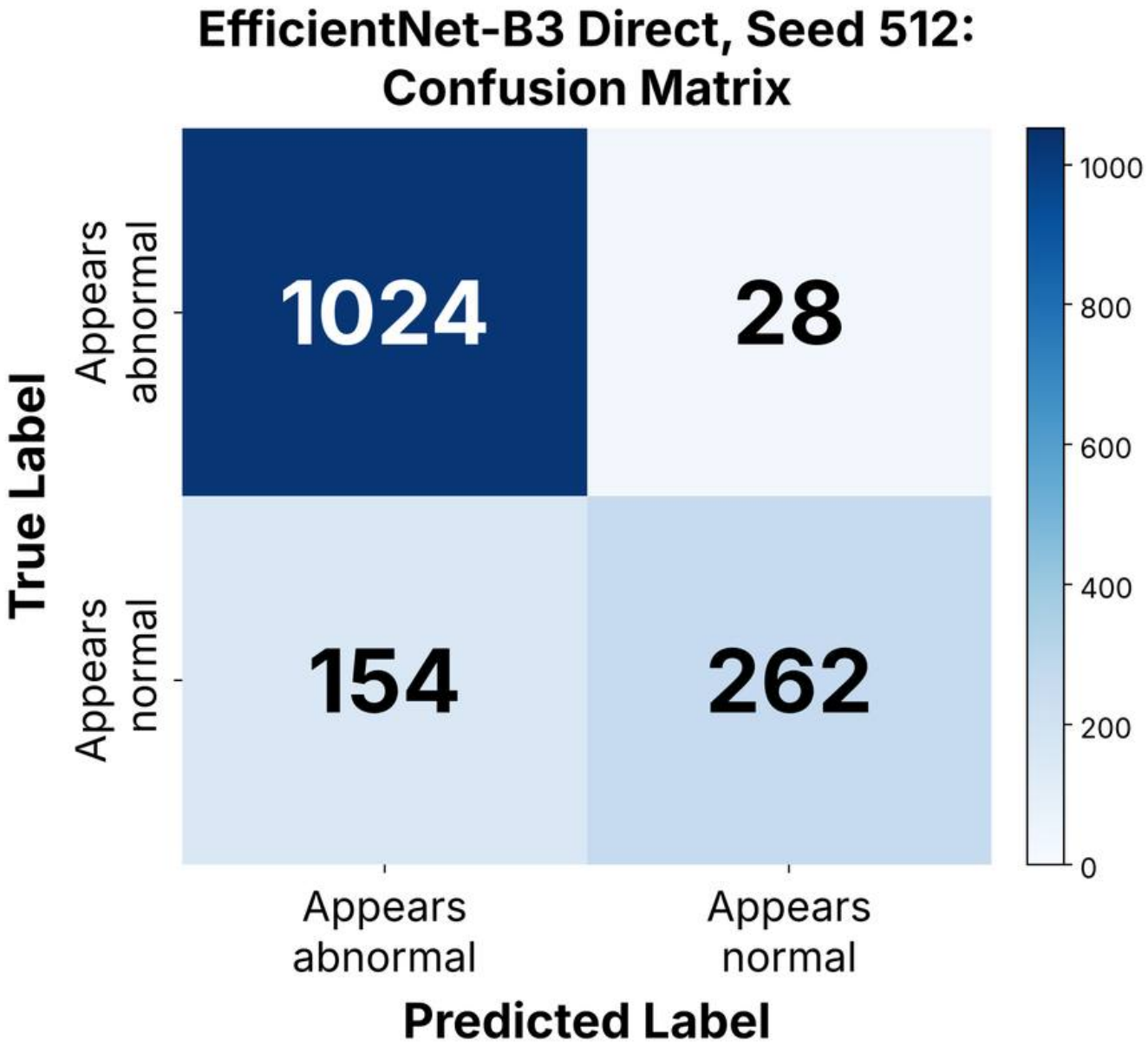


11. EfficientNet-B3 — Direct (fine-tuned CNN) — Seed 512 (seed 2 of 3)

Source: 3_Seed_exp.ipynb, cell 8. Re-run independently in *(seeded) EfficientNet B3 with Adaboost.ipynb, cell 4, seed 512* (same numbers for that seed).

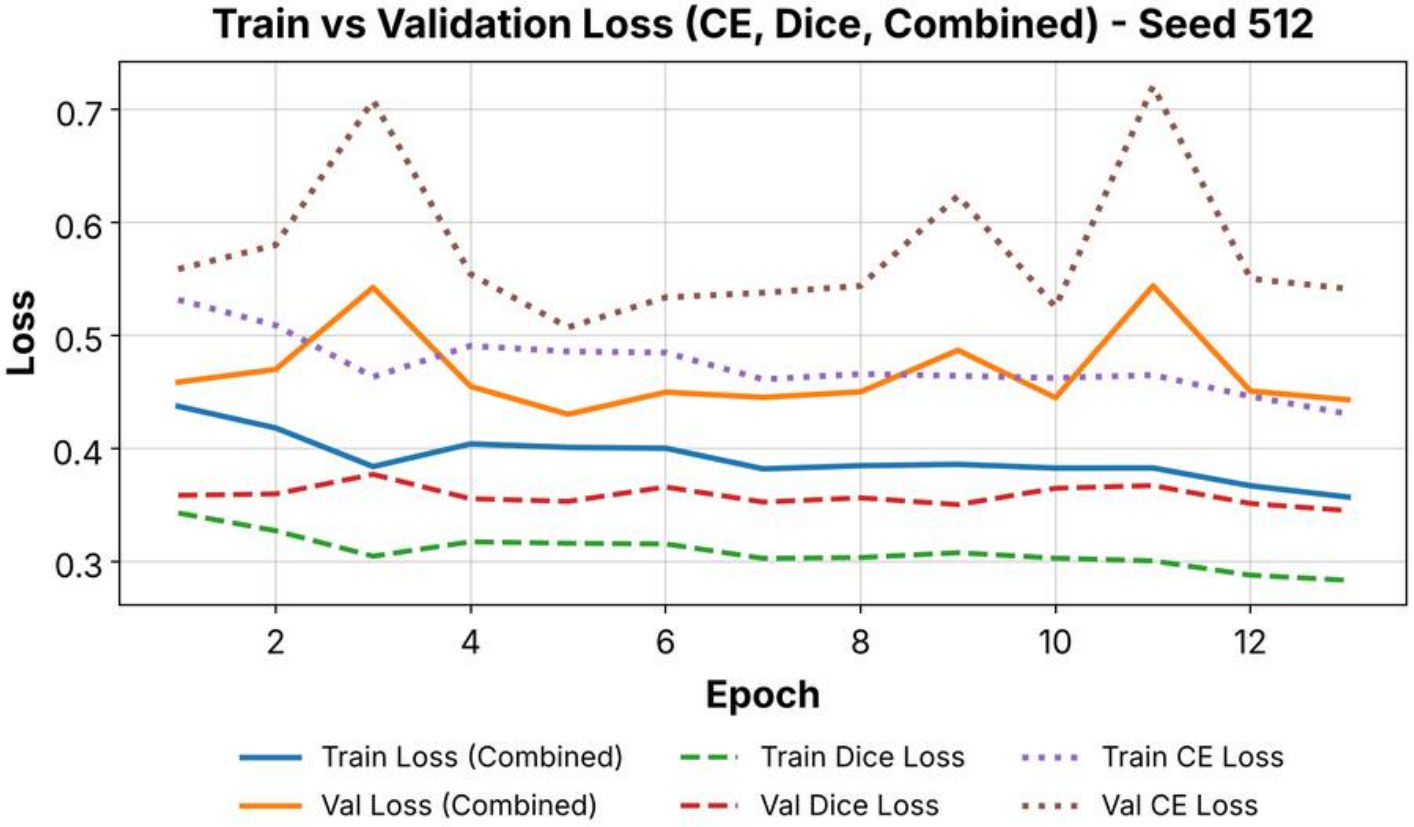
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	90.05%	**87.60%**	89.71%	89.12 ± 1.33 %
Precision (weighted)	90.10%	**87.90%**	89.84%	89.28 ± 1.21 %
Recall (weighted)	90.05%	**87.60%**	89.71%	89.12 ± 1.33 %
F1-score (weighted)	89.69%	**86.85%**	89.27%	88.60 ± 1.53 %
Test Dice loss	0.5196	0.5218	0.5247	0.5220 ± 0.0025
Best validation acc. (CNN)	76.46%	75.62%	72.50%	74.86 ± 2.09 %

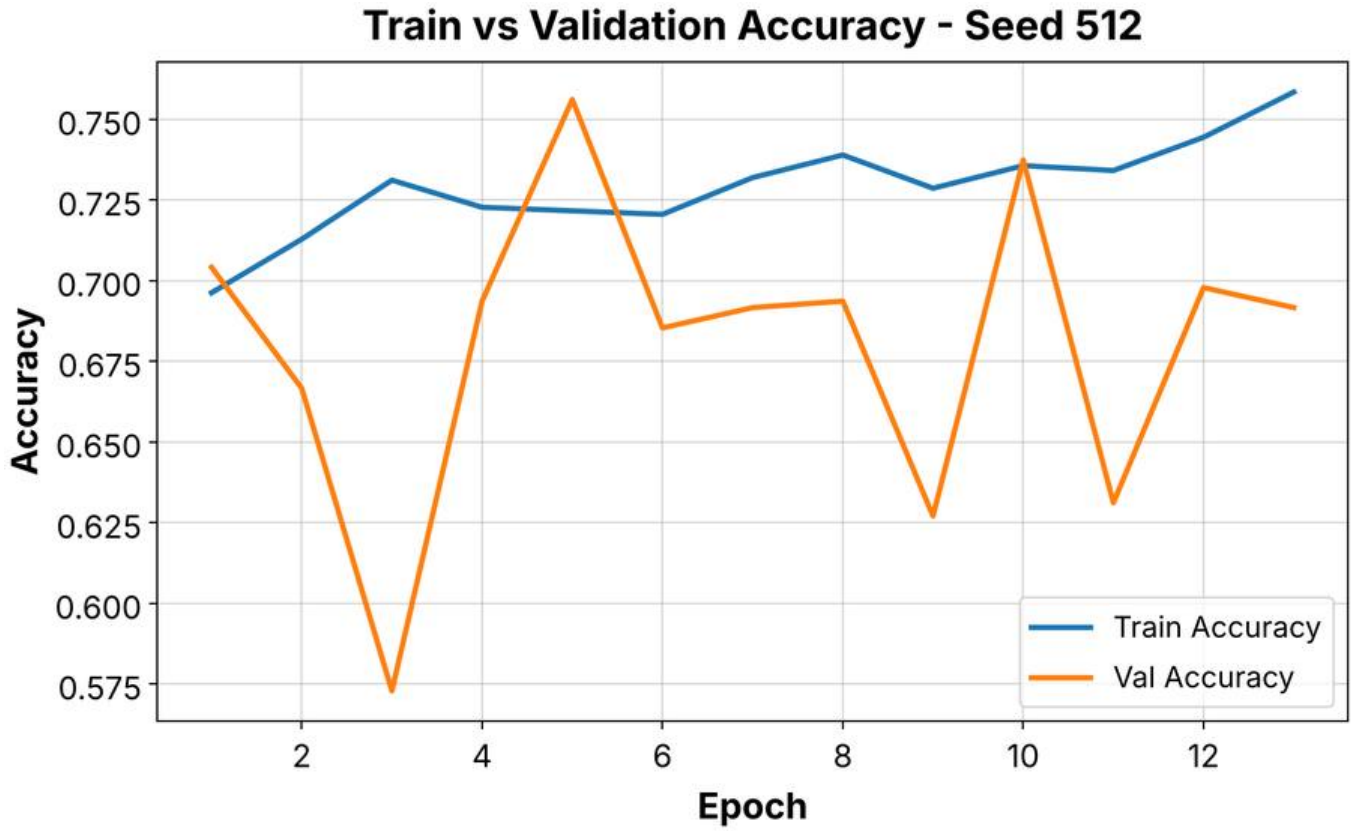
Confusion Matrix



11. EfficientNet-B3 — Direct (fine-tuned CNN), Seed 512 — graphs

Training / Validation Graphs



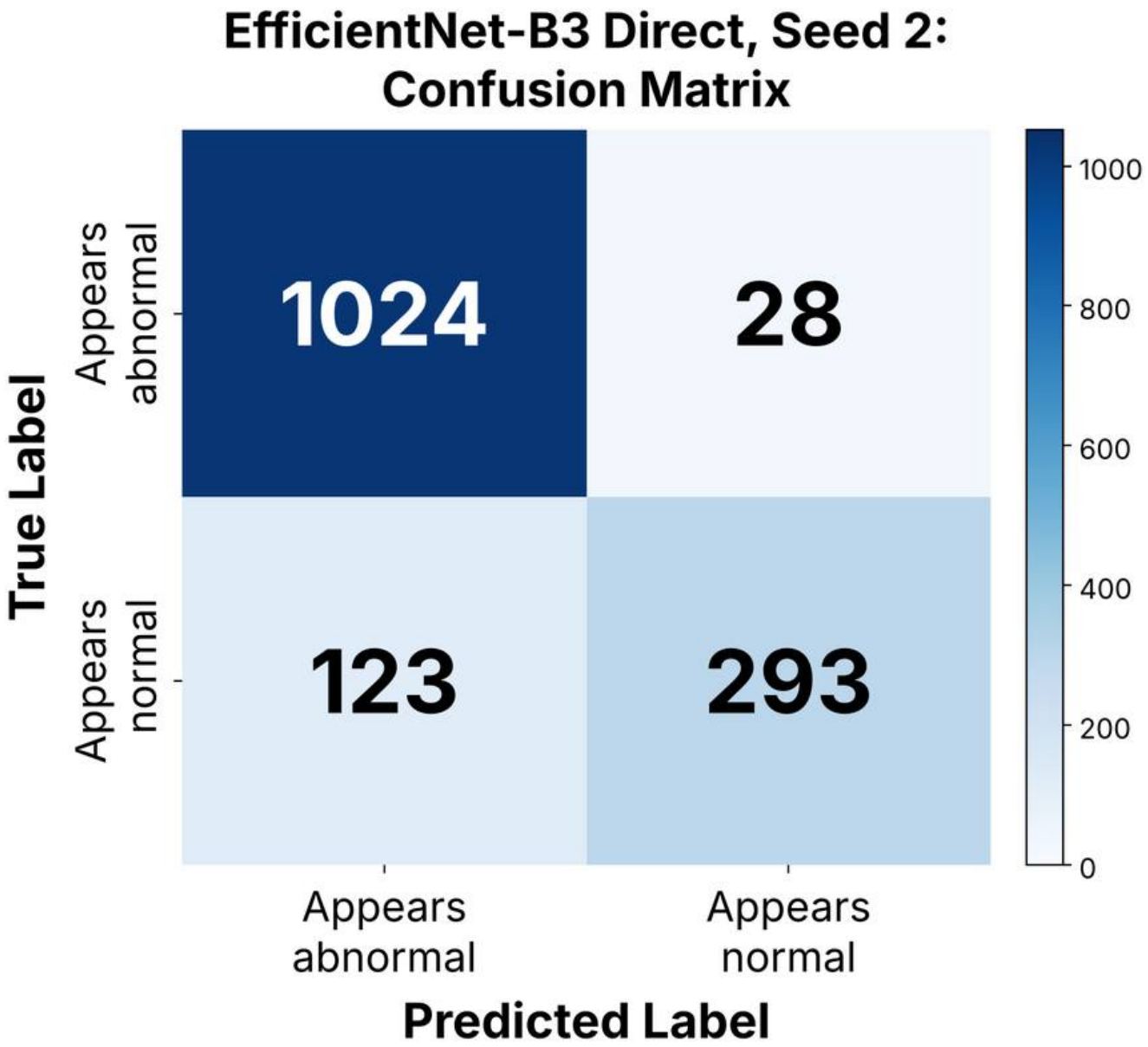


11. EfficientNet-B3 — Direct (fine-tuned CNN) — Seed 2 (seed 3 of 3)

Source: 3_Seed_exp.ipynb, cell 8. Re-run independently in *(seeded) EfficientNet B3 with Adaboost.ipynb, cell 4, seed 512* (same numbers for that seed).

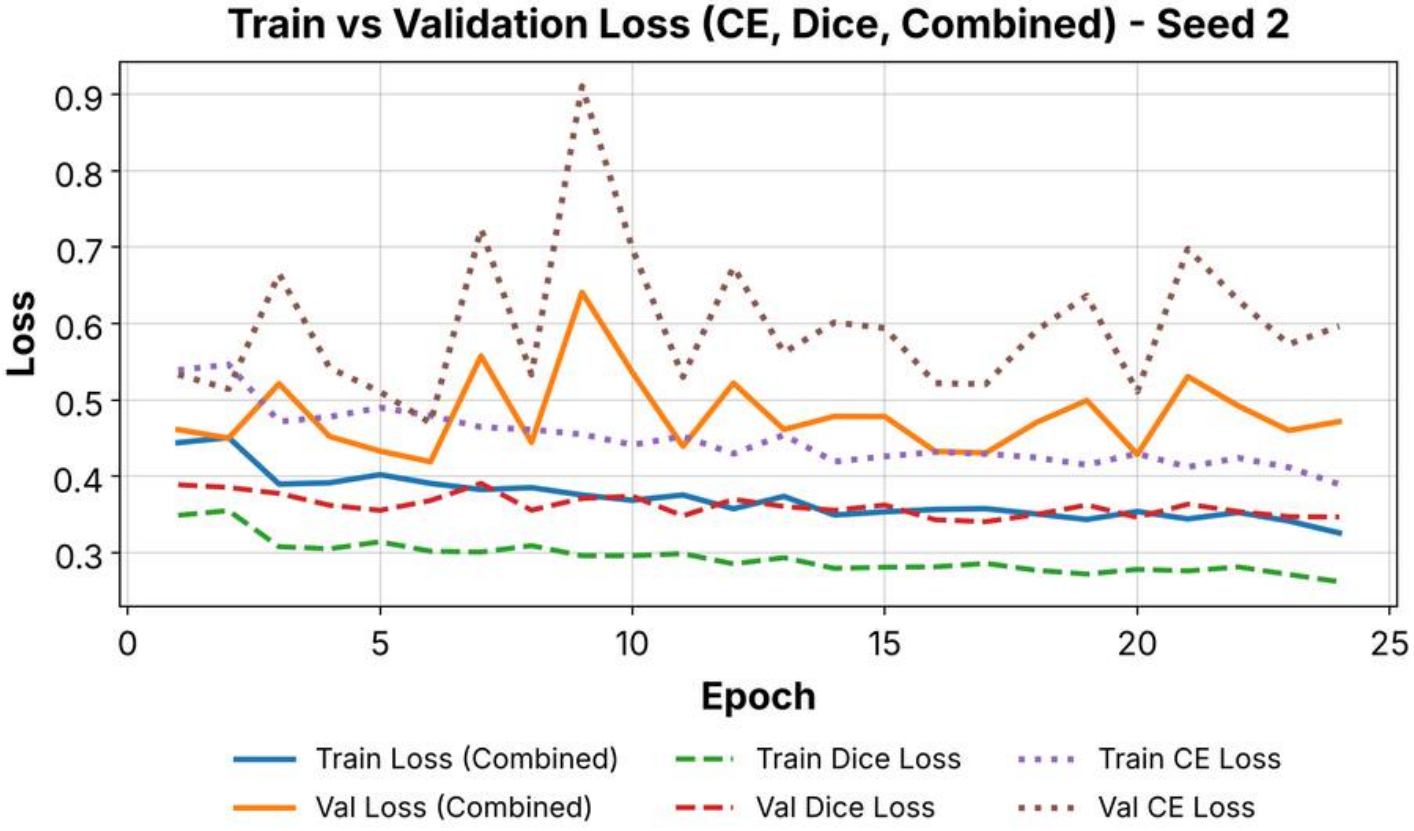
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	90.05%	87.60%	**89.71%**	89.12 ± 1.33 %
Precision (weighted)	90.10%	87.90%	**89.84%**	89.28 ± 1.21 %
Recall (weighted)	90.05%	87.60%	**89.71%**	89.12 ± 1.33 %
F1-score (weighted)	89.69%	86.85%	**89.27%**	88.60 ± 1.53 %
Test Dice loss	0.5196	0.5218	0.5247	0.5220 ± 0.0025
Best validation acc. (CNN)	76.46%	75.62%	72.50%	74.86 ± 2.09 %

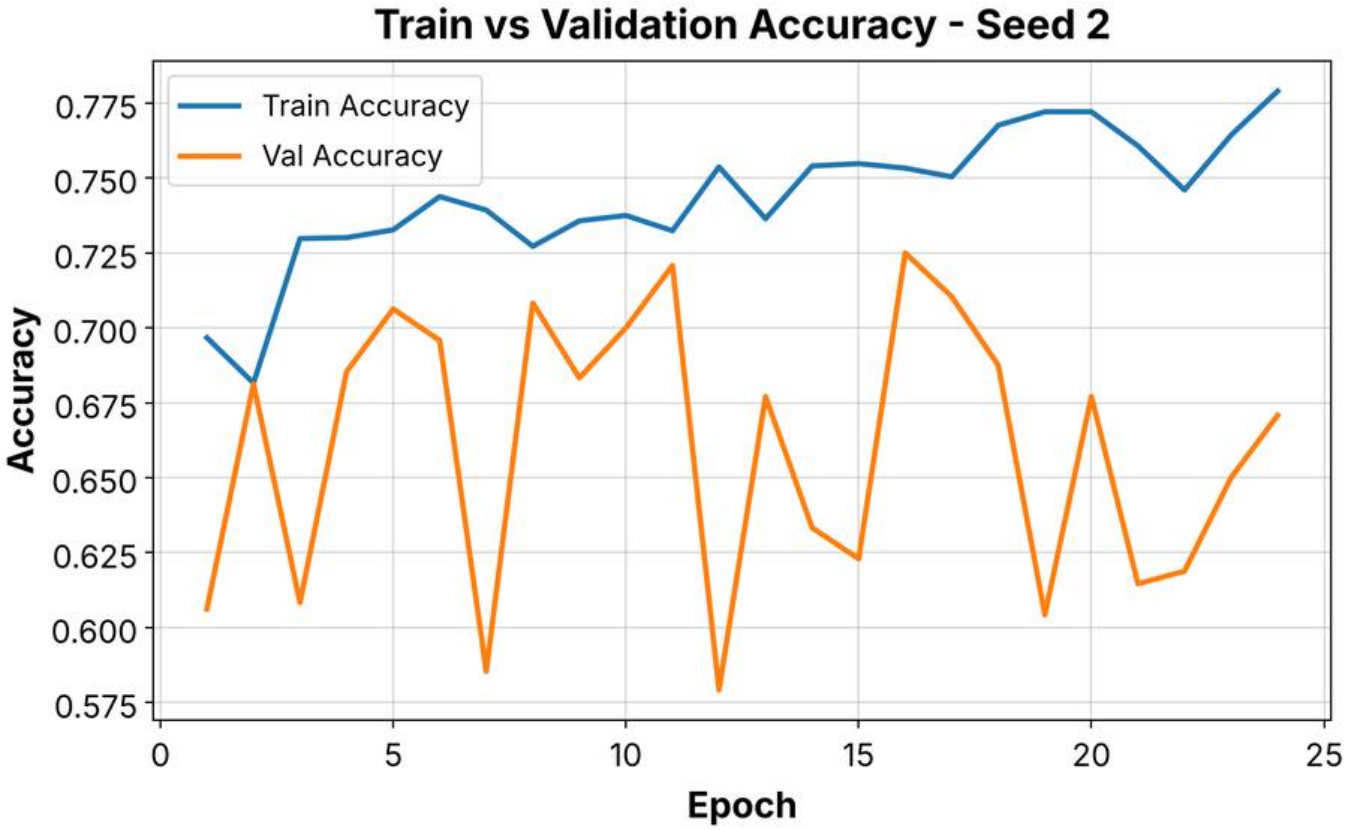
Confusion Matrix



11. EfficientNet-B3 — Direct (fine-tuned CNN), Seed 2 — graphs

Training / Validation Graphs



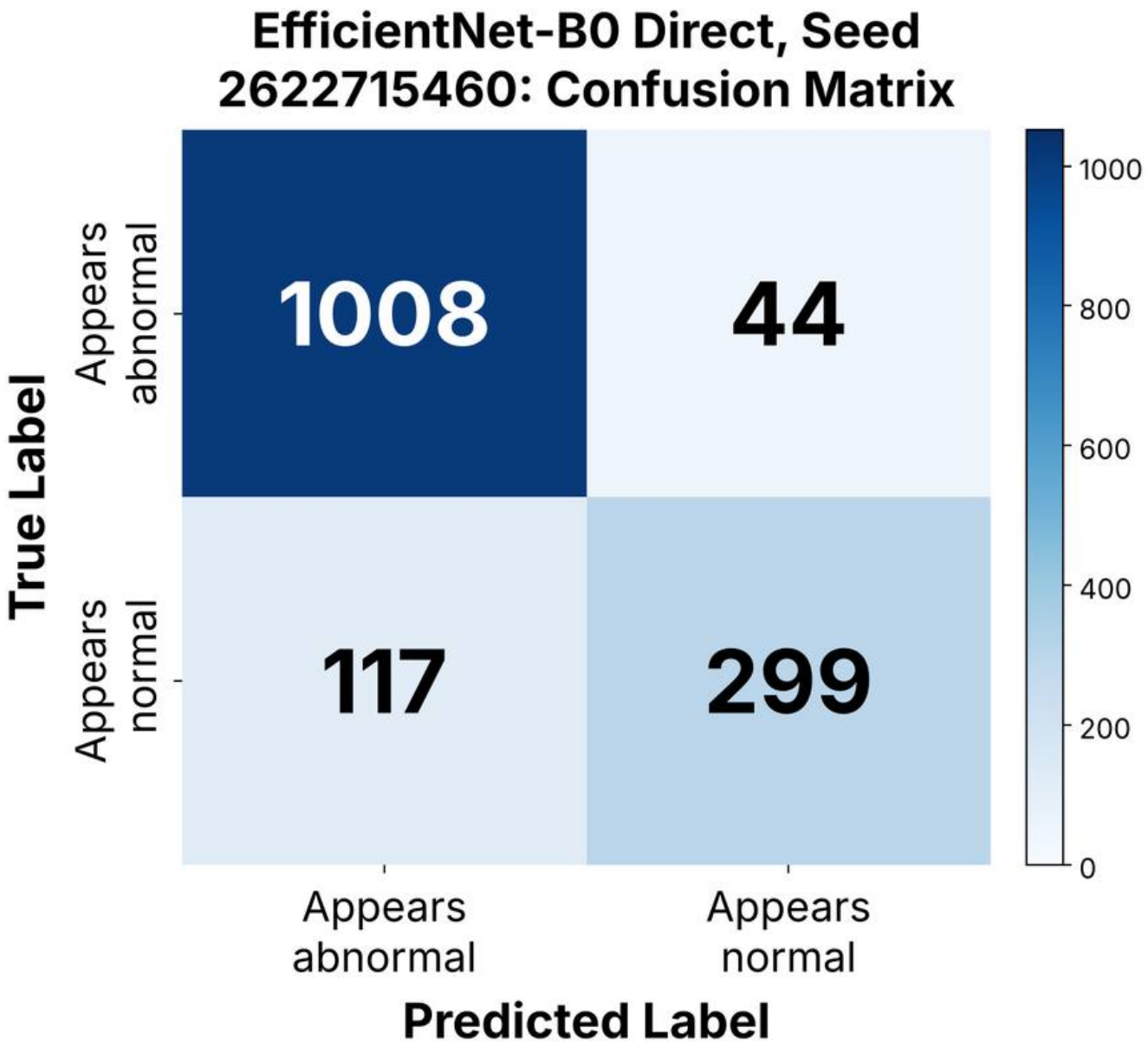


12. EfficientNet-B0 — Direct (fine-tuned CNN) — Seed 2622715460 (seed 1 of 3)

Source: 3_Seed_exp.ipynb, cell 12. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 8, seed 2622715460* (same numbers for that seed).

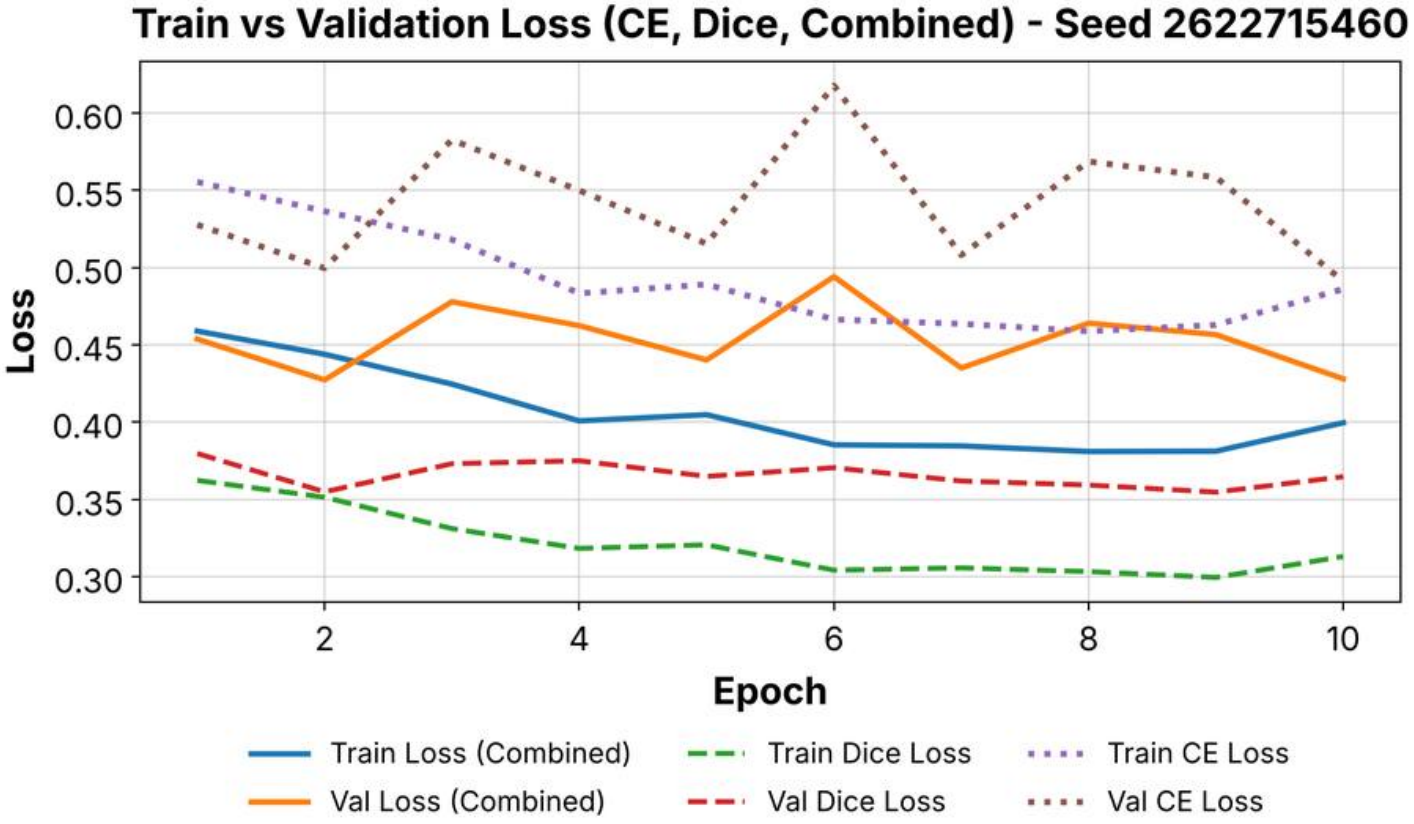
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**89.03%**	89.58%	88.42%	89.01 ± 0.58 %
Precision (weighted)	**88.91%**	89.52%	88.68%	89.04 ± 0.44 %
Recall (weighted)	**89.03%**	89.58%	88.42%	89.01 ± 0.58 %
F1-score (weighted)	**88.69%**	89.23%	87.78%	88.57 ± 0.73 %
Test Dice loss	0.5193	0.5218	0.5244	0.5218 ± 0.0026
Best validation acc. (CNN)	70.42%	75.62%	75.62%	73.89 ± 3.01 %

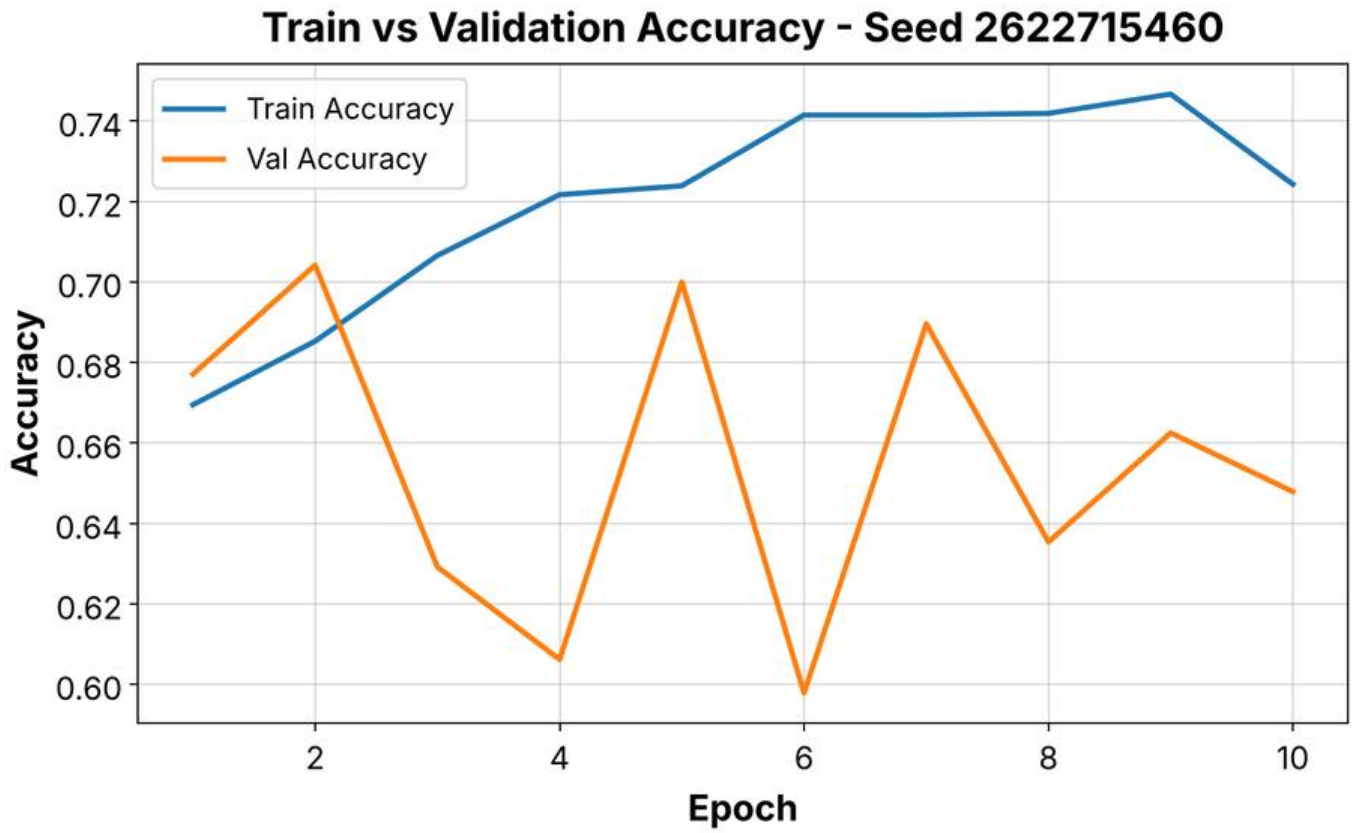
Confusion Matrix



12. EfficientNet-B0 — Direct (fine-tuned CNN), Seed 2622715460 — graphs

Training / Validation Graphs



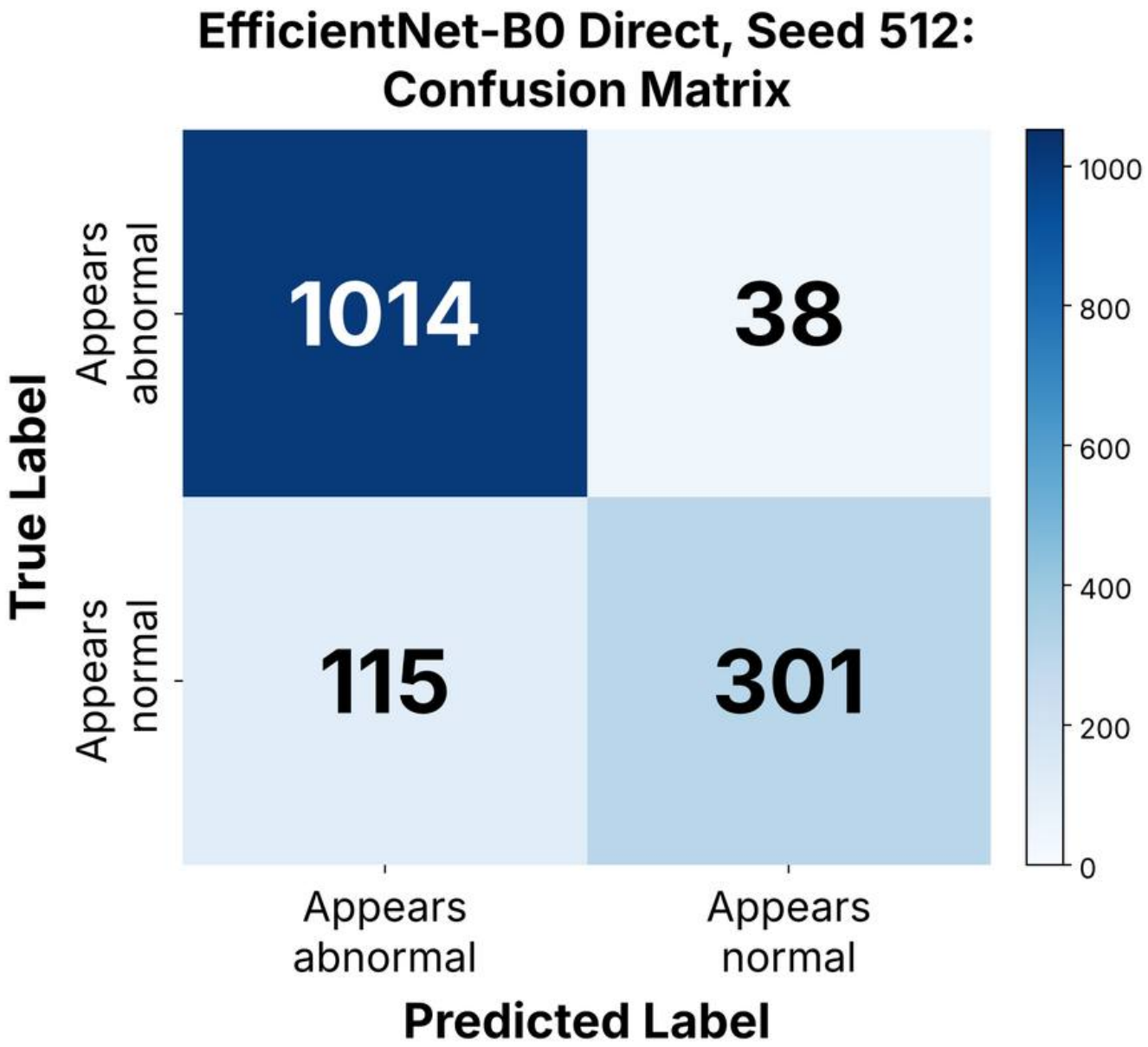


12. EfficientNet-B0 — Direct (fine-tuned CNN) — Seed 512 (seed 2 of 3)

Source: 3_Seed_exp.ipynb, cell 12. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 8, seed 2622715460* (same numbers for that seed).

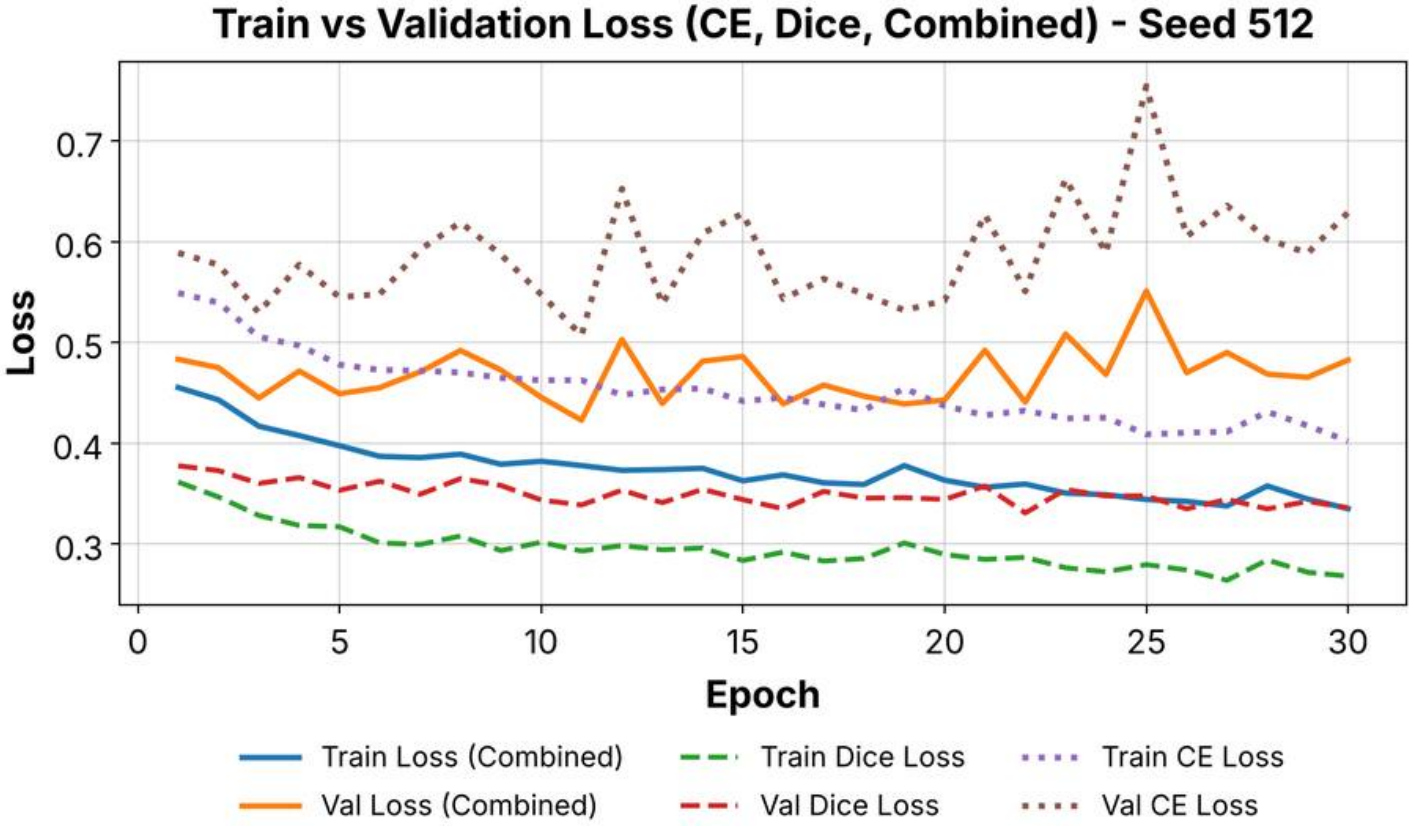
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	89.03%	**89.58%**	88.42%	89.01 ± 0.58 %
Precision (weighted)	88.91%	**89.52%**	88.68%	89.04 ± 0.44 %
Recall (weighted)	89.03%	**89.58%**	88.42%	89.01 ± 0.58 %
F1-score (weighted)	88.69%	**89.23%**	87.78%	88.57 ± 0.73 %
Test Dice loss	0.5193	0.5218	0.5244	0.5218 ± 0.0026
Best validation acc. (CNN)	70.42%	75.62%	75.62%	73.89 ± 3.01 %

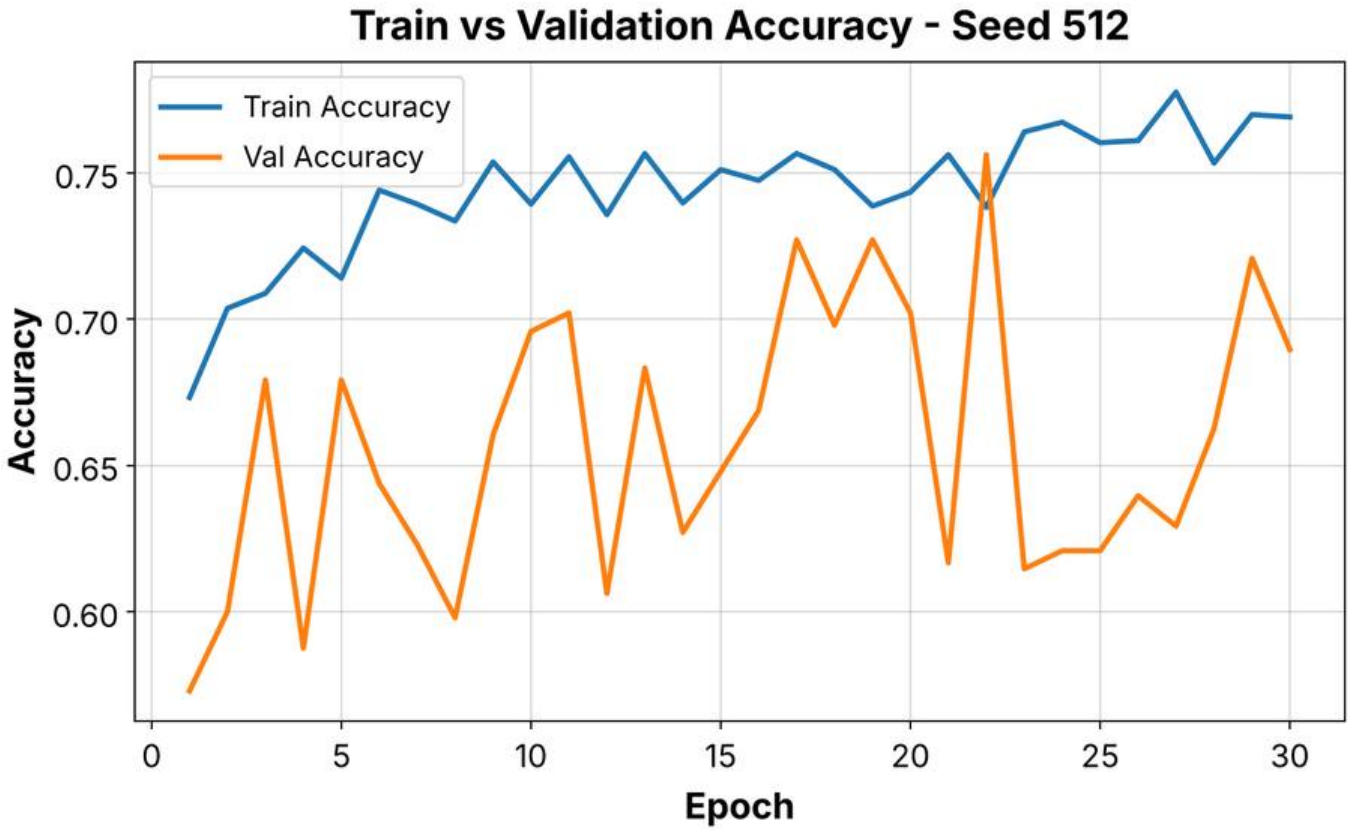
Confusion Matrix



12. EfficientNet-B0 — Direct (fine-tuned CNN), Seed 512 — graphs

Training / Validation Graphs



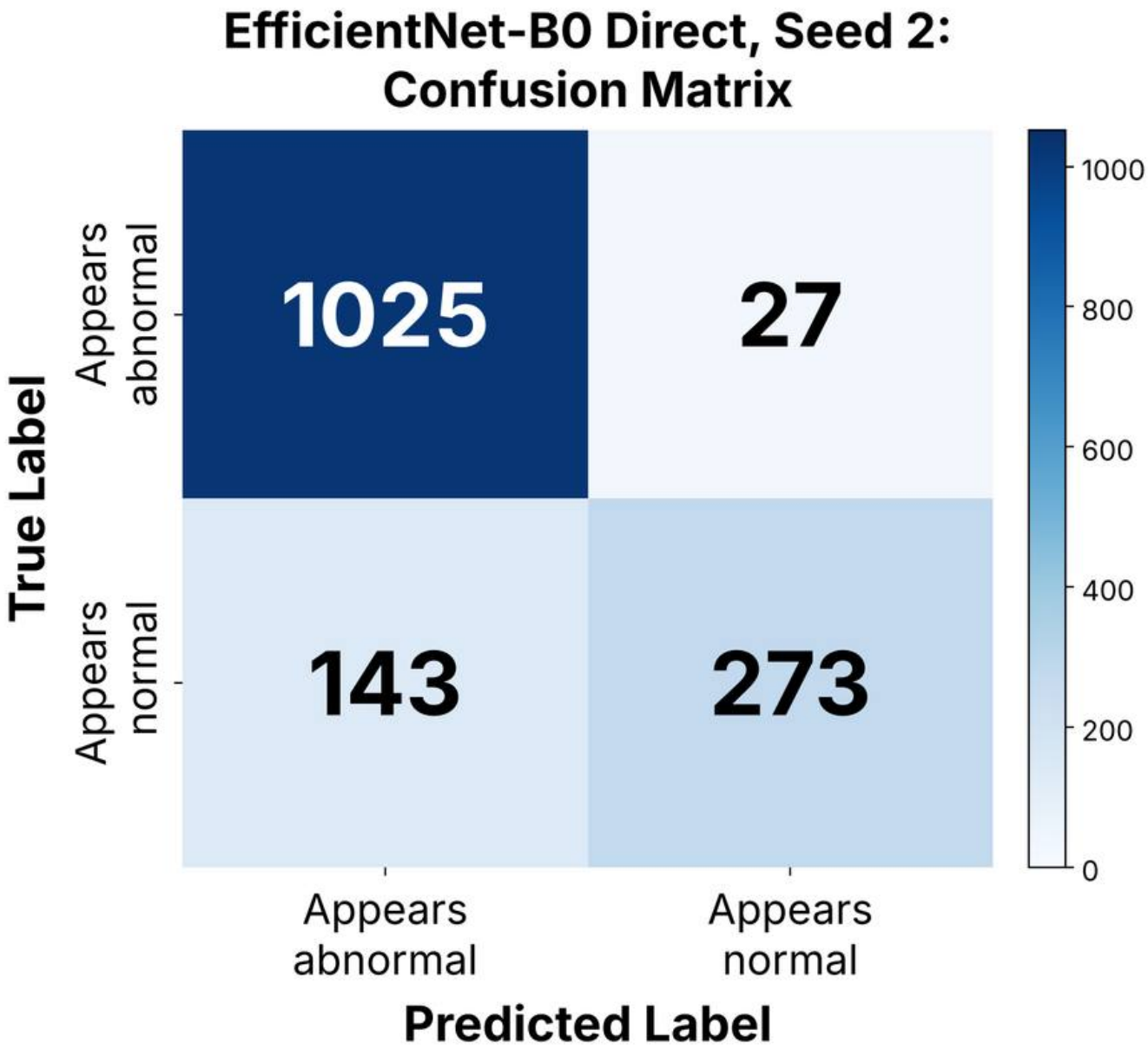


12. EfficientNet-B0 — Direct (fine-tuned CNN) — Seed 2 (seed 3 of 3)

Source: 3_Seed_exp.ipynb, cell 12. Re-run independently in *(seeded) MobileNet and EfficientNet B0-Adaboost.ipynb, cell 8, seed 2622715460* (same numbers for that seed).

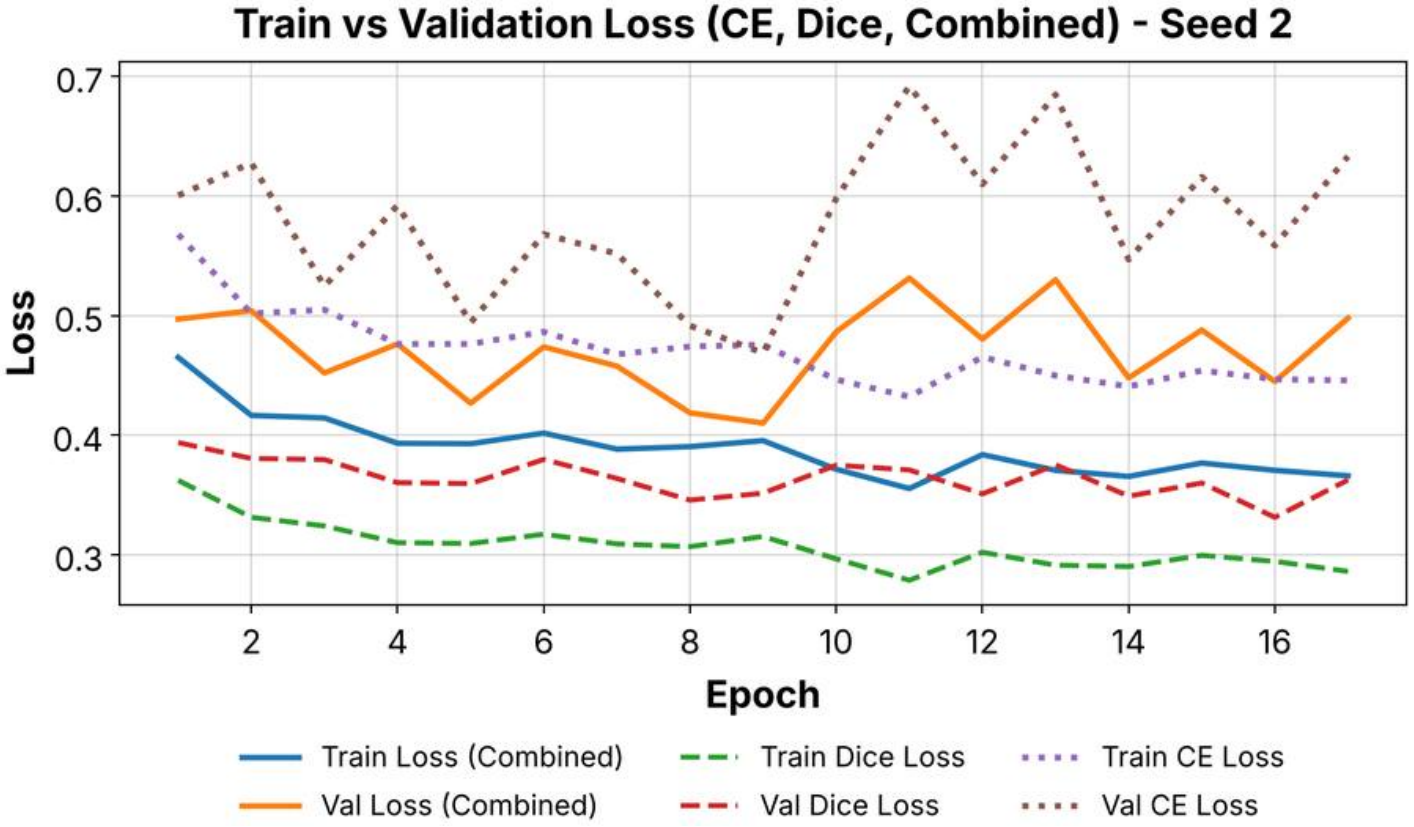
Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	89.03%	89.58%	**88.42%**	89.01 ± 0.58 %
Precision (weighted)	88.91%	89.52%	**88.68%**	89.04 ± 0.44 %
Recall (weighted)	89.03%	89.58%	**88.42%**	89.01 ± 0.58 %
F1-score (weighted)	88.69%	89.23%	**87.78%**	88.57 ± 0.73 %
Test Dice loss	0.5193	0.5218	0.5244	0.5218 ± 0.0026
Best validation acc. (CNN)	70.42%	75.62%	75.62%	73.89 ± 3.01 %

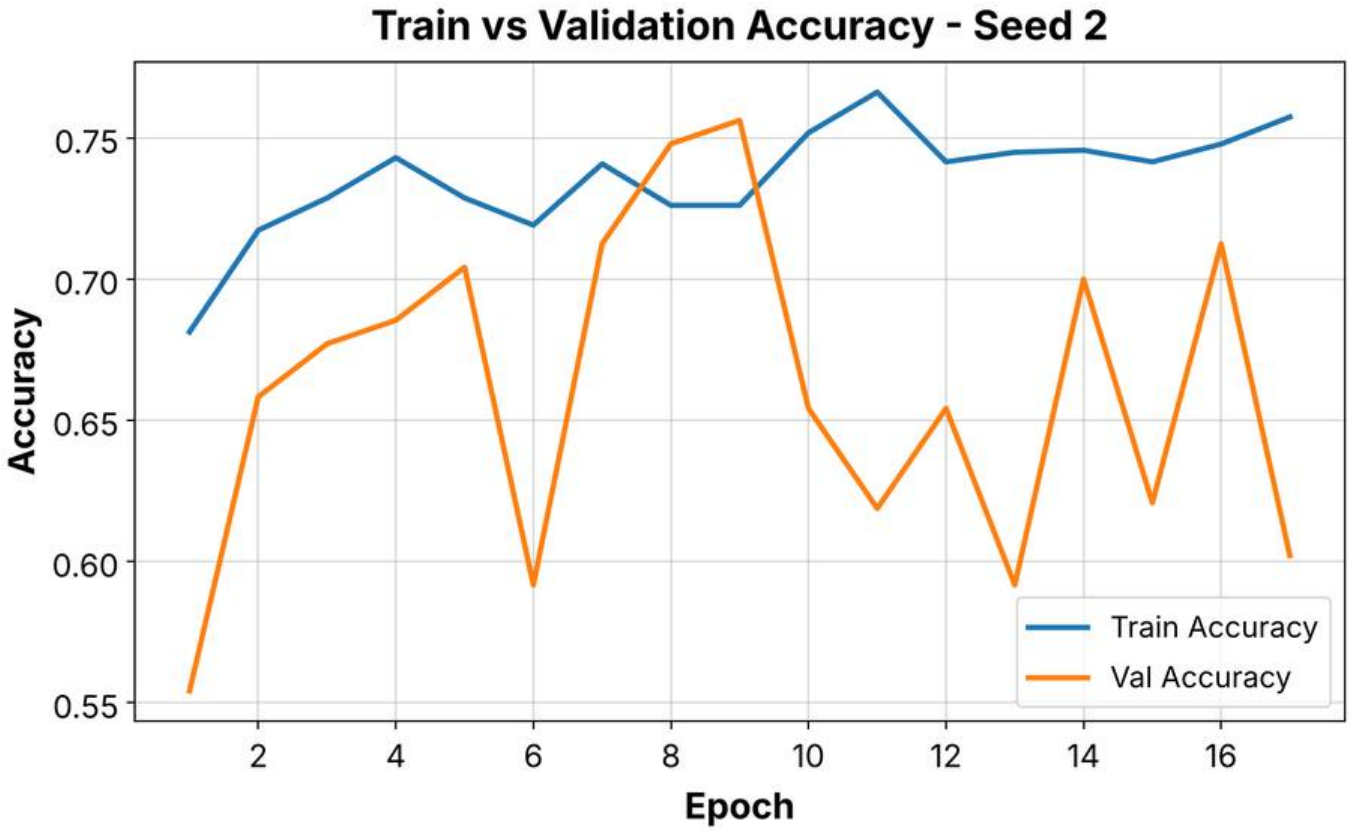
Confusion Matrix



12. EfficientNet-B0 — Direct (fine-tuned CNN), Seed 2 — graphs

Training / Validation Graphs





Section 3 — Seeded ensembles

Ensemble.ipynb trains MobileNetV2, EfficientNet-B0 and EfficientNet-B3 for each of the three seeds, then averages the members' AdaBoost test probabilities image by image (soft voting) and predicts the class with the highest averaged probability. Same seeds, same train / validation / test split, same test set as Section 2.

Rank	Model	Classifier	Test Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Source notebook (.ipynb)
13	MobileNetV2 + EfficientNet-B0	AdaBoost ensemble (soft vote)	92.44 ± 1.03 **2622715460: 93.53** 512: 92.30 2: 91.49	92.38 ± 1.06 **2622715460: 93.49** 512: 92.26 2: 91.38	92.44 ± 1.03 **2622715460: 93.53** 512: 92.30 2: 91.49	92.33 ± 1.03 **2622715460: 93.43** 512: 92.15 2: 91.40	Ensemble.ipynb (cell 11)
14	MobileNetV2 + EfficientNet-B0 + Efficient Net-B3	AdaBoost ensemble (soft vote)	92.37 ± 1.16 **2622715460: 93.46** 512: 92.51 2: 91.14	92.31 ± 1.19 **2622715460: 93.41** 512: 92.47 2: 91.05	92.37 ± 1.16 **2622715460: 93.46** 512: 92.51 2: 91.14	92.23 ± 1.21 **2622715460: 93.37** 512: 92.35 2: 90.96	Ensemble.ipynb (cell 11)
15	MobileNetV2 + EfficientNet-B3	AdaBoost ensemble (soft vote)	92.12 ± 0.73 **2622715460: 92.57** 512: 92.51 2: 91.28	92.07 ± 0.70 **2622715460: 92.50** 512: 92.45 2: 91.26	92.12 ± 0.73 **2622715460: 92.57** 512: 92.51 2: 91.28	91.97 ± 0.80 **2622715460: 92.50** 512: 92.37 2: 91.05	Ensemble.ipynb (cell 11)
16	EfficientNet-B0 + EfficientNet-B3	AdaBoost ensemble (soft vote)	91.76 ± 0.62 2622715460: 91.89 **512: 92.30** 2: 91.08	91.67 ± 0.66 2622715460: 91.80 **512: 92.27** 2: 90.96	91.76 ± 0.62 2622715460: 91.89 **512: 92.30** 2: 91.08	91.62 ± 0.62 2622715460: 91.78 **512: 92.14** 2: 90.94	Ensemble.ipynb (cell 11)

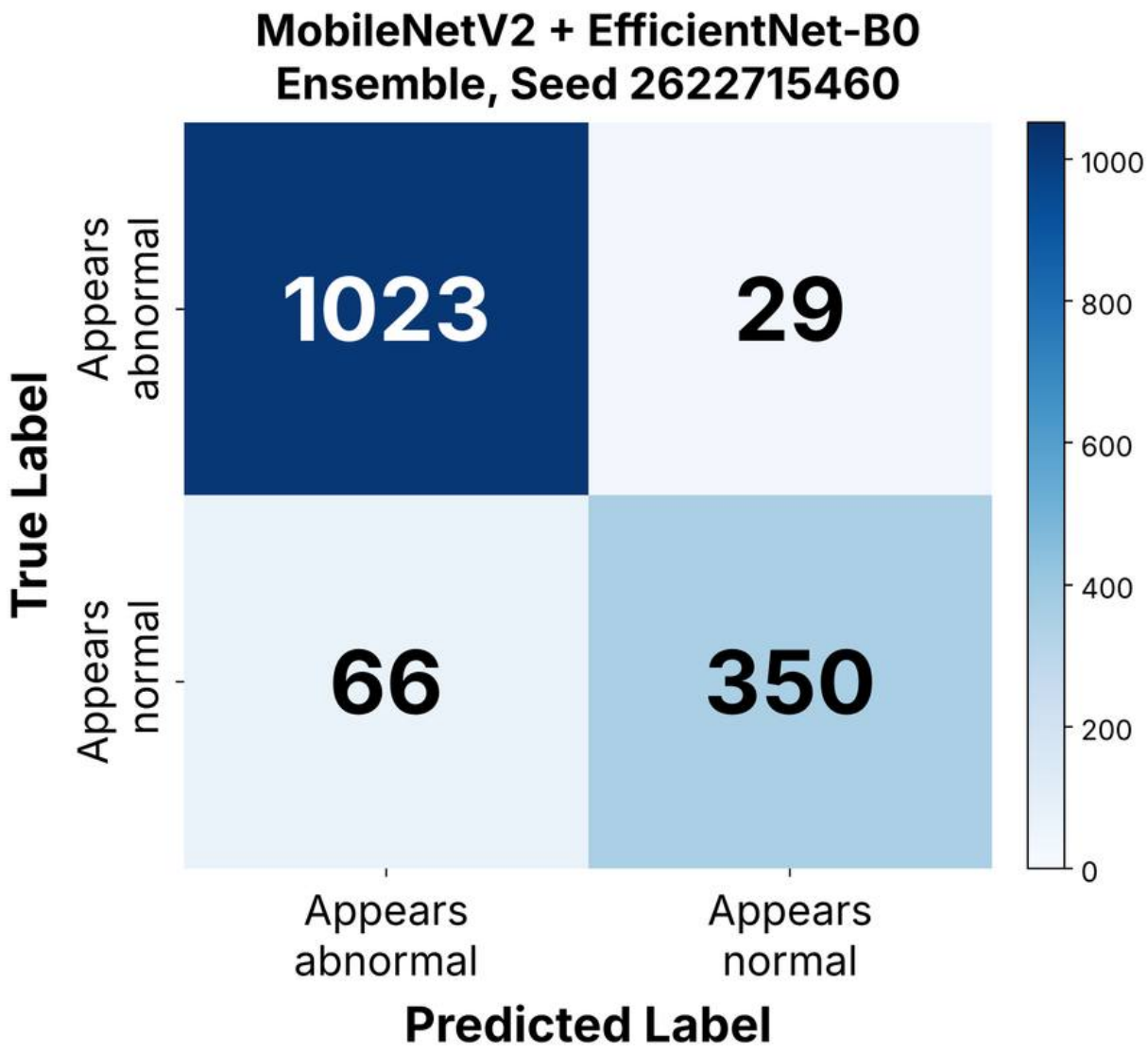
13. Ensemble — MobileNetV2 + EfficientNet-B0 (AdaBoost, soft voting) — Seed 2622715460 (seed 1 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**93.53%**	92.30%	91.49%	92.44 ± 1.03 %
Precision (weighted)	**93.49%**	92.26%	91.38%	92.38 ± 1.06 %
Recall (weighted)	**93.53%**	92.30%	91.49%	92.44 ± 1.03 %
F1-score (weighted)	**93.43%**	92.15%	91.40%	92.33 ± 1.03 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 7, EfficientNet-B0 = row 9.

Confusion Matrix



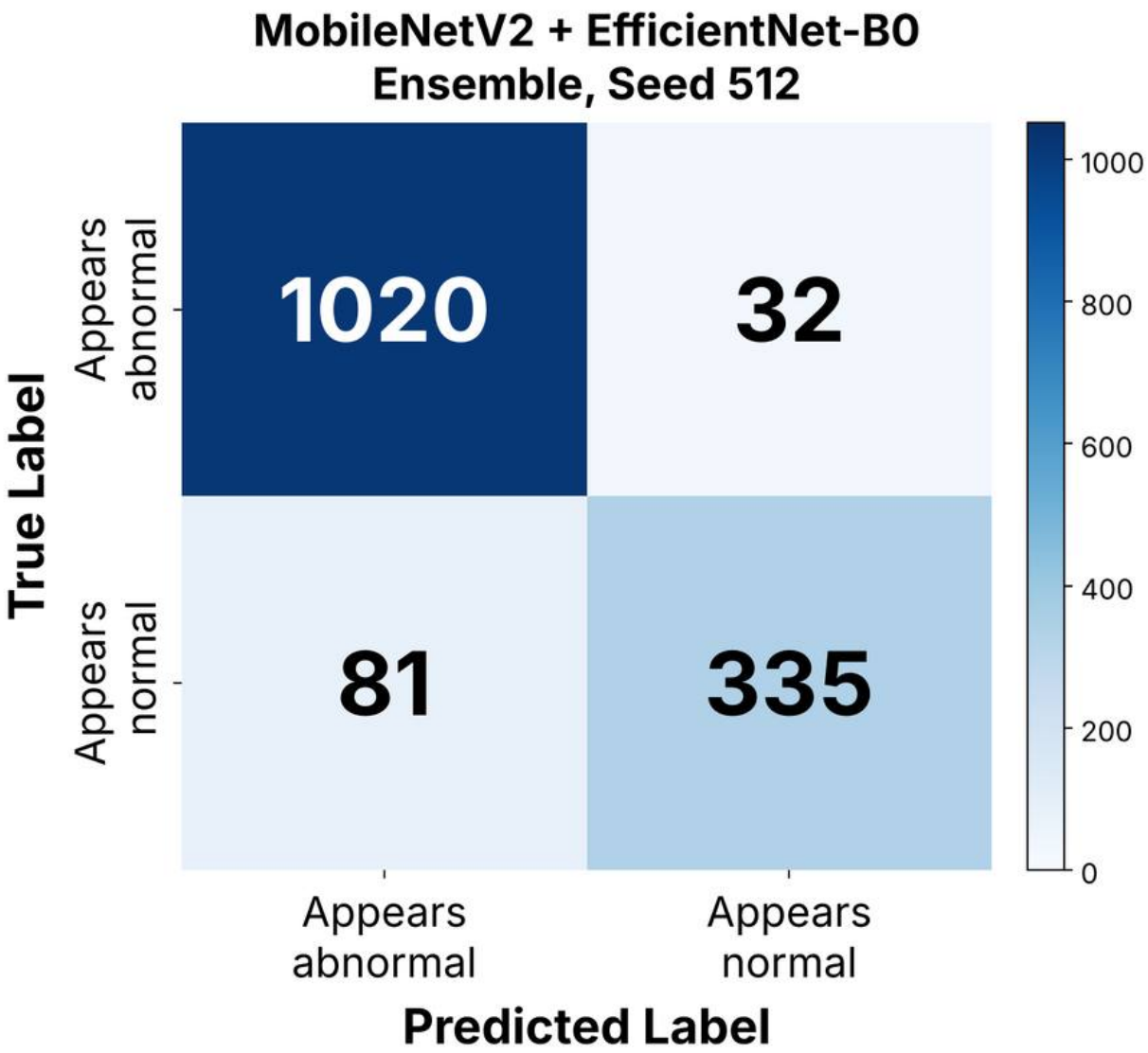
13. Ensemble — MobileNetV2 + EfficientNet-B0 (AdaBoost, soft voting) — Seed 512 (seed 2 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	93.53%	**92.30%**	91.49%	92.44 ± 1.03 %
Precision (weighted)	93.49%	**92.26%**	91.38%	92.38 ± 1.06 %
Recall (weighted)	93.53%	**92.30%**	91.49%	92.44 ± 1.03 %
F1-score (weighted)	93.43%	**92.15%**	91.40%	92.33 ± 1.03 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 7, EfficientNet-B0 = row 9.

Confusion Matrix



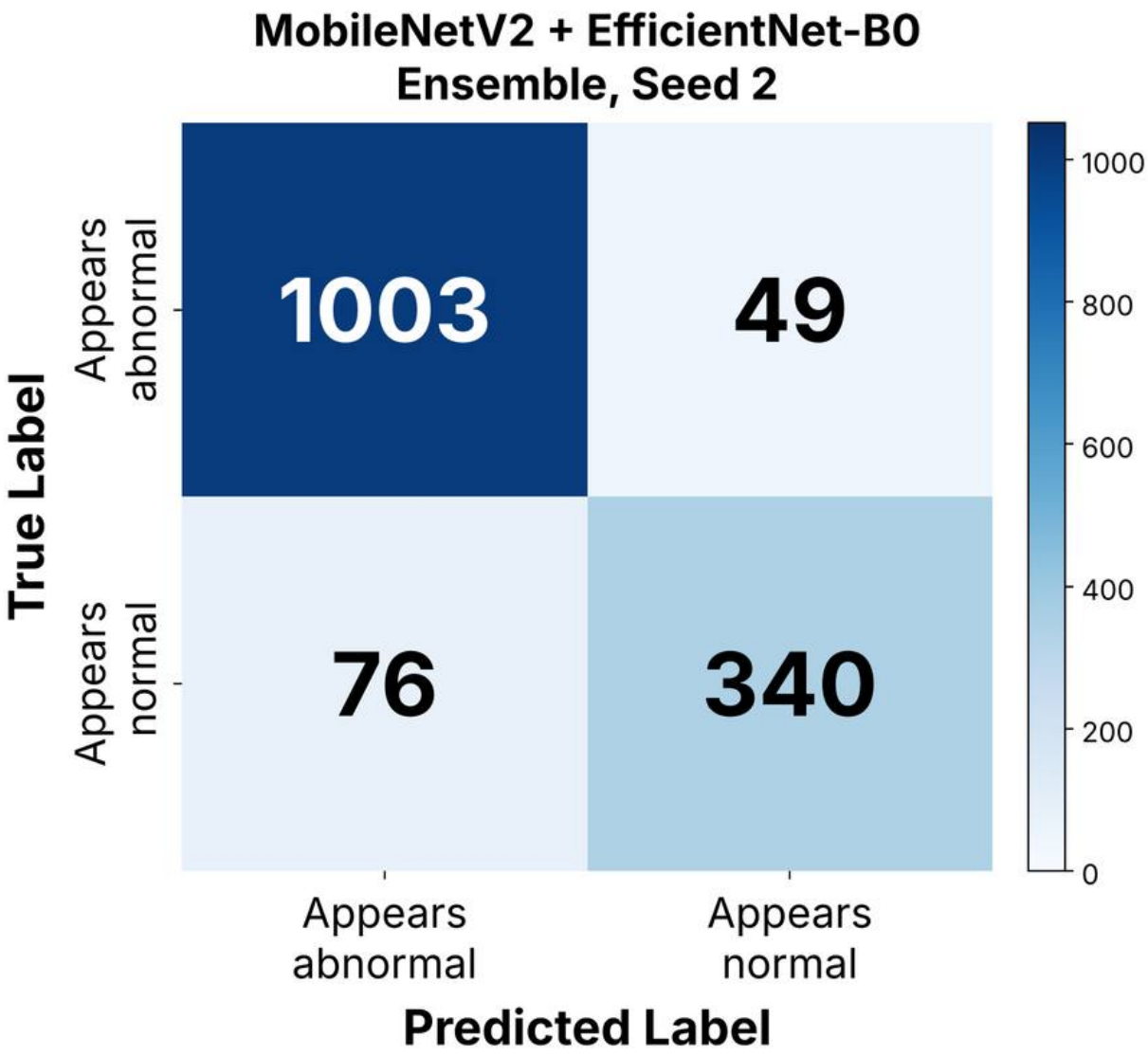
13. Ensemble — MobileNetV2 + EfficientNet-B0 (AdaBoost, soft voting) — Seed 2 (seed 3 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	93.53%	92.30%	**91.49%**	92.44 ± 1.03 %
Precision (weighted)	93.49%	92.26%	**91.38%**	92.38 ± 1.06 %
Recall (weighted)	93.53%	92.30%	**91.49%**	92.44 ± 1.03 %
F1-score (weighted)	93.43%	92.15%	**91.40%**	92.33 ± 1.03 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 7, EfficientNet-B0 = row 9.

Confusion Matrix



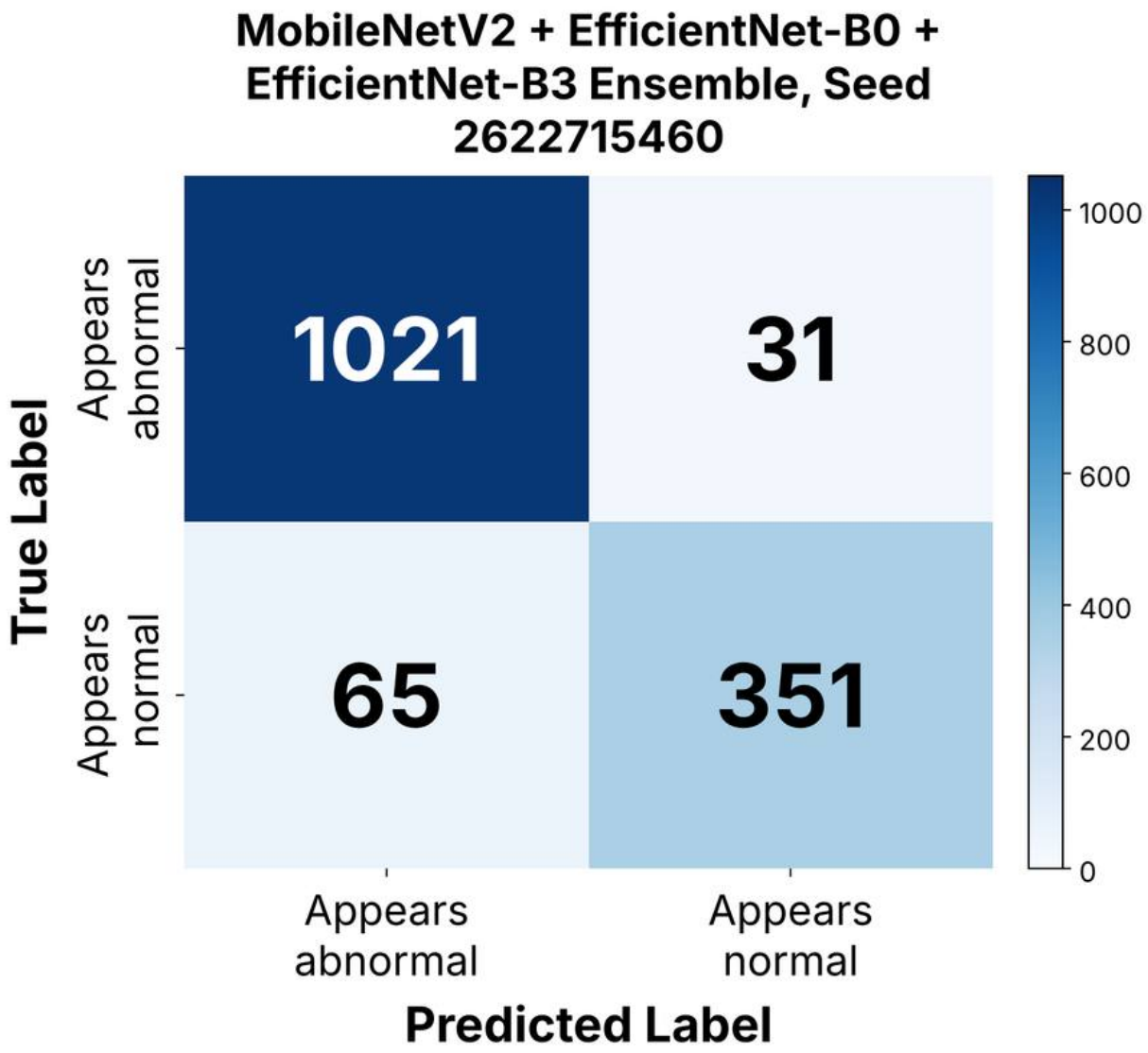
14. Ensemble — MobileNetV2 + EfficientNet-B0 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 2622715460 (seed 1 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**93.46%**	92.51%	91.14%	92.37 ± 1.16 %
Precision (weighted)	**93.41%**	92.47%	91.05%	92.31 ± 1.19 %
Recall (weighted)	**93.46%**	92.51%	91.14%	92.37 ± 1.16 %
F1-score (weighted)	**93.37%**	92.35%	90.96%	92.23 ± 1.21 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 7, EfficientNet-B0 = row 9, EfficientNet-B3 = row 8.

Confusion Matrix



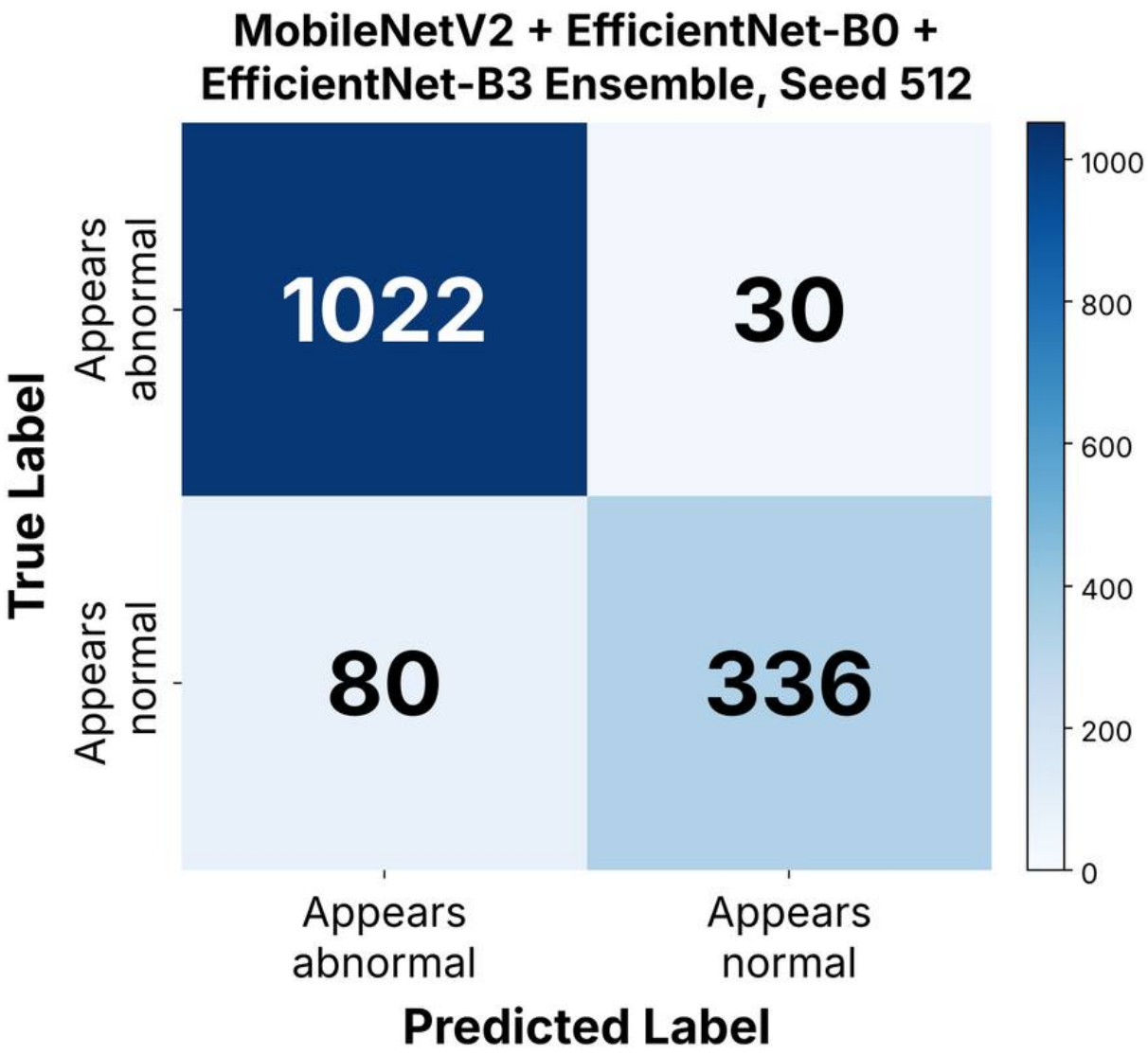
14. Ensemble — MobileNetV2 + EfficientNet-B0 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 512 (seed 2 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	93.46%	**92.51%**	91.14%	92.37 ± 1.16 %
Precision (weighted)	93.41%	**92.47%**	91.05%	92.31 ± 1.19 %
Recall (weighted)	93.46%	**92.51%**	91.14%	92.37 ± 1.16 %
F1-score (weighted)	93.37%	**92.35%**	90.96%	92.23 ± 1.21 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 7, EfficientNet-B0 = row 9, EfficientNet-B3 = row 8.

Confusion Matrix



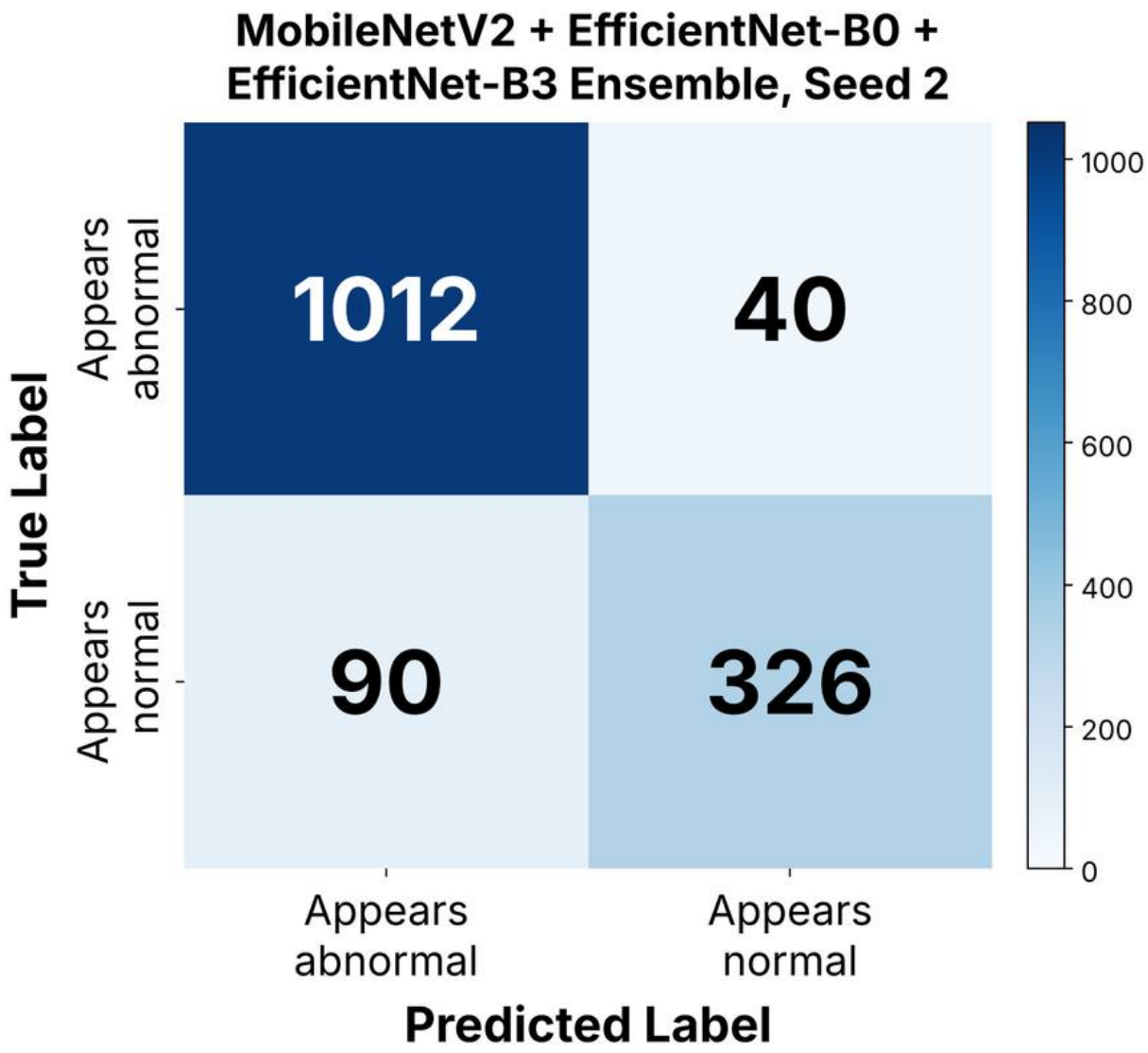
14. Ensemble — MobileNetV2 + EfficientNet-B0 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 2 (seed 3 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	93.46%	92.51%	**91.14%**	92.37 ± 1.16 %
Precision (weighted)	93.41%	92.47%	**91.05%**	92.31 ± 1.19 %
Recall (weighted)	93.46%	92.51%	**91.14%**	92.37 ± 1.16 %
F1-score (weighted)	93.37%	92.35%	**90.96%**	92.23 ± 1.21 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 7, EfficientNet-B0 = row 9, EfficientNet-B3 = row 8.

Confusion Matrix



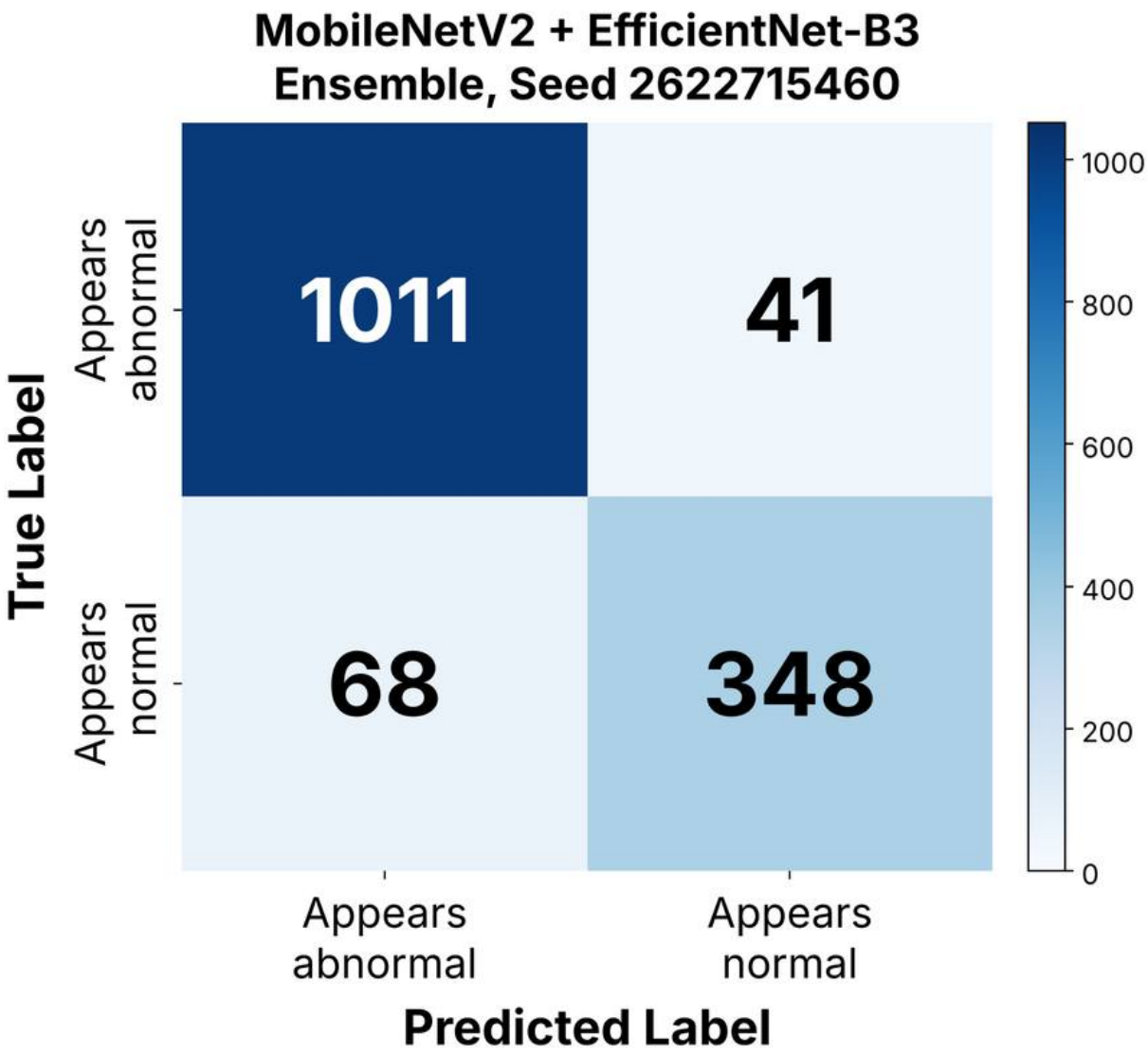
15. Ensemble — MobileNetV2 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 2622715460 (seed 1 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**92.57%**	92.51%	91.28%	92.12 ± 0.73 %
Precision (weighted)	**92.50%**	92.45%	91.26%	92.07 ± 0.70 %
Recall (weighted)	**92.57%**	92.51%	91.28%	92.12 ± 0.73 %
F1-score (weighted)	**92.50%**	92.37%	91.05%	91.97 ± 0.80 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 7, EfficientNet-B3 = row 8.

Confusion Matrix



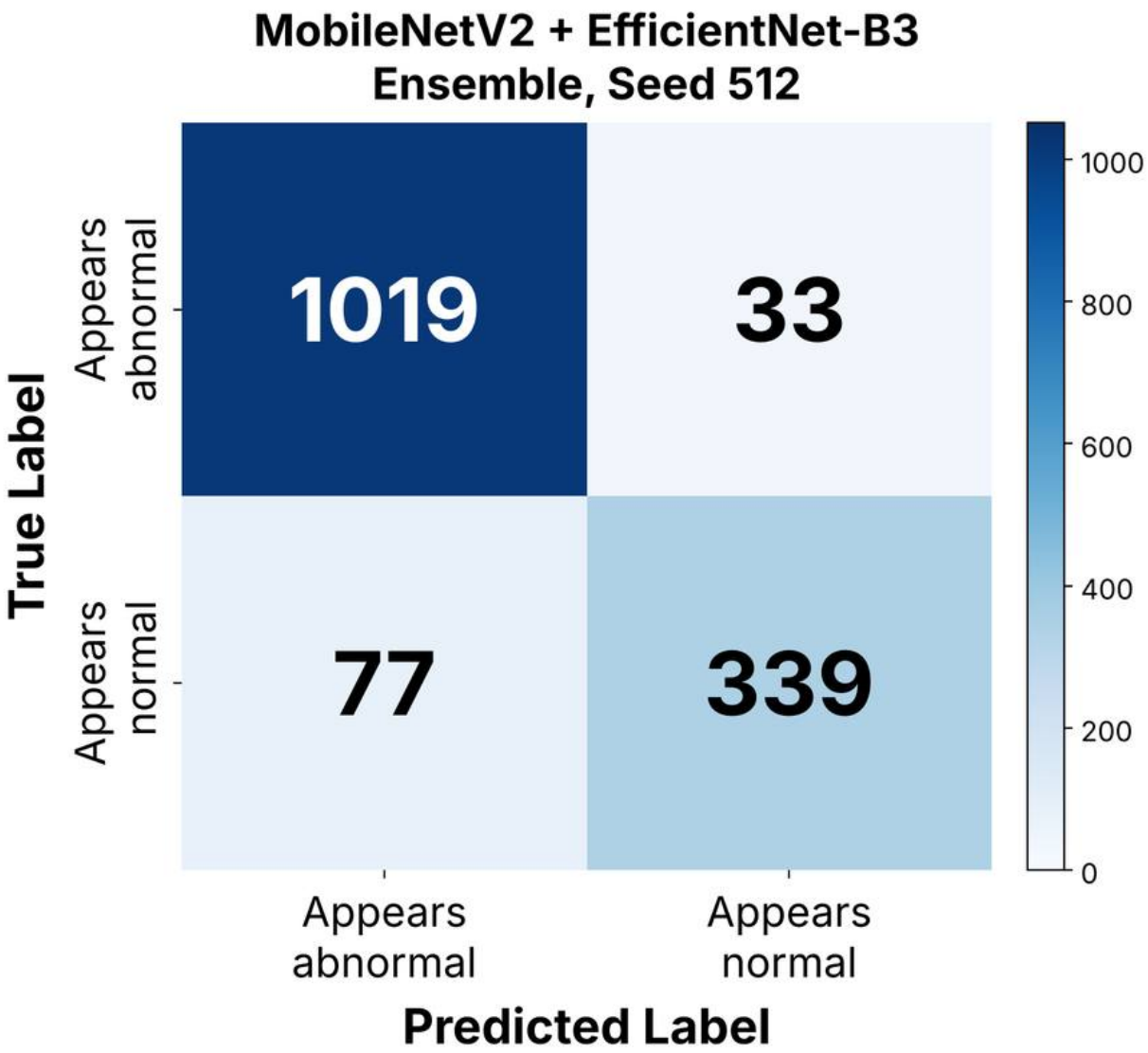
15. Ensemble — MobileNetV2 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 512 (seed 2 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	92.57%	**92.51%**	91.28%	92.12 ± 0.73 %
Precision (weighted)	92.50%	**92.45%**	91.26%	92.07 ± 0.70 %
Recall (weighted)	92.57%	**92.51%**	91.28%	92.12 ± 0.73 %
F1-score (weighted)	92.50%	**92.37%**	91.05%	91.97 ± 0.80 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 7, EfficientNet-B3 = row 8.

Confusion Matrix



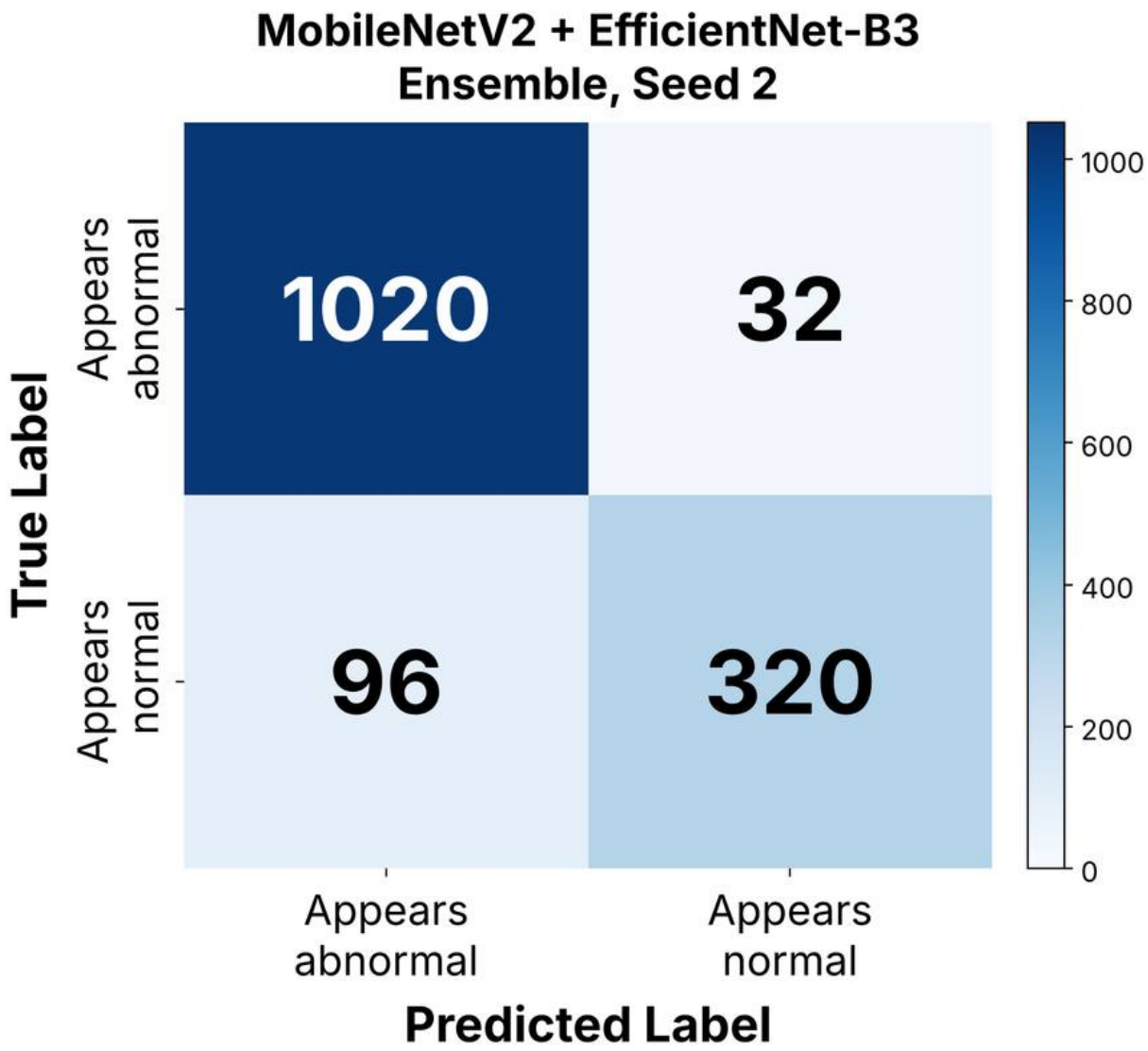
15. Ensemble — MobileNetV2 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 2 (seed 3 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	92.57%	92.51%	**91.28%**	92.12 ± 0.73 %
Precision (weighted)	92.50%	92.45%	**91.26%**	92.07 ± 0.70 %
Recall (weighted)	92.57%	92.51%	**91.28%**	92.12 ± 0.73 %
F1-score (weighted)	92.50%	92.37%	**91.05%**	91.97 ± 0.80 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: MobileNetV2 = row 7, EfficientNet-B3 = row 8.

Confusion Matrix



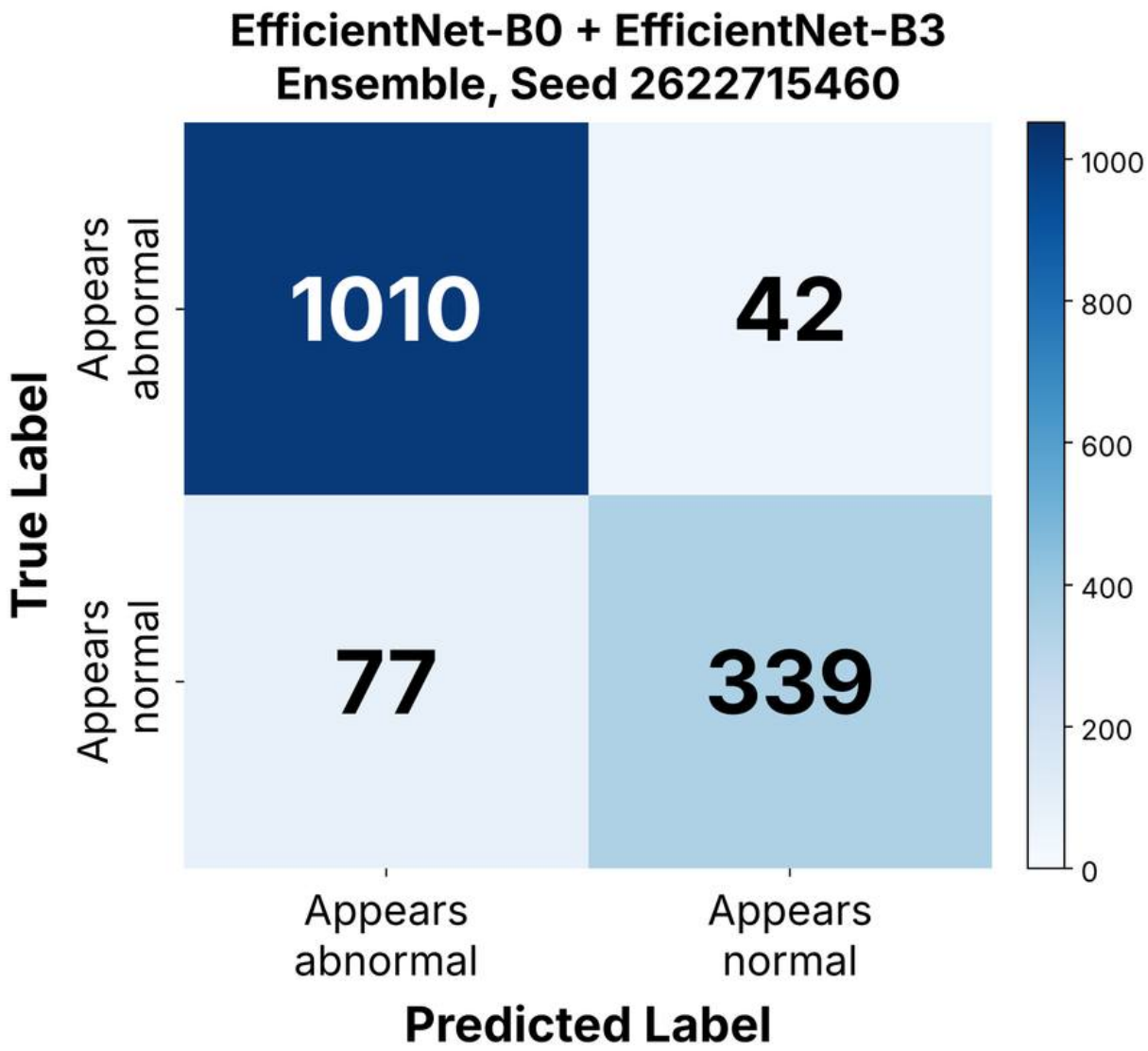
16. Ensemble — EfficientNet-B0 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 2622715460 (seed 1 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	**91.89%**	92.30%	91.08%	91.76 ± 0.62 %
Precision (weighted)	**91.80%**	92.27%	90.96%	91.67 ± 0.66 %
Recall (weighted)	**91.89%**	92.30%	91.08%	91.76 ± 0.62 %
F1-score (weighted)	**91.78%**	92.14%	90.94%	91.62 ± 0.62 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: EfficientNet-B0 = row 9, EfficientNet-B3 = row 8.

Confusion Matrix



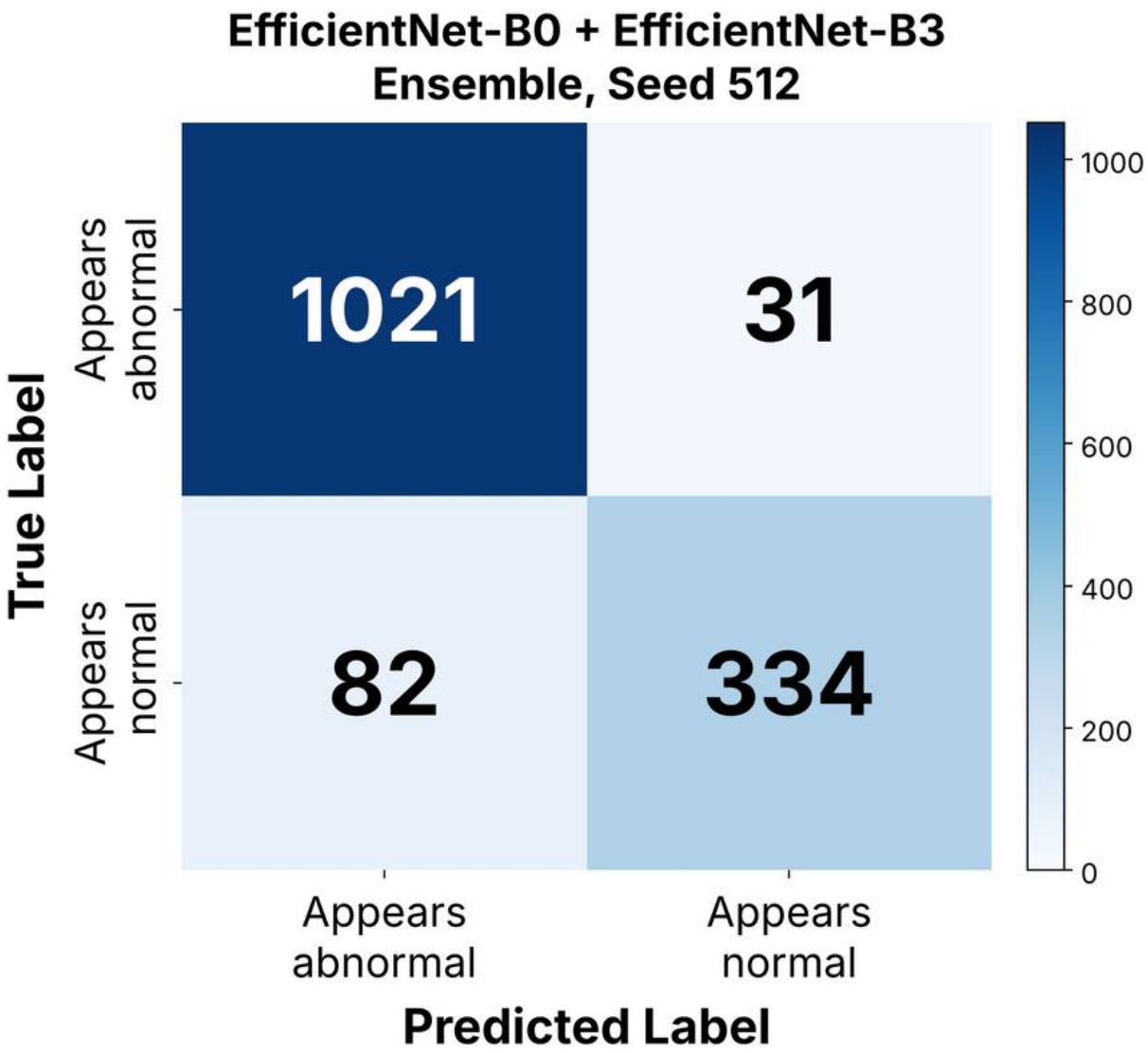
16. Ensemble — EfficientNet-B0 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 512 (seed 2 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	91.89%	**92.30%**	91.08%	91.76 ± 0.62 %
Precision (weighted)	91.80%	**92.27%**	90.96%	91.67 ± 0.66 %
Recall (weighted)	91.89%	**92.30%**	91.08%	91.76 ± 0.62 %
F1-score (weighted)	91.78%	**92.14%**	90.94%	91.62 ± 0.62 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: EfficientNet-B0 = row 9, EfficientNet-B3 = row 8.

Confusion Matrix



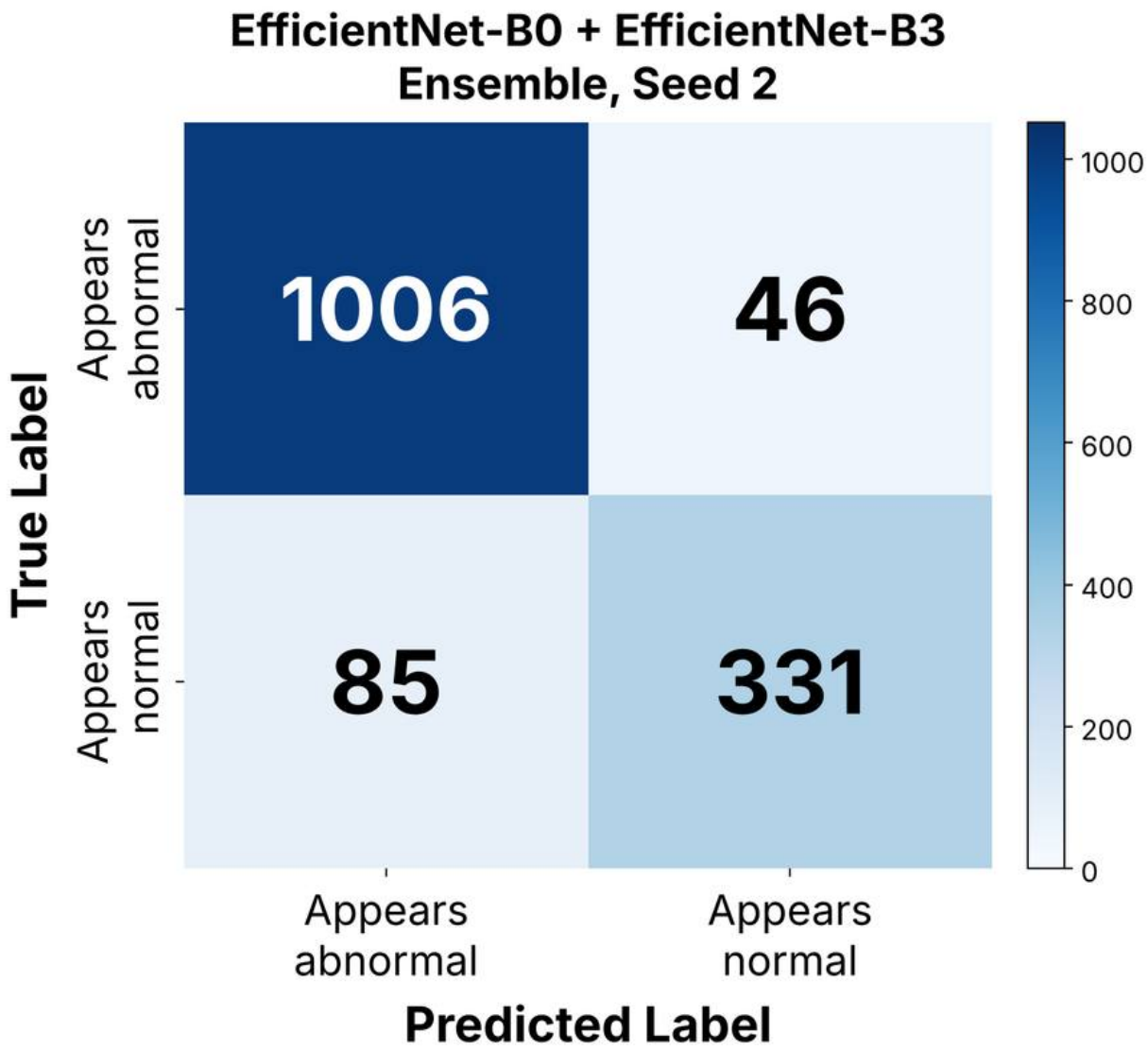
16. Ensemble — EfficientNet-B0 + EfficientNet-B3 (AdaBoost, soft voting) — Seed 2 (seed 3 of 3)

Source: (seeded) Ensemble.ipynb, cell 11 (soft voting: AdaBoost test probabilities of the members are averaged; same seed, split and test set).

Metric (test set)	Seed 2622715460	Seed 512	Seed 2	Mean ± SD (3 seeds)
Test Accuracy	91.89%	92.30%	**91.08%**	91.76 ± 0.62 %
Precision (weighted)	91.80%	92.27%	**90.96%**	91.67 ± 0.66 %
Recall (weighted)	91.89%	92.30%	**91.08%**	91.76 ± 0.62 %
F1-score (weighted)	91.78%	92.14%	**90.94%**	91.62 ± 0.62 %

No training graphs for the ensemble itself (it combines already-trained models). The training graphs of the member models are in the AdaBoost rows of the same seeds: EfficientNet-B0 = row 9, EfficientNet-B3 = row 8.

Confusion Matrix



Notes

- Sources: 3_Seed_exp.ipynb, “(seeded) MobileNet and EfficientNet B0-Adaboost .ipynb”, “(seeded) EfficientNet B3 with Adaboost.ipynb” and the executed “(seeded) Ensemble.ipynb”. Every value was read from the notebooks’ saved outputs; nothing was re-run for this report.
- Seeds: 2622715460, 512 and 2 are global seeds (random, numpy, torch, cudnn deterministic); the train / validation / test split comes from random_split after the seed is set, so each seed has its own split and the three models share it.
- Section 1 results are identical to the matching seed in Section 2 (checked value by value); the single-seed notebooks are the independent re-runs that confirm the 3-seed notebook for those seeds. Where the 3-seed notebook’s saved text log skips epochs (MobileNetV2, seed 2622715460: epochs 8–16), the complete 19-epoch log comes from the single-seed notebook and agrees exactly on every epoch both contain.
- Mean ± SD uses the sample standard deviation (n = 3). Precision, Recall and F1-score are class-weighted averages; weighted recall equals accuracy.
- Ensembles have no training graphs of their own because they only combine already-trained models; the graphs of the members are in the member rows of Section 2.
- Confusion matrices: copied from the notebook output where it prints a matrix; where only an image or per-class report exists, the counts were recovered exactly from per-class recall × support (1,052 / 416) and cross-checked with precision and accuracy. All matrices use the same colour map and the same fixed colour scale (0–1,052 images).
- Graphs: only values printed in the notebooks’ training logs are plotted — nothing is estimated, smoothed or interpolated. Where the notebook printed a complete epoch log the graph is redrawn from it (with Train/Val CE curves added); the notebook’s own original figures are shown wherever they exist. “CE” is torch.nn.CrossEntropyLoss; the combined loss is 0.5 × CE + 0.5 × Dice loss. In all scripts the validation images carry the training augmentations, so validation accuracy is lower and noisier than test accuracy.